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# LAUMED

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# HEALTH & LIFESTYLE

GAMING DISORDER IN ADOLESCENTS PG. 6 |

TRANSIENT SMARTPHONE BLINDNESS PG. 25 |

LAUMED JOURNAL CLUB

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*...more is definitely better!*

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**Greetings.**

It is a new world. And we have, for the last few months, been rendered vulnerable to the effects of Covid-19, a widespread global disease that had our world grounded. Even in a new world, change is constant, change is key. So, due to the pandemic, we made the difficult yet necessary decision to move our publication this year from print to electronic form on screens. This is very important as we continue to seek for sustainable ways to live our lives.

It is our second issue and counting. Here, we take a careful look into the many ways in which our lifestyles impact on our health. What we present is not an attempt at a guesswork but a careful examination of cause and effect. Where are our decisions taking us? What are the variable possibilities that come with our choices, how differently can we live?

Many thanks to all our members, the contributors, the editor-in-chief and our editorial consultants who have slaved over the production of this issue. Together we continue to strengthen the LAUMED Journal by creating an environment of learning through collaboration. While we may not have been able to see in person for a while, a time like this is a true litmus test of our solidarity and resolve as a team. We must continue to work together so we can attain notoriety in scientific writing and be better in more ways. Please, take care of yourself & yours and stay safe. More is definitely better.

Go very well, people.

**Ayotunde OJO,**

Pioneer Editor-in-Chief LAUMED Journal.



# Editorial

The beginning of this year spoke of joy, only to be crushed by the weight of the inevitable challenges that came upon us. From the burning fire of hope we gathered the ashes of hopelessness. COVID-19 has been a universal solvent. On 11 March, 2020, the WHO Director-General declared COVID-19 a pandemic. We suddenly realized we are fighting an enemy that knows neither country boundary nor colour. The enemy distorted our stratifications in life; rich and poor, white and black, intelligent and dull. The rich battles the invisible enemy the same way the poor does. Personal relationships have been altered, events have been cancelled and we have all adopted new ways of life. It has indeed been a pandemic.

For LAUMED JOURNAL CLUB, the pandemic posed a unique challenge. We knew that the desire to progress is eternal but in a year laden with jolting challenges; one which compelled our dreams to holiday and made our visions somnambulate as the Earth rotated on its axis, desire will be shown to be insufficient in the journey of progress. The maiden edition of this journal, published November of last year, opened two doors. The door to merriment led us into a moment of euphoria over the successful publication and after that we were faced with the challenge of continuity. But here we are with the second edition of the journal published against all odds.

Health is influenced by a complex interplay of physical, social, economic, mental, cultural, and environmental factors. This highlights the need for it to be approached with bold and efficient dynamism . The current pandemic has shown that there is a need to re-examine what lies at the heart of health, which is the health and well being of people, in Nigeria and the world as a whole. Health system and services is gradually becoming overly disease focused, biomedical oriented, technology driven and doctor dominated. However, the current pandemic provides a unique opportunity to be reminded of the core focus of any health system – the people. Health is a resultant effect of people's lifestyle and health in end affects people's lifestyle. This edition provides a perfect backdrop of how different lifestyle patterns affect health and the feedback mechanism of health on lifestyle. This edition resonates a message that is important for everyone.

My heartfelt gratitude goes to Dr Akintunde A.A, the chief editorial consultant, who carefully reviewed all the articles; our supervising consultants, who supervised our works meticulously; Dr Kayode Valentine, the pioneer chief editorial consultant, for his love and support; Dr Egbewale for his fatherly advice; Dr Olagunoye for guiding us through our choice of the theme for this edition; Dr Ajufo, the immediate past editor in chief, for her constant care and golden advice; Dr Ojo, the pioneer editor in chief, for his unwavering support; past board members, sponsors, medical doctors and students who stood by us through this year. To the current board members, I am immensely grateful for the effort you have put in, indeed, team work makes the dream work.

Enjoy your read.

**ADERINTO NICHOLAS O.**
EDITOR IN CHIEF,
DECEMBER 2020



# Foreword

**Dr. Akintunde Adeseye A.**
**(MBChB, FWACP, FMCP, Dr.med., FACC, FESC)**

I will like to congratulate the editorial team under the able leadership of Aderinto Nicholas for sustaining the vision of the founders of this laudable project. Keeping fate at a time like this when Covid-19 has since March 2020 affected every facet of life is simply a commendable work. The dedication of the various writers and the supervising consultants are also thankfully acknowledged.

The place of active research in any society determines the future of such people and any group of people that is not constantly applying research methods to evaluate present challenges and scientifically solve them is set to dissolve into a future of no relevance and extinction. This is classically reminiscent of the worldwide effort at critical research into covid-19 vaccine by dedicating a critical mass of resources since the pandemic started which have since yielded at least three successful vaccine and still counting with a possibility of turning the pandemic into a 'flu-like' condition in the nearest future. It is therefore expedient that students including medical students with rigid and extensive training in clinical activities are encouraged to participate in active research to solve human problems with a view to creating a critical mass of problem-solvers for the future. This is often the driving force for my involvement for the younger population and I will like to commend the LAUTECH Medical Students' Association for bringing this to fore and keeping the dream alive.

This second edition is looking at an all-time important discuss on the issue of health and lifestyle. It comes at a point in time when the burden of diseases of lifestyle otherwise called non-communicable diseases including heart diseases, stroke, kidney diseases, diabetes mellitus etc. are on the rise especially in developing world like Nigeria where we are. The increased burden is predicated on the changing lifestyle on a vulnerable genetically predisposed population. The burden of obesity, hypertension, dyslipidaemia and other risk factors are on a dangerous level with some reports of more than 40% of adult population having at least each of these risk factors. The advent of technology has also led to various issues which enhance the development of these risk factors. There is therefore a need to reappraise our health care and health behaviour to encourage imbibing healthy lifestyles that can mitigate the development of cardiovascular and non-cardiovascular diseases.

Without reservation therefore, I commend this work to the readership for an enlightening session and wish that it will engender appropriate behavioral shift and increased health awareness for all and sundry.

Congratulations and happy reading.

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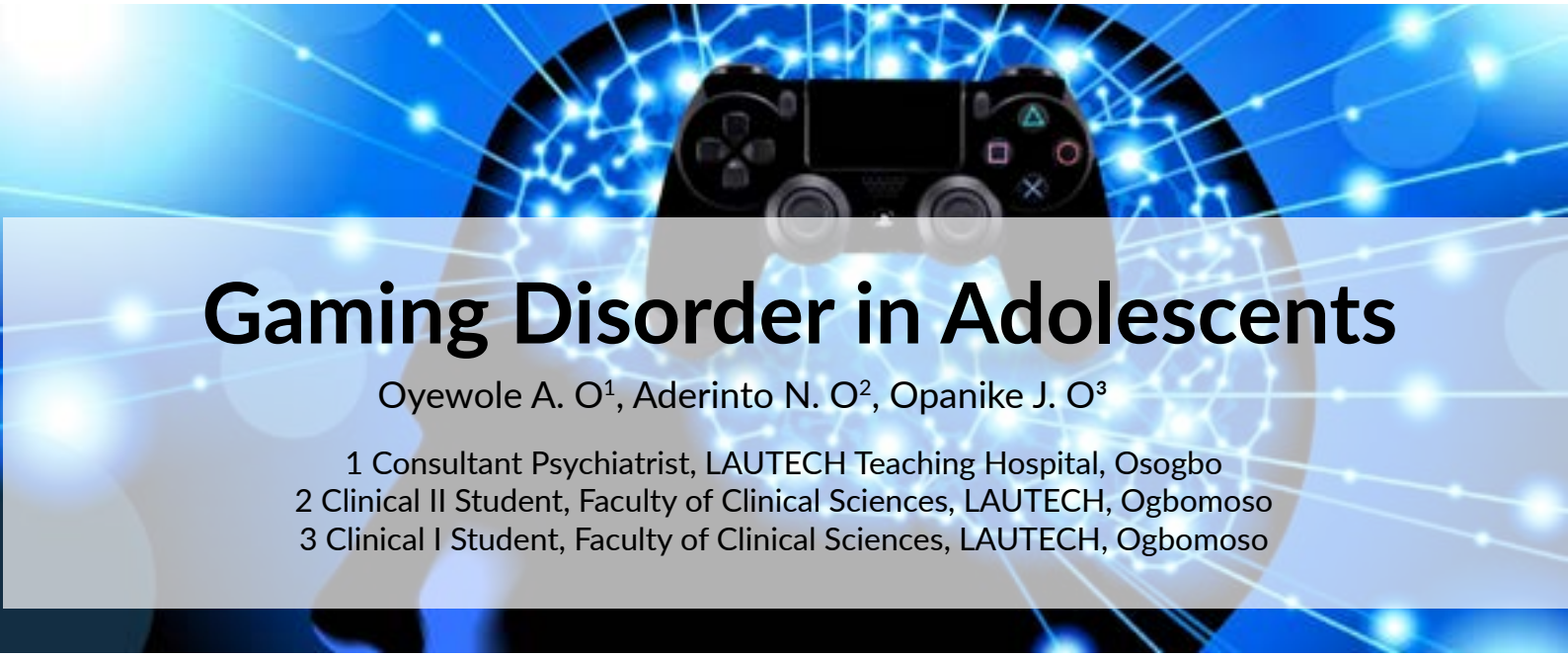

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# Gaming Disorder in Adolescents

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## INTRODUCTION

The advent of computers and its evolutionary transformation over centuries has had significant impact on human lives. It is particularly interesting that an innovation that seeks originally to achieve an easier way of doing things has increasingly become a necessity for life; from information to banking to entertainment to health to spiritual worship. Truly the limits of its prowess are still emerging.

The idea to create and design simple games began in the 1950s by academic scientists as part of their research or simply for fun and it didn't reach the mainstream until the 1970s and 1980s1. The evolution of gaming over the past decades parallels that of technological advancements in computing and hardware. There has been a major leap from expensive video consoles of the 1980s-1990s to relatively cheap games accessible on smartphones both online and offline from the 2000s till date2. As online games become more and more popular, gaming clans began to emerge all over the world. A clan, guild or faction is an organized group of players that regularly play together in multiplayer games. Such groups range from a few friends to thousands of players with a broad range of organizational structure, goals and members. There are forums where clans are rated against one another2 thereby opening up a community for diverse social interactions.

Truly, the gaming industry is quite lucrative for both players and companies. There are 2.5billion gamers all over the world3 in an industry projected to produce $196 billion in revenues by 20224. What's more, video games are challenging the definition of sports as teenage competitors are becoming more popular than professional athletes4 creating an emerging career path in the increasingly virtual world we live in.

However, gaming is more than just a booming entertainment business. It is reshaping the way gamers interact with their world and beginning to create a cankerworm of challenges for clinicians, researchers and policy makers. Generally, associated mental health phenomena with problematic use of digital media have been researched and studied since the late 20th century. Although there are no standardized terms used to refer to compulsive digital media use, digital addiction or dependencies are used to describe the disorders associated with internet and smartphone use. Common digital addiction disorders are (but unlimited to) online gambling, cyber bullying and online gaming disorders also called internet gaming disorders or for short, gaming disorders.

## GAMING DISORDER

In 2018, Gaming Disorder (GD) was defined in the International Classification of Diseases (ICD) publication of the WHO as a pattern of gaming behavior characterized by impaired control over other activities to the extent that gaming takes precedence over other interests and daily activities and continuation of gaming despite the occurrence of negative consequences.5 Many countries are recognizing GD as a public health issue and policies have been put in places by countries such as China, Switzerland and Iran to deal with it.6 Many clinics have been established in many Asia countries where clinical services for GD

are most developed. Clinics are also being developed in North America and Europe.7 Gaming Disorder is a topic of controversies. This is mainly due to lack of consensus on diagnostic criteria. There are questions on whether GD should be considered a disorder in the face of insufficient evidence because there could be a premature application of diagnosis in the medical community and treatment of false positives leading to stigmatization of heavy gamers who may not be unwell8. The Diagnostic and Statistical Manual of mental disorders, DSM (a publication of the American Psychiatry Association) in its fifth edition (DSM-5) in 2013 recognized GD as a condition for further research acknowledging that GD is a real issue but that criteria for classification needed to be sufficient to classify GD as a unique mental disorder9.

## THE ADOLESCENT PICTURE

Usually it is expected of gaming to be more popular with young people. In the US, 61% of gamers were 35 years and younger and a third of these are under 18 years10. The definition of an adolescent is a young person in the process of developing from a child to an adult. They are also referred to as teenagers. WHO categorizes individuals in the 10-19 year age group as adolescents and according to WHO, they comprise 20% of the world's population which is quite significant. The categorization based on age is only a rough marker that leaves no room for the physical, psychological or cultural expression which may begin at an earlier age or end at a later one. It is increasingly being accepted that the adolescence is a stage requiring special care and attention. This is seen in the emergence of such specialized care as Adolescence medicine or Transitional Medicine.

The adolescent stage is a period of unprecedented development of the body's anatomy and physiology, and of social interaction. The boy or girl becomes 'aware' of his environments - the sociocultural, economic and interpersonal elements, and begins to learn to respond to it in a broad range of ways. The adolescent suddenly finds himself in a complex world and tries to make sense of his role in it. This is also a stage of cognitive development and the adolescent goes ahead to make decisions for himself even when supposed consequences of such actions have been mentioned or depicted to him before. The adolescent explores his world beginning to recognize his passions and the things he doesn't want to do. He is able to visualize the future and put himself there, something adults have no luxury of doing and which children didn't even know existed. It is a time of discovering self and the opinions and values such an individual makes his own at this time defines their adulthood.

While development is ongoing, adolescents constantly seek a safety net to slip into when overwhelmed by the challenges they face. This protection could be provided by supportive atmosphere in the family, school and the bigger community. The absence of a well-structured support system is capable of exposing the adolescent to a number of psychological issues such as mood disorders, anxiety disorders, eating disorders and addictive disorders whose outcomes would now depend on a variety of factors chief of which is the media. Media influence and increased access to the use of technology can exacerbate the disparity between an adolescent's physical reality and his perception or aspirations for the future11 and this is what makes matters worse and is the very inglorious role of Gaming Disorder in the lives of adolescents.

## PREVALENCE

The worldwide prevalence of Gaming Disorder is 0.7-27.5% with the highest prevalence recorded in Asia12. A study conducted in Thailand indicated a prevalence of 5.4% amidst secondary students aged 12-18 years13. While these values seem insignificant, the increasing prevalence of GD say otherwise. There is a higher male prevalence for GD12 and a particular study showed differences in neuroimaging of male and female subjects. Females are most likely to use internet for social media than for gaming.

Despite the increase of studies on GD and technology related disorders around the world, this topic remains understudied in Africa14. In Nigeria which ranks first in internet connectivity on the continent with 123 million subscribers has a prevalence of 26.9% for GD14,15. While no exact data currently exists for prevalence of GD amidst adolescents in Nigeria, studies have shown that sociocultural differences may not be associated with GD13 which means that identified statistics, risk factors and clinical features from non-localized studies could apply.

Identified risk factors for GD include an unstable home environment, underlying personality and psychological problems. Adolescence is characterized by a need for autonomy and competence in decision making and an overbearing or dictatorial parental approach could have negative consequences with adolescents not opening up to their parents and channeling their emotional and psychologically

deprived minds to gaming16. The increased rate of divorce and separation which leaves children at the center of a war between parents is also a contributing factor.

The male sex is also a risk factor.

## CLINICAL FEATURES

According to the DSM-5, gaming must cause significant impairment or distress in several aspect of a person's life before a diagnosis of GD can be made9. There are a wide range of symptoms but generally, patients with GD are unable to limit the amount of time they spend gaming and continue to play despite negative consequences which could include lower academic achievement and consequent dropping out of school, aggression and social phobia with a low level of productivity and in some cases more insidious health challenges like clot formation due to inactivity.7 Proposed symptoms of GD by DSM-5 include the following:

- Preoccupation with gaming
- Withdrawal symptoms when gaming is taken away or not possible
- Tolerance, the need to spend more time gaming to satisfy the urge
- Inability to reduce playing, unsuccessful attempts to quit gaming
- Deceiving family members or others about the amount of time spent on gaming
- The use of gaming to relieve negative moods.
- Risk having jeopardized or lost a job or relationship due to gaming9

GD is associated with stress, anxiety and depression with the relationship going either way. A stressed individual may turn to gaming to escape these feelings. In the other way, GD may cause depression, stress and anxiety due to its impact on an individual's neuro-circuitry13. Organic disease involving the visual system were reported mainly due to the inability of the human optic system to tolerate prolonged viewing of digital images from electronic devices17. Musculoskeletal problems such as pain and tendinosis are characteristic presenting complaints of chronic gamers. Pathological entities such as Wii fracture and PlayStation thumb have also been reported17.

It's important here to mention that gaming has been found to have various effects on the academic performance of adolescents based on the time spent gaming and the diversity of games played with gamers of long hours and narrow diversity recording poor performance18. Likewise, implications of gaming based on weekday or weekend playing on academic performance are two different sides of a coin with weekday gaming having substantial negative effects on academic performance19. This is basically due to attention deficits and considerably low study time as most of the time has been spent gaming. While gaming could be a catalyst for poor academic performance, the relationship remains difficult to define due to the complex nature of student life and academic performance20. Noteworthy of mention is the application of gaming to stimulate interest and improve academic performance. The widespread success of such intervention are yet to be seen.

In Nigeria, there has been an escalation of the worrisome trend of poor academic performance of adolescents in recent years. Top on the list of causes is a lack of interest in academic activities and an abundance of distractions that make it difficult for young students to focus on academic activities. These are closely associated with the perception that sterling academic performance doesn't guarantee financial success. In a world that is increasingly materialistic, academic performance keeps sinking down the priority list of young people. The boom of technology has not helped matters. While electronic devices could pass as great study tools, many young people are actually distracted by them with only a few ever tapping into the numerous resources they provide. While gaming may not directly correlate with poor academic performance among Nigerian adolescents, its influence on cognitive performance makes it an important subject of consideration.

## DIAGNOSIS

There are a number of diagnostic tools currently in use for assessment of GD but lack of consensus renders many of them invalidated:

- The Internet Gaming Disorder Test (IGD-20)
- Internet Gaming Disorder Scale – Short Form (IGDS9-SF)
- The Internet Gaming Disorder Scale (IGDS)
- The Ten-Item Internet Gaming Disorder Test (IGDT-10)

- Clinical Assessment Tool (C-VAT 2.0)
- The Personal Internet Gaming Disorder Evaluation-9 (PIE-9)[21]

## MANAGEMENT

Treatment of GD is quite challenging because of the ubiquity of the internet. Approaches tend to focus on getting patient to recognize their addiction and to carefully pull them from their virtual world to reality. A mix of cognitive behavioral therapy, social skills development and treatment program emphasizing physical activity was used in a specialized clinic in Japan. Another clinic also in Japan used psychotherapy to reconnect patients with themselves, their life objectives and their social environment7.

Apart from clinical approach to treatment, parental roles have been identified. General parental monitoring including knowledge of adolescent's activities and setting of rules on internet and electronic device use can potentially reduce gaming disorder22. An authoritarian approach could lead to apathy to and rebellion which could complicate matters. Therefore, communication is the best strategy22. Parents need to create time out of their busy schedules to engage with their children, not monotonously but as a desirable relationship. This could be quite challenging for parents in the western world whose children have developed an early awareness and pursuit of independence. Parents in more traditional societies have the leverage of reference and respect for adults as an advantage.

Also, a sound father-child relationship is beneficial and seems to be more important than mother-child relationship in preventing and dealing with gaming disorder22. Children that grow up without their fathers are usually more likely to experience behavioral disturbances than those who did. This is possibly due to the father's unique contributions in attending to the child's activities and (in the case of gaming disorder), the provision of rules and supervision of internet use22. The mother's contributions of emotion warmth and communication are also valuable.

Apart from parents, school teachers are an important influence on adolescents. The ability of the teacher to provide firm but loving guidance for adolescents will go a long way. Many adolescents who treat their parents with disrespect actually worship their teachers. A careful exploitation of this will provide adolescents with troubled homes a safety net to land on.

## CONCLUSION

The internet, like alcohol and drugs creates a false sense of security for adolescents who are going through a transition period and are overwhelmed. In countries like Nigeria where they tend to be neglected, efforts should be trumped to create a safe enabling environment for their survival.

With a worldwide prevalence ranging from 0.7-27.5% and as more details of GD continue to surface, researchers, clinicians and policy makers need to focus on this emerging condition with the aims of obtaining relevant data, developing relevant clinical tools and designing public health policies to manage the disorder with regards to the local population's needs.

## REFERENCES

1. History of Video Games. https://en.m.wikipedia.org/wiki/history-of-video-games

2. Riad Chikhani. The History of Gaming: an evolving community. https://techrunch.com/2015/10/31/the-history-of-gamin-an-evolving-community

3. Tom Wijuan. The Global Games Will Generate $152.1 billion in 2019 as the US Overtakes China as the biggest market. 2019

4. Kevin Webb. Video Games Industry Reports. Business Insider. Oct 1, 2019

5. Online Q&A. Gaming Disorder. https://www.who.int. Sept 2018

6. John B. Saunders. Journal of Behavioural Addiction. J. Behav Addict. 2017 Sept; 6(3):271-279

7. Bulletins of the WHO. Vol 97, Num 6, June 2019 377-440. 2019

8. Espen Aarseth. Journal of Behavioral Addiction. J. Behav Addict. 2017 Sept; 6(3); 267-270

9. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders,5th disorder. 2013

10. Christina Gough. Average age of U.S video game players in 2019. Statista. 2019

11. https://www.who.int/news-room/fact-sheets/detail/adolescent-mental-health

12. Satoko Mihara BA; Susumu Higuchi MD, PhD. Cross Sectional and Longitudinal Epidemiological Studies of Internet Gaming Disorder: A Systematic Review of the Literature. Psychiatry and Clinical Neurosciences/ Vol 71, Issue 7. 24 April 2017

13. Pawan Taechoyotin et al. Prevalence and associated factors of internet gaming disorder among Secondary School Students in Rural Community, Thailand. BMC research notes 13, Article number:11(2020)

14. F.A Etindele Sosso et al. Insomnia, Sleepiness, Anxiety and Depression among different types of Gamers in African Countries.

15. J. Clement. Africa: number of internal users in selected countries 2019. https://www.statista.com. Jan 7, 2020.

16. Vasileios Stavropoulos et al. A preliminary cross-cultural study of hikikoori and Internet Gaming Disorder: The moderating effects of game playing-times and living with parents. Addictive Behaviour Reports. https://doi.org/10.1016/j.abrep.2018.10.001. 2019.

17. Ayenigbara IO (2018) Gaming Disorder and Effects of Gaming on Health: An Overview. J Addict Med Ther Sci 4(1): 001-003. DOI: http://doi.org/10.17352/2455-3484.000025

18. Matthew Ventura, Valerie Shute, Yoon Jeon Kim. Video gameplay, personality and academic performance. Computers& Education. Vol 58, Issue 4. https://doi.org/10.1016/j.compedu.2011.11.022. May 2012.

19. Andree Hartano, Wei Xing Toh, Hwajin Yang. Context counts: The different implications of weekday and weekend video gaming for academic performances in mathematics, reading and science. Computers& Education. Vol 120, pages 64-74. May 2018. https://doi.org/10.1016/j.compedu.2017.12.007.

20. Vivek Anand. A study of Time Management: The Correlation between Game Usage and Academic Performance Markers. CyberPscychology& Behaviour, Vol. 10, No. 4. August 2007 https://doi.org/10.1089/cpb.2007.9991

21. Hally M. Pontes. Current Practices in the Clinical and Psychometric assessment of internet gaming disorder in the era of the DSM-5: A mini review of existing assessment tools. Mental Health and Addiction Research. 2016

22. Binyuan Su et al. Father-Child Longitudinal Relationship: Parental Monitoring and Internet Gaming Disorder in Chinese Adolescents. Front.Psycho 9:95 https://doi.org/10.3389/fpysg.2018.00095. 2018

# SLEEP, HEALTH AND WELLNESS

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## 1.0 INTRODUCTION

Sleep plays an important role in health and wellness. Alteration in the normal sleep pattern as required by an individual leads to a distorted day. Decreased duration and quality of sleep, whether due to sleep disorders or lack of proper sleep patterns, have been linked to cardiovascular disease risk factors such as hypertension, dyslipidemia and obesity. [1] The terms Health and wellness are often used interchangeably, but each has distinct origin and meaning. As established by the World Health Organization (WHO) in the 1940s, health is referred to as "a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity". Although this definition has been criticized for being overly inclusive and unattainable, especially as it relates to the word "complete," very importantly, it broadens the medical definition of health beyond the absence of disease. [2] Can stress affect health? Absolutely. It affects the physical, mental, and social well-being. Can stress affect wellness? Yes, but only to a point, it affects one's perspective or dampens one's energy in turn negatively influencing one's commitment to self-care and the aim to achieve one's full potential. Wellness is always a matter of choice. [3] Sleep in most people occupies between 20% and 40% of the day. Prehistoric evidence suggests the importance of sleep in human life [4] which is consistent with archaeological and historical accounts of sleep having a prominent and important role in early human society. Sleep is a universal phenomenon that is inescapable and thus was incorporated into social structures. In this way, sleep became not just a set of physiologic processes, but one represented in sociocultural structures. [5]

## 1.1 HOW SLEEP INTERACTS WITH HEALTH AND WELLNESS

Aside sleep, many factors interact with our health in negative ways. Irregular exercise, emotional stress and poor eating habit are examples. Sleep appears to affect many processes in the body including energy metabolism, immune system function, learning/ memory, appetite regulation, and gene expression. [6] But these do not undermine the catastrophic effect and damage poor sleep pattern causes. In the complex relationship between sleep, health and wellness, sleep impairment acts as a modulator. Sleep disturbances may be both the cause and the consequence of reduced wellness and therefore set a cascade of events with consequences affecting productivity and in the long run health. [7] Studies have shown that people who sleep fewer than 5 hours per night has an increased risk for type 2 diabetes. [8] The interaction between sleep and immune system also increases the levels of many inflammatory mediators. [9] Short sleep duration and other aspects of poor sleep habit are growing in modern society in both children and adults. Literature has established that short sleep duration and other dimensions of poor sleep are associated with obesity and appear to predict obesity risk and rate of weight gain. [10] Despite a clear understanding that sleep is vital to health, the exact role played by sleep remain controversial. [11] Among the leading theories for the role of sleep is that it provides a time of relative cortical quiescence to allow brain energy stores to rebuild, that sleep allows for synaptic pruning which facilitates neural plasticity allowing memory and learning [12] and that sleep enhances clearance of neural waste products through the glymphatic system which is activated during sleep. [13] In addition to brain health, there is growing interest in the role sleep may play in somatic health. Growth hormone secretion during sleep may provide one mechanism for the restoration of periphery and regulation of immune system function. Studies have shown that short sleep duration is associated with poor mental health. Sleep disruptions are common diagnostic features of mental health disorders. [14] Patients with mood disorders and anxiety disorders frequently experience short sleep duration. Sleep duration has also been identified as a suicide risk

factor. [15] In the general population, overall mental health has been identified as the leading predictor of self-reported insufficient sleep. [16]

## 1.2 NEUROANATOMY OF SLEEP

Sleep is a dynamic process involving complex neural activation. Neurons of the reticular system are central to the regulation of wakefulness. There are two ascending pathways. A dorsal route from the cholinergic laterodorsal and pedunculopontine segmental nuclei which activates thalamic neurons to promote electroencephalogram activity via glutamatergic thalamocortical projections. [17] The ventral route through the hypothalamus includes the aminergic arousal system that originates from the brainstem with serotogenic , noradrenergic, histaminergic and dopaminergic neurons. [18] Sleep timing, depth, and duration in humans are regulated by anatomical sites such as the anterior hypothalamus, reticular activating system, suprachiasmatic nucleus and the pineal gland. This is governed by interactions between two processes; process S and process C. [19] Sleep is also regulated by homeostatic and circadian factors, and the regulation of sleep shares many molecular properties with the rest state of sub-mammalian species. Several brain structures take part in wakefulness; the basal forebrain, posterior and lateral hypothalamus, and nuclei in the tegmentum and pons. [20] Waking is associated with neuronal activity in several chemically defined ascending arousal systems. This includes monoaminergic neurons in the brainstem and posterior hypothalamus, cholinergic neurons in the brainstem and basal forebrain, and hypocretin (orexin) neurons in the lateral hypothalamus. Collectively, these systems impart tonic activation to their neuronal targets in the diencephalon and neocortex that is reflected in the low-voltage fast-frequency electroencephalogram patterns of wakefulness. [21] Some biochemical substances (such as adenosine, prostaglandin D2 and some cytokines) that accumulate in the brain during wakefulness stimulate sleep-promoting GABAergic neurons located in the pre-optic region. In turn, these GABAergic neurons inhibit the aminergic (wake-promoting) neurons located in the tubero-mammillary (histaminergic) nucleus, in the dorsal raphe (serotoninergic neurons) and locus coeruleus (noradrenergic neurons). Inhibition of the wake-promoting neurons leads to drowsiness and favours the onset of sleep. When adenosine, PGD2 and other somnogenic substances decrease during sleep, aminergic neurons are released from the inhibitory control of GABAergic neurons, thus favouring awakening. [22]

## 1.3 NEUROPHYSIOLOGY OF SLEEP

Neurophysiology of sleep is the study of the neuroscientific and physiological basis of the nature of sleep and its functions. Structures in the brainstem use various neurotransmitters to influence higher brain structures in the midbrain and cortex. An active cortex that exhibits a characteristic pattern of desynchronized EEG manifests wakefulness. During wakefulness, there is increased sympathetic tone and decreased parasympathetic. [23] Extensive interactions have been identified between sleep and the endocrine system. Numerous endocrine factors can affect sleep duration and quality. Studies have shown a profound effect of sleep behaviour on overall endocrine function and stability. [24] For the endocrine system, the circadian system creates "biological day" and "biological night". Hormones and other biological properties have been shown to fluctuate on a 24-h cycle according to stereotypical patterns; lesions of the circadian system have been shown to disrupt the daily fluctuation of hormone levels such as growth hormone [25], cortisol and leptin. [26] The evidence for sleep as a homeostatic process is well-established. Like other homeostatic systems, sleep pressure responds in an analogy pattern to the relative distance from its homeostatic mean. In essence, sleep pressure directly increases with wake time. Recently identified glymphatic system may provide a mechanism for control of brain microenvironment homeostasis that is sleep-dependent. The glymphatic system is a fluid-dynamic model of cerebrospinal (CSF) and interstitial (ISF) fluid flow around the brain and through the brain parenchyma. This flow has been shown to be important for clearance of metabolites and other waste products including amyloid beta protein from the brain. [27] Steroid receptors, particularly for estrogen and progesterone, are present throughout the brain and prevalent on multiple sleep-regulating nuclei such as the hypothalamus[28] and basal forebrain. Insulin is another hormone entrained rhythmically. Insulin, a pancreatic hormone, has a principal function of promoting the absorption of glucose from the blood. However, in experiments of clamping glucose concentration, insulin still rises in the late phase of the sleep cycle, most likely to restrain hepatic glucose production in order to prevent a glycaemic peak. [29]

## 2.0 SLEEP

Despite being among life's common behavioural states, sleep remains a phenomenon that resists easy explanation. Sleep is generally characterized by an increased threshold for response to sensory input, a decrease in motor function, and a lack of consciousness. It has come increasingly clear, however, that no matter how hectic lives may be, we cannot afford to ignore what research is telling us about the importance of sleep for mental and physical well-being. Sleep serves multiple functions. [30] One of the simplest aspects of sleep is duration; the total amount of time spent asleep. For children and adolescents, the American Academy of Sleep Medicine recommends eight to sixteen hours per 24-hour period with age-specific recommendations decreasing as age increases. [31]

## 2.1 STAGES OF SLEEP

Between 1953 and 1957, Kleitman's laboratory in Chicago made a discovery that revolutionized sleep physiology and gave birth to sleep medicine. Kleitman and his co-workers discovered that there are two types of sleep: one is identified by EEG tracings alone since the thirties and characterized by high amplitude EEG waves (sleep spindles and slow or delta waves) while the other identified by recording eye movements had been ignored because its EEG features (low amplitude fast EEG activity) could not be distinguished from those characterizing active wakefulness and drowsiness. [32] Sleep associated with eye movements and vivid dream activity was called REM sleep (sleep with Rapid Eye Movements). Sleep associated with eye movements and vivid dream activity was called REM sleep (sleep with Rapid Eye Movements). Between 1958 and 1962, Michel Jouvet, a French physiologist, demonstrated that the executive mechanisms responsible for REM sleep are located in the rostral part of the pontine reticular formation. [33] He called REM sleep "paradoxical sleep" (PS) because it represented a strange neuropsychological situation: a waking (dreaming) brain in a "sleeping" body (skeletal muscles were functionally blocked during REM sleep). During sleep, our body cycles through 4 different stages consisting of both rapid-eye-movement (REM) and non-rapid eye movement (NREM) sleep. The body usually cycles through these stages on average 4 to 6 times, averaging 90 minutes in each stage. As the night progresses, fewer NREM stages occur, and the duration of REM sleep episodes increases. [34] Sleep is broken down into 5 phases: wake, N1, N2, N3, and R. Stages N1 to N3 have considered non-rapid eye movement sleep, each progressively going into deeper sleep. The majority of sleep is spent in the N2 stage. [35]

Wake

The first stage is the wake stage or stage W, which further depends on whether the eyes are open or closed. During eye-open wakefulness alpha and beta waves present, predominantly beta. As individuals become drowsy, and the eyes close, the alpha rhythm is the predominant pattern. [36]

**N1**

Stage 1 is the lightest stage of NREM sleep. Often defined by the presence of slow eye movements, this drowsy sleep stage can be easily disrupted causing awakenings or arousals. Muscle tone throughout the body relaxes and brain wave activity begins to slow from that of wake. Occasionally people may experience hypnic jerks or abrupt muscle spasms and may experience sensation of falling while drifting in and out of Stage 1. [37]

**N2**

This stage represents deeper sleep as the heart rate and body temperature drop. It is characterized by the presence of sleep spindles and K-complexes or both. The K-complexes show a transition into a deeper sleep. They are single, long delta waves lasting for a second. Stage 2 sleep lasts around 25 minutes in the initial cycle and lengthens with each successive cycle eventually making about 50% of total sleep.

**N3**

This is considered the deepest stage of sleep and is characterized by a slower frequency with high amplitude signals known as delta waves. This stage is the most difficult to be awaken from, and for some people, even loud noises (over 100 decibels) will not wake them. Although this stage has the greatest arousal threshold, if someone is awoken during this stage there will be transient phase of mental fogginess known as sleep inertia. This is the stage when the body repairs and regrows its tissues, builds bone and muscle, and strengthens the immune system.

**REM**

REM sleep, also known as rapid eye movement, is most commonly known as the dreaming stage. Eye movements are rapid, moving from side to side and brain waves are more active than in Stages 2 & 3 of sleep. Interestingly, the EEG is similar to an awake individual but the skeletal muscles are atonic and without movement. [38] This stage usually starts 90 minutes after sleep, and REM cycles gets longer throughout the night. The first period typically lasts 10 minutes, and the final one can last up to an hour. [39]

## 2.2 EFFECT OF LIGHT ON SLEEP

Light is necessary for life. Artificial light improves visual performance and safety, but there is an increasing concern of the potential health and environmental impacts of light. To understand the effects of light on the human physiology, it is important to understand light. According to Merriam-Webster, light is an electromagnetic radiation of any wavelength that travels in a vacuum with a specific speed. Two effects of light have been interrogated extensively in human circadian and sleep research: (1) the acute suppression of melatonin in response to light exposure and (2) the ability of light exposure to shift circadian phase. [40] A night of light exposure during sleep acutely impacts insulin resistance, and chronic overnight exposure may have long-term effects on metabolic function. Light exposure at night is known to impact circadian rhythms such as melatonin and sleep, and disturbances in this rhythm have been shown to affect cardio metabolic functions. [41] Exogenous administration of melatonin has also been associated with sleep induction occurring mostly during period of insufficient sleep drive. [42] Sleep timing, duration, and structure have been linked to both evening and morning single-day bright light exposure with morning light advancing sleep timing and leading to a reduction of sleep duration at the expense of rapid eye movement (REM) sleep. [43] Some field studies have gone further in assessing how light exposure influences sleep when individuals have the freedom to habitually sleep and conduct their usual daily routines. Objective actigraph and subjective field measurements have shown that early sleep timing is related to more daytime light exposure, either with [44] or without imposed sleep schedules. [45] Early light timing also resulted in early sleep phase and repeated bright daytime light exposure has been observed to lead to earlier sleep phase in Alzheimer's disease patients in real-life settings. [46] Exposure to higher levels of maximal light was related to less REM in subsequent sleep. Light exposure in the evening and at night can suppress and delay (phase-shift) the normal melatonin secretion. In addition to melatonin secretion, light exposure affects sleep-wakefulness cycle. Thus, as rapid-eye-movement (REM) sleep timing is controlled by the master circadian clock, the circadian rhythm disruption can also be detected by measuring REM sleep parameters with electroencephalogram (EEG). [47] A circadian rhythm can be determined by measuring melatonin levels. Melatonin is a hormone that signals night to the human body. Melatonin secretion is cyclic throughout the day; its secretion is high at night (in the dark) and low during daytime. The level of high melatonin secretion at night is not equal for all people, but it varies depending on age and sex. [48]

## 2.3 LATE NIGHT SLEEP

Bedtime is frequently delayed by many factors. Homeostatic response to the delay may compensate partly for shortened duration of sleep. Sleep duration becomes shorter with age and women complain of disturbed sleep more often than men. [49] Sudden demands in life can force delay in normal

bedtime and reduce sleep duration. This has immediate effects on fatigue/alertness seen in increased physiological and self-reported sleepiness. Sleep quality is perceived as good when it is easy to fall into sleep and wake up, and when sleep is continuous and long enough. Thus, both subjective and objective measures of sleep are needed to analyse the perception and the architecture of the sleep, respectively. [50] However, vigorous late-night exercise is mentioned by the American Academy of sleep as one possible practice that may produce increased arousal and lead to inadequate sleep hygiene. Lifestyle factors such as diet (heavy meals, tyramine/tryptophan rich foods), alcohol, caffeine, drug use, and exercise may contribute to sleep disturbances.

## 2.4 SLEEP DEPRIVATION

Sleep deprivation is common among university students and has been associated with poor academic performance and physical dysfunction. The effect

of sleep deprivation on cognitive performance has also been documented previously with a correlation between sleep quality and grade point average in first year university students. [51] According to the National Institute of Health, sleep deficiency is a broad concept that occurs (a) if an individual does not get enough sleep (sleep deprivation); (b) if an individual's sleeping habits are out of sync with the body's natural circadian rhythm (sleeping during the wrong time of the day) and (c) if the quality or duration of sleep is diminished due to sleep disorders or external factors. [52] There are four variations of sleep deprivation: insomnia, acute total sleep deprivation (TSD), partial sleep deprivation (PSD), and night shift workers. Acute TSD refers to the avoidance of sleep for a period of at least one night. PSD or sleep restriction refers to the reduction in the total sleep time relative to one's usual baseline during a 24-hour period. PSD is the most common form of sleep deprivation encountered in everyday life in modern societies. [53] Insomnia is defined as predominant complaint of dissatisfaction with sleep duration or quality associated with one or more of the following symptoms; difficulty in initiating sleep, difficulty in maintaining sleep characterized by frequent awakenings or problems returning to sleep after awakenings, or early morning awakenings with inability to return to sleep. A shift worker is anyone who follows a work schedule that is outside the typical "9am to 5pm" business day schedule. Sleep deprivation results in poor memorizing, schematic thinking (which yields wrong decisions) and emotional disturbances such as deteriorated interpersonal responses and increased aggressiveness. The symptoms are accompanied by brain tissue hypo-metabolism particularly in the thalamus, prefrontal, frontal and occipital cortex and motor speech centres. Sleep deprivation intensifies muscle tonus and coexisting tremor. Speech performance becomes monotonous and unclear with sensitivity to pain becoming high. Sleeplessness also relates to the changes in the immune response and the pattern of hormonal secretion of the growth hormone. The risk of obesity, diabetes and cardiovascular disease increases. [54] A study conducted at James Madison University worked with 124 college students. Results from the research revealed that over 50 percent of the students reported high levels of stress that was related to academic workload and time management which was linked to unhealthy behaviours such as decreased quantity of sleep. [55] Sleep has an integral role in learning and memory consolidation, for memory formation of learned information, thus enabling students to recall information.

## 3.0 HEALTH AND WELLNESS

People often think about wellness in terms of physical health — nutrition, exercise, weight management, etc. But wellness is a holistic integration of physical, mental, and spiritual well-being, fuelling the body, engaging the mind, and nurturing the spirit (56). Although it always includes striving for health, it's more about living life fully, and is "a lifestyle and a personalized approach to living life in a way that allows you to become the best kind of person that your potentials, circumstances, and fate will allow" (57). Wellness encompasses eight mutually interdependent dimensions: physical, intellectual, emotional, social, spiritual, vocational, financial, and environmental. Attention must be given to all the dimensions, as neglect of any one over time will adversely affect the others, and ultimately one's health, well-being, and quality of life. [58] Wellness promotion augments health which, in turn, can lead to the enrichment of one's quality of life. An individual's quintessential quality of life is reflective of an optimal state of health. Notwithstanding, the phenomenon of wellness signifies the relationship between quality of life and health. [59] Although some authors do not distinguish between health and wellness, [60] others differentiate between the terms, advocating that health is a broader concept [61] still others believe that wellness is more comprehensive. [62] From the definitions, one can safely conclude that both terms are multidimensional and each influences the other. An individual can be healthy yet ill (a person who exercises compulsively, neglecting relationships, spiritual endeavours, and intellectual pursuits). The opposite is possible. A person can be well yet diseased (patients and clients treated by physical therapists for chronic disease and disability but are productive and fulfilled). The important message for physical therapists and other health care providers is that both health and wellness encompass more than physical health, and efforts that focus primarily on physical health (e.g, physical activity, weight management, nutrition), although important, fail to recognize the influence of the other dimensions to overall well-being.

## 3.1 EFFECT OF POOR SLEEP ON HEALTH

Sleep is a biologic process that is essential for life and optimal health. Sleep plays a critical role in brain function and systemic physiology including

metabolism, appetite regulation, functioning of immune and hormonal systems. [63] Healthy sleep is characterized by sufficient duration, good quality, appropriate timing and regularity, and the absence of sleep disturbances and disorders. [64] The effects of sleep disorders on the body are numerous and widely varied across the multiple body systems. As a result of the physiologic changes associated with poor sleep, numerous health consequences have been reported. Short-term consequences of poor sleep include increased stress responsivity; somatic problems; reduced quality of life; emotional distress; mood disorders and other mental health problems; cognition, memory, and performance deficits; and behaviour problems in otherwise healthy individuals. Poor sleep may also diminish the health-related quality of life of children and adolescents with underlying medical conditions. Poor sleep alters cognition and performance in many domains, including attention/vigilance, executive function, emotional reactivity, memory formation, decision-making, risk-taking behaviour, and judgment. [65] An experimental study showed that poor sleep resulted in slower or impaired information processing, impaired sustained attention, less precise motor control, and erroneous implementation of well-practiced actions. [66] Poor sleep quality negatively affects the emotional valence of memories. [67] Among primary care physicians, disrupted sleep is associated with high burnout levels. [68] The increased activity of the sympathetic nervous system that is associated with sleep deprivation has substantial long-term consequences for adults and adolescents.[69] Adults who experienced sleep disruption have elevated blood pressure [70] and an increased risk of developing hypertension. [71] The interdependent relationship between sleep and the immune system may be a factor in the effect of sleep abnormalities on common gastrointestinal disorders. Poor sleep may worsen symptoms of inflammatory bowel disease, irritable bowel syndrome, and gastroesophageal reflux disease. [72] Poor sleep is a pervasive problem, with numerous contributing factors from lifestyle and environmental factors to psychosocial issues and iatrogenic effects. Sleep is vital to major physiologic processes and as such poor sleep has vast potential for adverse short- and long-term health consequences in otherwise healthy individuals as well as those with underlying medical conditions. In healthy individuals, short-term consequences include a heightened stress response; pain; depression; anxiety; and cognition, memory, and performance deficits. In adolescents and children, disrupted sleep can lead to poor school performance and behaviour problems. Long-term consequences for otherwise healthy individuals include hypertension, dyslipidaemia, CVD, weight gain, metabolic syndrome, and T2DM. There is also evidence that sleep disruption may increase the risk of certain cancers and death in males and suicidal adolescents. [73]

## 3.2 MALNUTRITION AND OBESITY

Obesity is prevalent worldwide and has been projected to surpass tobacco use as the most economically important modifiable risk factor in public health and disease. [74] Multiple genetic, environmental and behavioural factors contribute to the increasing trend in obesity [75]. The increased availability of low-cost, high-calorie, nutrient-poor foods over the past four decades is a key component to the rise in obesity worldwide [76]. The impact of obesity on health is well documented. Being overweight makes people susceptible to an array of non-communicable or chronic illnesses including cardiovascular disease, type 2 diabetes and some cancers. The total number of people with diabetes has increased nearly fourfold, from 108 million worldwide in 1980 to 422 million adults in 2014. In 2010, studies estimated that overweight and obesity caused 3.4 million deaths. The term 'malnutrition' has no universally accepted definition. It has been used to describe a deficiency, excess or imbalance of a wide range of nutrients resulting in a measurable adverse effect on body composition, function and clinical outcome. [77] Although malnourished individuals can be under- or over-nourished, 'malnutrition' is often used synonymously with 'under-nutrition', as in this article. Malnutrition is a common, under-recognised and undertreated problem facing patients and clinicians. It is a cause and consequence of disease and exists in institutional care and the community. Malnutrition in developed countries is unfortunately commoner in situations of poverty, social isolation and substance misuse. However, most adult malnutrition is associated with disease and may arise due to: a) reduced dietary intake b) reduced absorption of micro/macronutrients c) increased energy expenditure. The consequences of malnutrition on physiological function have an important impact on clinical outcome. In the 1930s surgeons observed that patients who were starved or underweight had a higher incidence of postoperative complications and mortality. A large number of studies have subsequently supported this original observation. Malnourished surgical patients have

complication and mortality rates three to four times higher than normally nourished patients with longer hospital admissions, incurring up to 50% greater costs. Similar findings have also been described in medical patients, particularly the elderly. [78]

## CONCLUSION

While sleeping well is no guarantee of good health it does help to maintain many vital functions. One of the most important of these functions may be to provide cells and tissues with the opportunity to recover from the wear and tear of daily life. Major restorative functions in the body such as tissue repair, muscle growth, and protein synthesis occur almost exclusively during sleep. Many other conclusions about the role sleep plays in maintaining health have come from studying what happens when humans and other animals are deprived of the sleep they need. In the meantime, sleep experts say there is ample evidence that shows that when people get the sleep they need, they will not only feel better, but will also increase their odds of living healthier and more productive lives.

## REFERENCES


1. Michelle Kohansieh, Amgad N. Makarysus (2015) Sleep deficiency and deprivation leading to cardiovascular disease

2. St Claire L, Watkins CJ, Billinghurst B. Differences in meanings of health: An exploratory study of general practitioners and their patients.

3. Debbie L. Stoewen (2015) Health and wellness

4. Park DA. The fire within the eye: a historical essay on the nature and meaning of light.

5. Michael A. Grandner. Sleep, health and society.

6. Mark R. Zielinski et al. (2016) Function and Mechanism of sleep

7. Nicola Magnavita, Sergio Garbarino (2017) Sleep, health and wellness: A scoping Review.

8. Knutson KL, et al. Role of Sleep Duration and Quality in the Risk and Severity of Type 2 Diabetes Mellitus

9. Luciana Besedovsky et al. (2019) The sleep-Immune Crosstalk in Health and Disease.

10. Rachael P. Ogilvie, Sanjay R. Patel (2017) The epidemiology of sleep and obesity.

11. Philip Hunter (2008) To sleep, perchance to live. Sleeping is vital for health, memory, cognitive function and long life

12. Andy R. Eugene, Jolanta Masiak (2015) The Neuroprotective Aspect of Sleep.

13. Giulio Tononi, Chiara Cirelli (2014) Sleep and the price of plasticity: From synaptic and cellular homoestasis to memory consolidation and integration.

14. Andrew D. Krystal (2012) Physchiatry disorders and sleep

15. Subhajit Chakravoty et al. (2015) Sleep duration and insomnia symptoms as risk factor for suicidal ideation in a nationally representative sample.

16. Vijay Kumar Chattu et al (2019) The global problem of insufficient sleep and its serious public health implications.

17. Alessandro Stefani et al. (2013) The Serendipity Case of the Pedunculopontine Nucleus Low-Frequency Brain Stimulation: Chasing a Gait Response, Finding Sleep, and Cognition Improvement

18. Ritchie E. Brown (2012) Control of sleep and wakefulness.

19. JJ Hauw (2011) Neuropathology of sleep disorders: A review.

20. Arguinchona JH, Tadi P. Neuroanatomy, Reticular activating system.

21. Ronald Syzmusiak, Dennis McGinty (2008) Hypothalamic regulation of sleep and arousal.

22. Micheal D. Schwartz, Thomas S. Kilduff (2015) The Neurobiology of sleep and wakefulness.

23. Anne M. Fink et al (2018) Autonomic regulation during sleep and wakefulness: a review with implications for defining the Pathophysiology of neurological disorders.

24. Philip C. Smith, Jessica A. Mong (2019) Neuroendocrine control of sleep.

25. Davide Gnochi et al (2017) Circadian Rhythms and Hormonal Homoestasis: Pathophysiological Implications.

26. Born J, Wagner U (2004) Memory consolidation during sleep: role of cortisol feedback.

27. Benveniste H et al. (2019) The Glymphatic system and waste clearance with brain aging: A Review.

28. Cintron D, Lipford M, Larrea-Mantilla L, Spencer-Bonilla G, Lloyd R, Gionfriddo MR et al. (2017) Efficacy of menopausal hormone therapy on sleep quality: systematic review and meta-analysis.

29. Lisa Morselli et al. (2010) Role of sleep duration in the regulation of glucose metabolism and appetite.

30. Susan L. Worley (2018) The extraordinary importance of sleep.

31. Shalini Paruthi et al. (2016) Consensus Statement of the American Academy of Sleep Medicine on the Recommended Amount of Sleep for Healthy Children: Methodology and Discussion

32. Aserinsky E, Kleitman N (1953). Regularly occurring periods of eye motility, and concomitant phenomena, during sleep.

33. Jouvet M, et al (1959). On a stage of rapid brain electrical activity during physiological sleep.

34. Patel AK et al. Physiology, sleep Stages.

35. Malik J, Lo YL, Wu HT. Sleep-wake classification via quantifying heart rate variability by convolutional neural network.

36. Varga B, Gergely A, Galambos Á, Kis A. Heart Rate and Heart Rate Variability during Sleep in Family Dogs (Canis familiaris). Moderate Effect of Pre-Sleep Emotions.

37. Jocelyn Zakri (2020) Stages of sleep and sleep cycles.

38. Della Monica C, Johnsen S, Atzori G, Groeger JA, Dijk DJ. Rapid Eye Movement Sleep, Sleep Continuity and Slow Wave Sleep as Predictors of Cognition, Mood, and Subjective Sleep Quality in Healthy Men and Women, Aged 20-84 Years.

39. Ferri R, Rundo F, Silvani A, Zucconi M, Bruni O, Ferini-Strambi L, Plazzi G, Manconi M. REM Sleep EEG Instability in REM Sleep Behavior Disorder and Clonazepam Effects.

40. Christine Blume et al. (2019) Effects of Human Circadian rhythms, sleep and mood.

41. Mason Grimaldi et al. (2018) Impact of light exposure during sleep on cardiometabolic function.

42. Nava Zisapel (2018) New perspective on the role of melatonin in human sleep, circadian rhythms and there regulation.

43. Tom Woelders et al. (2017) Linking light exposure and subsequent sleep: A field polysomnography study in humans.

44. Goulet G, Mongrain V, Desrosiers C, Paquet J, Dumont M. Daily light exposure in morning-type and evening-type individuals.

45. Okudaira N, Kripke DF, Webster JB. Naturalistic studies of human light exposure.

46. Wright KP, McHill AW, Birks BR, Griffin BR, Rusterholz T, Chinoy ED. Entrainment of the human circadian clock to the natural light-dark cycle.

47. Timo Partonen et al. (2018) Systematic review of light exposure impact on human circadian rhythm.

48. Alina Masters et al. (2014) Melatonin, the hormone of darkness: From Sleep promotion to Ebola treatment.

49. Mats Lekander et al. (2017) Effect of late night short sleep on in home polysomnography: relation to adult age and sex.

50. John R. Best et al. (2015) Measuring sleep quality in older adults: a comparison using subjective and objective methods.

51. Alice lee et al. (2017) Effects of sleep deprivation on cognitive and physical performance in university students.

52. NHLBI. What Makes You Sleep? 2014, http://www.nhlbi.nih.gov/health/health-topics/topics/sdd/whatmakes.html.

53. Doghramji K. Waking up to sleep deprivation. Medscape Neurology. 2005;7(1)

54. Jolanta Orzel Gryglewska. (2010) Consequences of sleep deprivation.

55. Britz, J. and E. Pappas, 2010. Sources and outlet of stress among university students: Correlations between stress and unhealthy habits. Undergraduate Research Journal for the Human Science.

56. 8 Dimensions of Wellness, (UMD) University of Maryland's Your Guide to Living Well.

57. Ardell DB. Definition of Wellness. Ardell Wellness Report.

58. Debbie L. Stoewen (2017) Dimension of wellness: Change your habits, Change your life.

59. Health to Wellness: A Review of Wellness Models and Transitioning back to health.

60. Myers JE, Sweeney TJ, Witmer M. The wheel of wellness counseling for wellness: a holistic model for treatment planning.

61. Corbin CB, Pangrazi RP. Towards a uniform definition of wellness: a commentary.

62. Thompson CR, Thorofare NJ: Prevention Practice. A Physical Therapist's Guide to Health, Fitness, and Wellness

63. Institute of Medicine, Committee on Sleep Medicine and Research, Board on Health Sciences Policy . Sleep Disorders and Sleep Deprivation: An Unmet Public Health Problem.

64. Watson NF, Badr MS, Belenky G, et al. Joint Consensus Statement of the American Academy of Sleep Medicine and Sleep Research Society on the recommended amount of sleep for a healthy adult: methodology and discussion.

65. McCoy JG, Strecker RE. The cognitive cost of sleep lost.

66. Groeger JA, Stanley N, Deacon S, Dijk DJ. Dissociating effects of global SWS disruption and healthy aging on waking performance and daytime sleepiness.

67. Tempesta D, De GL, Natale V, Ferrara M. Emotional memory processing is influenced by sleep quality.

68. Vela-Bueno A, Moreno-Jimenez B, Rodriguez-Munoz A, et al. Insomnia and sleep quality among primary care physicians with low and high burnout levels.

69. Meerlo P, Sgoifo A, Suchecki D. Restricted and disrupted sleep: effects on autonomic function, neuroendocrine stress systems and stress responsivity.

70. Sadeh A, Gruber R, Raviv A. Sleep, neurobehavioral functioning, and behavior problems in school-age children.

71. Phillips B, Mannino DM. Do insomnia complaints cause hypertension or cardiovascular disease?

72. Ali T, Choe J, Awab A, Wagener TL, Orr WC. Sleep, immunity and inflammation in gastrointestinal disorders.

73. Goran Medic et al. (2017) Short and long term health consequences of sleep disruption.

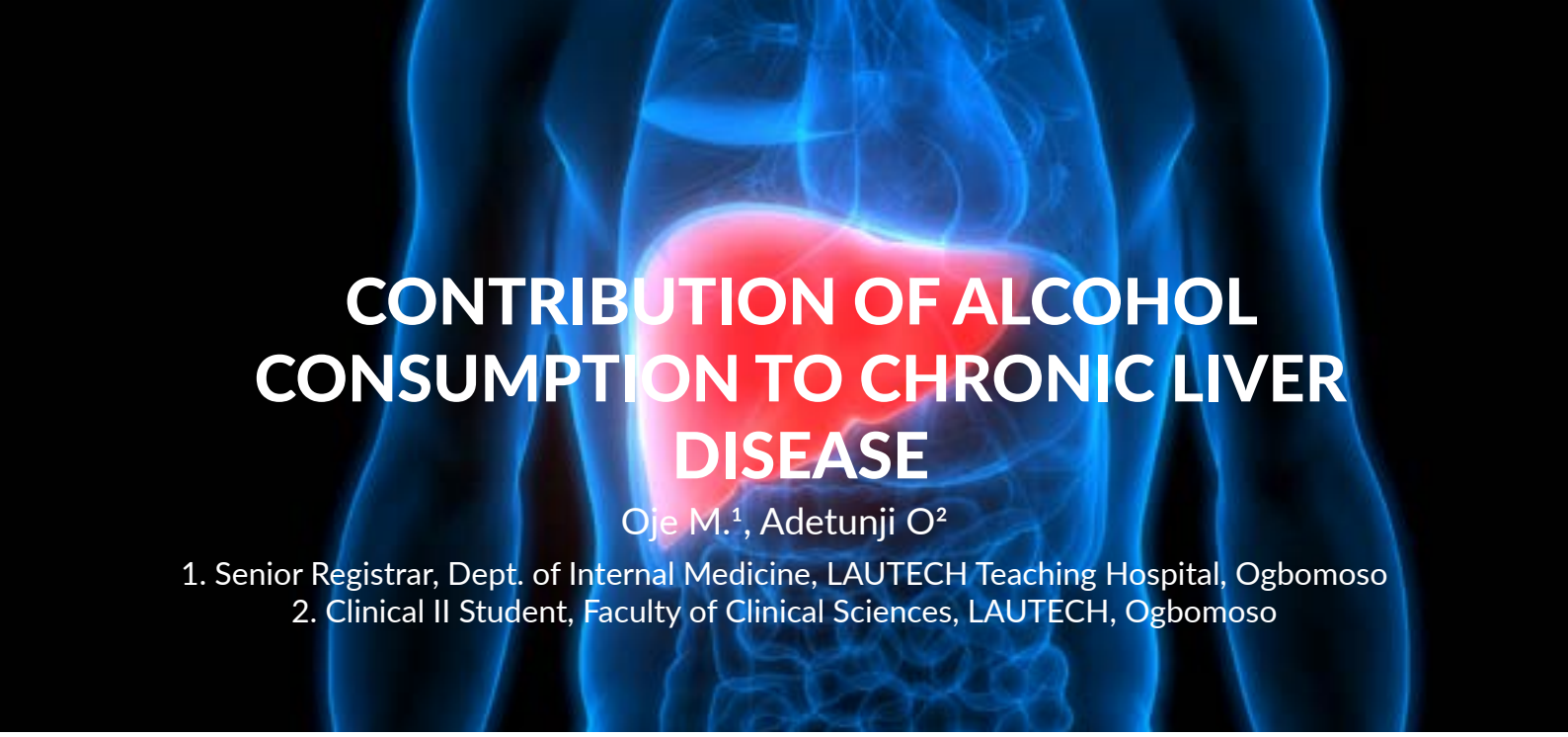
74. Hurt RT, Frazier TH, McClave SA, Kaplan LM. Obesity epidemic: overview, pathophysiology, and the intensive care unit conundrum. Journal of Parenteral and Enteral Nutrition.

75. Flegal KM, Carroll MD, Ogden CL, Curtin LR. Prevalence and trends in obesity among US adults, 1999–2008. Journal of the American Medical Association.

76. Swinburn BA, Sacks G, Hall KD, et al. The global obesity pandemic: shaped by global drivers and local environments.

77. Elia M, editor. Guidelines for detection and management of malnutrition. Malnutrition Advisory Group, Standing Committee of BAPEN.

78. Stratton RJ, King CL, Stroud MA, Jackson AA, Elia M. 'Malnutrition Universal Screening Tool' predicts mortality and length of hospital stay in acutely ill elderly.



# CONTRIBUTION OF ALCOHOL CONSUMPTION TO CHRONIC LIVER DISEASE

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## INTRODUCTION

The liver is a large, meaty organ that is located on the upper right quadrant of the abdomen. Weighing approximately 1.5 kg, the liver is reddish-brown in colour. The liver's main function is to filter the blood coming from the digestive tract, before passing it to the rest of the body. The liver also detoxifies chemicals and metabolizes drugs. It is responsible for the production of several important proteins in the body as well.

Liver diseases contribute markedly to the global burden of mortality and disease as well. Chronic liver disease in the clinical context is a disease process of the liver that involves a process of progressive destruction and regeneration of the liver parenchyma leading to fibrosis and cirrhosis1. Chronic liver disease refers to disease of the liver which lasts over a period of six months. It consists of a wide range of liver pathologies which include inflammation (chronic hepatitis), liver cirrhosis, and hepatocellular carcinoma. The entire spectrum need not be experienced.

## ALCOHOLIC LIVER DISEASE (ALD)

Alcoholic liver disease (ALD), also called alcohol-related liver disease (ARLD), is a term that encompasses the liver manifestations of alcohol overconsumption, including fatty liver, alcoholic hepatitis, and chronic hepatitis with liver fibrosis or cirrhosis2.

## EPIDEMIOLOGY

It has been estimated that 4% of the burden of disease and 3.2% of all deaths globally can be attributed to alcohol3. In the United States of America, alcohol is implicated in more than 50% of liver-related deaths and ALD accounts for about $3 billion annually4 . It has, however, been observed that whereas alcohol consumption is declining in developed countries the opposite is the case in developing countries where viral hepatitis is already prevalent. Also, it is known that alcohol and viral hepatitis B and/or C have synergistic effect on the progression of liver disease to cirrhosis and hepatocellular carcinoma (HCC)5.

## RISK FACTORS OF ALD

Quantity of alcohol taken: The actual amount of alcohol that will cause ALD has been the subject of some studies. Savolainen et al6 found that there was an increased risk of ALD in males drinking an average of 60g of alcohol daily. Another study showed that daily consumption of >50g of alcohol was generally significant for both sexes, specifically 40-80g for males and 20-40g for females7

Pattern of alcohol consumption: Drinking outside of meal times increases up to 3 times the risk of alcoholic liver disease8

Duration of alcohol consumption: Most patients with ALD drink more than 100 g/d9 (which corresponds to 6-7 drinks per day where one drink contains 13-15 g of alcohol), with 150-200 g per day being common10.

The typical patient has consumed alcohol heavily for two or more decades10, although in an occasional patient alcohol abuse may be for less than 10 years. However, clinicians should consider anyone drinking more than 30-50 g/d for more than 5-10 years at risk of developing ALD11.

Sex: women are more vulnerable to alcoholic liver damage and also develop more severe disease than men4. The postulated mechanisms for this gender difference in alcoholic liver damage include a relative deficiency of gastric alcoholic dehydrogenase in females, differences in alcohol bioavailability and other effects related to female sex hormones4.

Genetic factors: Genetic factors predispose both to alcoholism and to alcoholic liver disease. Both monozygotic twins are more likely to be alcoholics and to develop liver cirrhosis than both dizygotic twins. Polymorphisms in the enzymes involved in the metabolism of alcohol, such as ADH, ALDH, CYP4502E1, mitochondrial dysfunction, and cytokine polymorphism may partly explain this genetic component. However, no specific polymorphisms have currently been firmly linked to alcoholic liver disease.

Diet: Malnutrition, particularly vitamin A and E deficiencies, can worsen alcohol-induced liver damage by preventing regeneration of hepatocytes. This is particularly a concern as alcoholics are usually malnourished because of a poor diet, anorexia, and encephalopathy12.

Many alcoholics are malnourished, either because they ingest too little of essential nutrients (eg, carbohydrates, proteins, and vitamins) or because alcohol and its metabolism prevent the body from properly absorbing, digesting, and using those nutrients. As a result, alcoholics frequently experience deficiencies in proteins and vitamins, particularly vitamin A, which may contribute to liver disease and other serious alcohol-related disorders. Furthermore, alcohol breakdown in the liver, both by the enzyme alcohol dehydrogenase and by an enzyme system called the microsomal ethanol-oxidizing system (MEOS), generates toxic products such as acetaldehyde and highly reactive, and potentially damaging, oxygen-containing molecules. These products can interfere with the normal metabolism of other nutrients, particularly lipids, and contribute to liver cell damage. Nutritional approaches can help prevent or ameliorate alcoholic liver disease. For example, a complete balanced diet can compensate for general malnutrition. Administration of antioxidants (eg, precursors of the endogenous antioxidant glutathione) can help the body eliminate reactive oxygen molecules and other reactive molecules generated from abnormal lipid breakdown. New agents currently are being studied as promising nutritional supplements for alcoholics with liver disease.

The mechanism of ALD is not completely understood. 80% of alcohol passes through the liver to be detoxified. Chronic consumption of alcohol results in the secretion of pro-inflammatory cytokines (TNF-alpha, Interleukin 6 [IL6] and Interleukin 8 [IL8]), oxidative stress, lipid peroxidation, and acetaldehyde toxicity. These factors cause inflammation, apoptosis and eventually fibrosis of liver cells. Why this occurs in only a few individuals is still unclear. Additionally, the liver has tremendous capacity to regenerate and even when 75% of hepatocytes are dead, it continues to function as normal13.

## PATHOGENESIS

### Fatty change

Fatty change, or steatosis, is the accumulation of fatty acids in liver cells. These can be seen as fatty globules under the microscope. Alcoholism causes development of large fatty globules (macro-vesicular steatosis) throughout the liver and can begin to occur after a few days of heavy drinking14. Alcohol is metabolized by alcohol dehydrogenase (ADH) into acetaldehyde, then further metabolized by aldehyde dehydrogenase (ALDH) into acetic acid, which is finally oxidized into carbon dioxide and water15. This process generates NADH, and increases the NADH/NAD+ ratio. A higher NADH concentration induces fatty acid synthesis while a decreased NAD level results in decreased fatty acid oxidation. Subsequently, the higher levels of fatty acids signal the liver cells to compound it to glycerol to form triglycerides. These triglycerides accumulate, resulting in fatty liver.

### Alcoholic hepatitis

Alcoholic hepatitis is characterized by the inflammation of hepatocytes. Between 10% and 35% of heavy drinkers develop alcoholic hepatitis (NIAAA, 1993). While development of hepatitis is not directly related to the dose of alcohol, some people seem more prone to this reaction than others. This is

called alcoholic steato-necrosis and the inflammation appears to predispose to liver fibrosis. Inflammatory cytokines (TNF-alpha, IL6 and IL8) are thought to be essential in the initiation and perpetuation of liver injury by inducing apoptosis and necrosis. One possible mechanism for the increased activity of TNF-α is the increased intestinal permeability due to liver disease. This facilitates the absorption of the gut-produced endotoxin into the portal circulation. The Kupffer cells of the liver then phagocytose endotoxin, stimulating the release of TNF-α. TNF-α then triggers apoptotic pathways through the activation of caspases, resulting in cell death16.

**Cirrhosis**

Cirrhosis is a late stage of serious liver disease marked by inflammation (swelling), fibrosis (cellular hardening) and damaged membranes preventing detoxification of chemicals in the body, ending in scarring and necrosis (cell death). Between 10% to 20% of heavy drinkers will develop cirrhosis of the liver (NIAAA, 1993). Acetaldehyde may be responsible for alcohol-induced fibrosis by stimulating collagen deposition by hepatic stellate cells16. The production of oxidants derived from NADPH oxidase and/or cytochrome P-450 2E1 and the formation of acetaldehyde-protein adducts damage the cell membrane16. Symptoms include jaundice (yellowing), liver enlargement, and pain and tenderness from the structural changes in damaged liver architecture. Without total abstinence from alcohol use, cirrhosis will eventually lead to liver failure. Late complications of cirrhosis or liver failure include portal hypertension (high blood pressure in the portal vein due to the increased flow resistance through the damaged liver), coagulation disorders (due to impaired production of coagulation factors), ascites (heavy abdominal swelling due to build-up of fluids in the tissues) and other complications, including hepatic encephalopathy and the hepatorenal syndrome. Cirrhosis can also result from other causes than alcohol abuse, such as viral hepatitis and heavy exposure to toxins other than alcohol. The late stages of cirrhosis may look similar medically, regardless of cause. This phenomenon is termed the "final common pathway" for the disease. Fatty change and alcoholic hepatitis with abstinence can be reversible. The later stages of fibrosis and cirrhosis tend to be irreversible, but can usually be contained with abstinence for long periods of time.

Alcoholic liver disease (ALD) is typically diagnosed when, in addition to significant alcohol drinking, liver histology shows the characteristic features of centrilobular polymorphonuclear infiltrates, centrilobular hepatocyte swelling, ballooning degeneration, macrovesicular steatosis, Mallory bodies (hepatitis) and micronodular cirrhosis (cirrhosis)4.

## CONCLUSION

It is undeniable that alcohol consumption contributes to the development of liver damage and disease in no small way and so not drinking further alcohol is the most important part of treatment as well as prevention. The public should be made aware of the risk factors named above and the imminent danger in alcohol overconsumption to prevent further cases of morbidity and mortality as a result of liver disease.

## REFERENCE


1. "NHS Choices". Cirrhosis. Retrieved 6 October2015.

2. O'Shea RS, Dasarathy S, McCullough AJ (January 2010). "Alcoholic liver disease: AASLD Practice Guidelines" (PDF). Hepatology. 51 (1): 307–28.

3. Adrian R. Alcohol and the liver. Curr Opin Gastroenterol. 2006; 22:263 -271

4. Shah VH. Alcoholic liver disease. Scientific Sessions Handouts. Digestive Disease Week, Washington DC, 2007:105-113

5. DA Ndububa, OS Ojo, VA Adetiloye, AO Aladegbaiye, RA Adebayo, O Adekanle The contribution of alcohol to chronic liver disease in patients from South-west Nigeria. 2010.

6. Savolainen VT, Liesto K, Mannikko A, Penttila A, Karhunen PJ. Alcohol consumption and alcoholic liver disease: evidence of a threshold level of effects of ethanol. Alcoholism: Clinical and Experimental Research. 1993; 17:1112-1117.

7. Gramenzi A, Caputo F, Biselli M, Kuria F, Loggi E, Andreone P, et al. Alcoholic liver disease-pathophysiological aspects and risk factors. Aliment Pharmacol Ther. 2006; 24: 1151-1161.

8. Menon KV, Gores GJ, Shah VH (October 2001). "Pathogenesis, diagnosis, and treatment of

alcoholic liver disease" (PDF). Mayo Clin. Proc. 76(10): 1021–9. doi:10.4065/76.10.1021. PMID 11605686

9. Naveau S, Giraud V, Borotto E, Aubert A, Capron F, Chaput JC. Excess weight risk factor for alcoholic liver disease. Hepatology. 1997;25:108–111. [PubMed] [Google Scholar]

10. Mendenhall CL, Moritz TE, Roselle GA, Morgan TR, Nemchausky BA, Tamburro CH, Schiff ER, McClain CJ, Marsano LS, Allen JI. A study of oral nutritional support with oxandrolone in malnourished patients with alcoholic hepatitis: results of a Department of Veterans Affairs cooperative study. Hepatology. 1993;17:564–576. [PubMed] [Google Scholar]

11. Becker U, Deis A, Sørensen TI, Grønbaek M, Borch-Johnsen K, Müller CF, Schnohr P, Jensen G. Prediction of risk of liver disease by alcohol intake, sex, and age: a prospective population study. Hepatology. 1996;23:1025–1029. [PubMed] [Google Scholar]

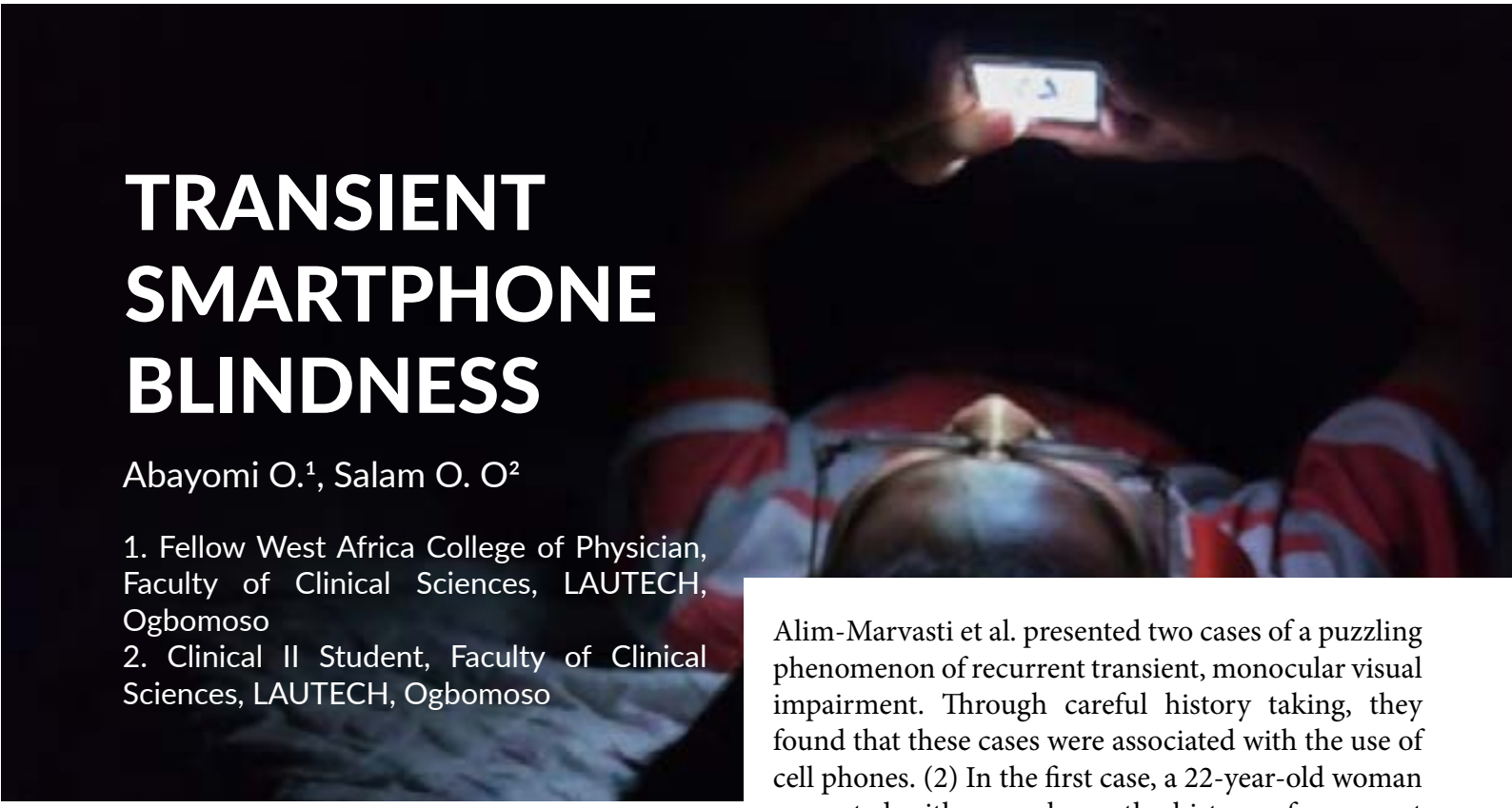
12. Alcohol Research & Health 27 (3), 220, 2003 Relationships between nutrition, alcohol use, and liver disease. Charles S Lieber

13. Longstreth, George F.; Zieve, David, eds. (18 October 2009). "Alcoholic Liver Disease". MedLinePlus: Trusted Health Information for You. Bethesda, MD: US National Library of Medicine & National Institutes of Health. Archived from the original on 22 January 2010. Retrieved 27 January 2010.

14. Inaba, Darryl; Cohen, William B. (2004). Uppers, downers, all arounders: physical and mental effects of psychoactive drugs (5th ed.). Ashland, Or: CNS Publications. ISBN 978-0-926544-27-7.

15. ^ Inaba & Cohen 2004, p. 185

16. Menon KV, Gores GJ, Shah VH (October 2001). "Pathogenesis, diagnosis, and treatment of alcoholic liver disease" (PDF). Mayo Clin. Proc. 76(10): 10219. doi:10.4065/76.10.1021. PMID 11605686



# TRANSIENT SMARTPHONE BLINDNESS

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Transient Smartphone Blindness is a temporary monocular loss of vision. It is a real phenomenon and may contribute to neurologic misdiagnosis in clinical practice and as such an important condition to be well understood by doctors as well as smartphone users.

## AETIOLOGY

The use of smartphone darkrooms in combination with lying on the side result in the emergence one-sided and temporary loss of vision. (1) In this century, the use of smartphones and other electronic gadgets have become inevitably important. Not only do we use our smartphones for our day to day activities, we also use it at night to catch up on one thing or the other, be it mails, messages or news.

## EPIDEMIOLOGY

The incidence of TSB is yet to be ascertained due to the clinical misdiagnosis associated with it. However, it has been predicted that the incidence is likely to rise with time due to the high increase in the number of smartphone users worldwide. According to Statista, the current number of smartphone users in the world this year is 3.5 billion, implying that 44.81% of the world's population owns a smartphone.

Alim-Marvasti et al. presented two cases of a puzzling phenomenon of recurrent transient, monocular visual impairment. Through careful history taking, they found that these cases were associated with the use of cell phones. (2) In the first case, a 22-year-old woman presented with several months history of recurrent impaired vision in her right eye at night. She had an extensive ophthalmic and systemic workup, and all findings were normal. The second case involved a 40-year-old woman with a 6-month history of recurrent monocular visual impairment upon waking that lasted up to 15 minutes. Investigations for a vascular source were normal, and aspirin therapy was started.

The patients were seen in the neuro-ophthalmic clinic, where a detailed history revealed that symptoms arose only after several minutes of viewing a smartphone screen while lying in bed in the dark (before going to sleep in the first case, and after waking up in the second case). The physicians asked both patients to experiment and record their symptoms. When they did, the patients reported that their symptoms were always in the eye contralateral to the side on which they were lying in the bed.

Likelihood of misdiagnosis of this condition is very common. The current investigators point out that their patient initially received a false diagnosis of multiple sclerosis (MS) and was advised to start a disease-modifying treatment. (3)

This phenomenon is certainly unusual. It's not known if it's underreported by patients or under recognized by physicians.

There are very few reports of transient smartphone blindness in the literature with none in a binocular

fashion.

## RISK FACTORS

As mentioned earlier on, the risk factors of transient smartphone blindness are not farfetched. They basically revolve round the use of smartphone and the how of the use.

These include:

* The current risk of developing the TSB is heightened by the prolonged use of the Smartphones, especially in a darkroom

* Lack of awareness of the effects of the smartphone by its users is also a point of consideration

* The intensified blue light emitted by the latest smartphones

* The use of smartphone while lying on one side of the body in a darkroom

## PATHOGENESIS

The loss of vision is usually one sided and this is due to the differential bleaching of photo-pigments, with the viewing eye becoming light-adapted while the other eye blocked by the pillow or whatever surface being laid on becomes darkness adapted. Subsequently, with both eye uncovered in the dark, the light adapted eye is perceived as 'blind'.

What happens invariably is that starring at a smartphone for a long time in a darkroom causes the light sensitive pigment in that eye to be bleached which makes them temporarily less sensitive to regular light than the other eye. The temporary loss of vision in the contralateral eye usually last a few minutes, which studies have shown to be 15 minutes on an average. (4)

## DIAGNOSIS AND INVESTIGATION

Detailed history taking and an understanding of retinal physiology can reassure both patient and doctor and can help in avoiding unnecessary anxiety and costly investigations.

It is a condition that involves the intervention of both the ophthalmologists and neurologists due to the high possibility of misdiagnosis.

"Misinterpretation of TSB symptoms, especially as a vascular or inflammatory disorder, can lead to unnecessary investigations and therapy", Saraniya Sathiamoorthi, BS, Mayo Medical School in Rochester, Minnesota, and Dean M. Wingerchuk, MD, Mayo Clinic in Scottsdale, Arizona. TSB is a condition with the potential for misdiagnosis as multiple sclerosis or ischemic vascular disease.

The disruptions in the sleep cycle by blue light exposure may also be exacerbated by smartphone usage in bed and thus promotes the development of TSB. The danger may arise from the initiation of inappropriate treatments following the misdiagnosis in the patients with ambiguous test results.

It is therefore crucial for the ophthalmologists and neurologists to include history of smartphone usage and the patients' daily practices before diagnosis.

Future investigation into the TSB should assess disease frequency and the severity in those with predisposing conditions such as vitamin A deficiency or hereditary congenital stationary night blindness.

## PROGNOSIS

While TSB is considered relatively benign and a short term pathology, the long-term implications arising from the disease are yet to be elucidated.

Blue light emission from the smartphone has the potential to contribute to various retinal pathologies, such as age-related macular degeneration.

## PREVENTION

TSB can be preventable. In order to prevent the TSB, people using smartphones should use them in well-lit rooms and in a position in which light does not enter only one eye.

One should avoid using smartphones before going to bed in order to avoid disturbing sleep cycles due to disruptive blue light emission.

Another important way to overcome the alterations in sleep cycle from blue light is the use of technology that automatically reduces blue light for certain duration of time, this is currently incorporated into certain smartphone types, e.g., eye care and blue light filter.

A number of lens manufacturers have introduced special glare-reducing anti-reflective coatings that also block blue light from digital devices.

Mobile phone manufacturers can also spread

awareness through public service messages warning users of possible side effects of using phones in bed and in poorly lit rooms.

The ophthalmologists can also create awareness about the TSB through counseling especially teens.

Further interventions may be provided in the form of mobile applications that monitor blue light exposure and offer prompts for prolonged or intense exposure.

## CONCLUSION

By and large, the condition, transient smartphone blindness (TSB) is a clinical condition that requires uttermost attention in order to avoid the transient blindness which may progress into permanent vision impairment.

It is also of important consideration because of the possibility of misdiagnosis and wrong treatment.

## REFERENCE

1. Transient smartphone blindness. Alim-Marvasti A, Bi W, Mahroo OA, et al. N Engl J Med. 2016;374:2

2. Transient Smartphone "Blindness" - American Academy of Ophthalmology - AAO.org https://www.aao.org/eyenet/article/transient-smartphone-blindness 502–2504. [PubMed] [Google Scholar]

3. TSB misdiagnosed as multiple sclerosis. https://www.medscape.com/viewarticle/875347 02/02/2017

4. Transient Smartphone "Blindness" | NEJM - New England Journal https://www.nejm.org › doi › full



# LIFESTYLE MODIFICATION FOR STROKE PREVENTION

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## ABSTRACT

*Non-communicable diseases like stroke are increasingly becoming major health problems in Africa. Stroke is the leading cause of disability and worse in Africa because of the relatively younger age of occurrence and severity of stroke in blacks relative to the white population. A review of the modifiable risk factors backed with available facts will in no measure be invaluable at this time. According to the World Health Organization, 15 million people suffer stroke worldwide each year. Of these, 5 million die, and another 5 million are permanently disabled. Eighty percent have at least one of the risk factors for heart and/or cerebrovascular disease: daily smoking, physical inactivity, being overweight, self-reported high blood pressure, or diabetes. High blood pressure contributes to more than 12.7million strokes worldwide.*

***Keywords:*** *stroke, hypertension, obesity, physical activity, smoking, low to moderate alcohol consumption, low-risk diet, stress.*

Stroke is classically characterized as a neurological deficit attributed to an acute focal injury of the central nervous system ( CNS) by a vascular cause, including cerebral infarction, intracerebral hemorrhage ( ICH), and subarachnoid hemorrhage (SAH), and is a major cause of disability and death worldwide. In 2013,the America Heart Association/ America Stroke Association (AHA/ASA) published an expert consensus statement updating the definition of stroke. This statement was endorsed and/or affirmed by the following organizations:

- American Association of Neurological Surgeons Congress of Neurological Surgeons
- American Academy of Neurology

The AHA/ASA classification broadly characterizes stroke into the following four broad subtypes

- Central nervous system (CNS) infarction (including ischemic stroke and silent infarction)
- Intracerebral hemorrhage (ICH)
- Subarachnoid hemorrhage (SAH)
- Stroke caused by cerebral venous thrombosis

CNS infarction is defined as brain, spinal cord, or retinal cell death attributable to ischemia, based on one of the following

- Pathological, imaging, or other objective evidence of cerebral, spinal cord, or retinal focal ischemic injury in a defined vascular distribution or
- Clinical evidence of cerebral, spinal cord, or retinal focal ischemic injury based on symptoms persisting > 24 hours or until death, and other etiologies excluded.

Stroke caused by intracerebral hemorrhage is defined

as rapidly developing clinical signs of neurological dysfunction attributable to a focal collection of blood within the brain parenchyma or ventricular system. The bleeding is not caused by trauma.

Stroke caused by subarachnoid hemorrhage is defined as rapidly developing signs of neurological dysfunction and/or headache because of bleeding into the subarachnoid space (the space between the arachnoid membrane and the pia mater of the brain or spinal cord). The bleeding is not caused by trauma.

Stroke caused by cerebral venous thrombosis is defined as infarction or hemorrhage in the brain, spinal cord, or retina because of thrombosis of a cerebral venous structure. Symptoms or signs caused by reversible edema without infarction or hemorrhage do not qualify as stroke.

Not all strokes will fit into one of those four categories. Stroke, not otherwise specified is defined as an episode of acute neurological dysfunction presumed to be caused by ischemia or hemorrhage, and persisting ≥24 hours or until death.

Globally, 70% of strokes and 87% of both stroke-related deaths and disability-adjusted life years occur in low- and middle-income countries. Over the last four decades, the stroke incidence in low and middle income countries has more than doubled. During these decades stroke incidence has declined by 42% in high income countries. On average, stroke occurs 15 years earlier in and causes more deaths of people living in low and middle-income countries when compared to those high-income countries. Patient management challenges, combined with inadequate rehabilitation services, lack of preventive measures, as well as poor understanding of the possible unique risk factors associated with stroke in low- and middle-income countries, may account for the disproportionately large stroke burden borne by these countries.

## RISK FACTORS FOR STROKE

Risk factors for stroke can be categorized as modifiable and non-modifiable. Hypertension, obesity, smoking, alcohol consumption, physical inactivity, diet are among some of the more commonly reported modifiable risk factors.

## HYPERTENSION

According to WHO, hypertension, also known as high or raised blood pressure, is a condition in which the blood vessels have persistently raised pressure?

Globally, an estimated 26% of the world's population (972 million people) has hypertension, and the prevalence is expected to increase to 29% by 2025, driven largely by increases in economically developing nations

Hypertension can cause stroke through many mechanisms. A high intraluminal pressure will lead to extensive alteration in endothelium and smooth muscle function in intracerebral arteries. The increased stress on the endothelium can increase permeability over the blood-brain barrier.

About 50% of people who suffer a first heart attack and about 66% of people experiencing a first stroke have blood pressure above 160/95 mm Hg. Also, high blood pressure is associated with a 2–3 times higher risk of developing heart failure.

Because the consequences of high blood pressure are so serious, early detection, treatment, and control are critical.

Blood pressure can be reduced by the following ways:

A. Salt Restriction

An excessive salt intake has been suggested to be related to high blood pressure by observational studies. The hypotensive effect of salt restriction has also been shown by many large-scale intervention studies in Western countries. According to the results of some clinical studies, blood pressure was not significantly decreased without reducing salt intake to 6 g day−1. On the basis of these results, Western guidelines recommend a salt intake of <6 g day−1. The recommendations of the American Heart Association (AHA) in 2006 and The European Society of Hypertension-European Society of Cardiology (ESH-ESC) in 2007 are that optimal salt intake should be 3.8 g day−1.

The effectiveness of salt restriction is considered to depend on its strictness (meta-analysis indicated a decrease in systolic blood pressure of about 1 mmHg with a decrease in salt intake of 1 g day), guidance should be given to gradually reduce salt intake over a long period. Recently, a follow-up study of the TOHP reported that salt restriction reduces the long-term risk of cardiovascular disease.

Other ways by which blood pressure can be reduced include;

b. Reduction in total and saturated fat

c. Reduction in Red meat consumption

d. Reduction in the intake of Sugar, sugary drinks

e. Reduction in the consumption of refined carbohydrates

f. Increased consumption of fruits, vegetables, whole grain

g. Increased consumption of Fish and poultry

h. Increased consumption of Low-fat dairy products

Previous intervention studies have provided evidence of a BP lowering effect of certain dietary patterns. The DASH(Dietary approaches to stop hypertension) randomized controlled trial showed that a diet rich in fruits, vegetables, legumes, nuts, seeds, low-fat dairy products, and fish, as well as a sodium intake restricted to 3g/day, can lead to a BP reduction of 5.5/3.0mm Hg

According to a Research carried out by the Food Security Research Center, Isfahan University of Medical Sciences, Isfahan, Iran, Meta-analysis showed that the DASH diet significantly reduced systolic blood pressure by 6.74 mmHg (95%CI: −8.25, −5.23, I2 = 78.1%) and diastolic blood pressure by 3.54 mmHg (95%CI: −4.29, −2.79, I2 = 56.7%). RCTs with the energy restriction and those with hypertensive subjects showed a significantly greater decrease in blood pressure

The results revealed the profitable reducing effect of the DASH-like diet on both systolic and diastolic blood pressure in adults; although there was a variation in the extent of the fall in blood pressure in different subgroups

## OBESITY

Overweight and obesity are defined as abnormal or excessive fat accumulation that may impair health. Body mass index (BMI) is a simple index of weight-for-height that is commonly used to classify overweight and obesity in adults. It is defined as a person's weight in kilograms divided by the square of his height in meters (kg/m2).

For adults, WHO defines overweight and obesity as follows:

- overweight is a BMI greater than or equal to 25; and
- obesity is a BMI greater than or equal to 30

Children aged between 5–19 years

Overweight and obesity are defined as follows for children aged between 5–19 years:

- overweight is BMI-for-age greater than 1 standard deviation above the WHO Growth Reference median; and
- obesity is greater than 2 standard deviations above the WHO Growth Reference median.

In 2016, more than 1.9 billion adults aged 18 years and older were overweight. Of these over 650 million adults were obese.

In 2016, 39% of adults aged 18 years and over (39% of men and 40% of women) were overweight.

Overall, about 13% of the world's adult population (11% of men and 15% of women) were obese in 2016.

The worldwide prevalence of obesity nearly tripled between 1975 and 2016.

The prevalence of obesity among adults in the United States (defined as a body mass index [BMI] ≥30 kg/m2) has increased from 13% to 34% over the past half-century.A similar trend has been observed in other countries although the absolute prevalence of obesity varies considerably.For example, compared with the United States, current obesity rates are lower in Canada (24%), Germany (23%), and China (4%).

In 2019, an estimated 38.2 million children under the age of 5 years were overweight or obese.

In Africa, the number of overweight children under 5 has increased by nearly 24% percent since 2000. Almost half of the children under 5 who were overweight or obese in 2019 lived in Asia.

Obesity can increase the risk of stroke due to inflammation caused by excess fatty tissue. This can lead to difficulty in blood flow and an increase risk of blockage, both of which can cause stroke

Waist circumference and waist-to-hip circumference ratio (WHR) provide a useful estimation of the proportion of abdominal or upper-body fat (Björntorp 1984; Björntorp 1985; Kissebah 1985). A WHR greater than 1.0 in men and 0.85 in women indicates abdominal fat accumulation; a high-risk waist circumference is thought to be 88 cm or greater for women and 102 cm or greater for men (Clinical Gdlns 1998). Overweight and obesity seems to be associated with higher morbidity from many conditions such as arterial hypertension, type 2 diabetes mellitus, coronary heart disease, stroke,

gallbladder disease, osteoarthritis, sleep apnoea and other respiratory problems and some types of cancer (for example, endometrium, breast, prostate, colon) (Clinical Gdlns 1998). The abdominal fat is assumed to better correlate with increased mortality and risk of disorders such as diabetes mellitus, hyperlipidaemia, hypertension and cardiovascular diseases (Donahue 1987; Ducimetiere 1986; Lapidus 1984; Larsson 1984; Ohlson 1985; Stokes 1985).

A weight loss of 5% to 15% of baseline weight, may be associated with significant improvements in lipid levels, glycaemic control, and blood pressure control (Douketis 2005). These effects could have the potential to reduce for example the incidence of cardiovascular morbidity and mortality. In this manner, weight loss may help control diseases worsened by overweight or obesity and may also decrease the likelihood of developing these diseases.

Overweight and obesity are important public health problems and are associated with many serious health conditions including stroke which is a leading cause of death and severe long-term disability. Being overweight strains the heart and blood vessels and is associated with high blood pressure. Obesity also predisposes a person to heart disease and diabetes, both of which increase the risk for stroke. Keeping your weight to recommended levels for your height and build is a prudent preventive measure. It appears logical that weight reduction in overweight or obese people should have positive health consequences lowering the number and consequences of strokes.

## SMOKING

Cigarette smoking is a well-established risk factor for all forms of stroke. While both the general public and the global health care system is aware of the vascular risks associated with smoking, the prevalence of tobacco use has remained largely unchanged over the last quarter of a century. Approximately one in five US adults are classified as regular smokers, with the initiation of smoking typically occurring during the teenage years.

Plausible mechanism by which primary and environmental tobacco smoke exposure can increase the risk of stroke and heart disease are numerous and include carboxyhemoglobinemia, increased platelet aggregability, increased fibrinogen levels, reduced HDL-cholesterol and toxic effects of compounds such as 1,3-butadiene, a vapor phase constituent of environmental smoke that has been to shown to accelerate atherosclerosis as measured by B-mode ultrasound of the carotid wall, as well as to early arterial damage as assessed by endothelium-dependent brachial artery dilation. Last, pathogenesis of increased stroke risk in populations exposed to cigarette smoke has been suggested to be that of advanced atherogenesis. Potentially in relation to chronic infection.

The evidence linking smoking to stroke is extremely convincing. Studies performed across various ethnicities and populations demonstrate a strong association between smoking and stroke risk, with current smokers having at least a two- to fourfold increased risk of stroke compared with lifelong nonsmokers or individuals who had quit smoking more than 10 years prior. In one study, the risk increased to sixfold when this population was compared with nonsmokers who had never been exposed to environmental tobacco smoke (i.e., second-hand smoke).

Among the included studies, 7 reported the details of the relationship between the number of cigarettes smoked and the incidence of stroke. In the dose-response meta-analysis, the non-linear association between cigarette smoking and stroke was non-significant (P for nonlinearity = .729). A linear elevation in stroke incidence was observed when cigarette consumption increased, and the risk of stroke was increased by 12% for each increment of 5 cigarettes of per day (95% Cl: 0.01, 0.02 p <.001) Linear dose-dependent relationship between number of cigarettes smoked per day and the relative risk of stroke among all subjects.

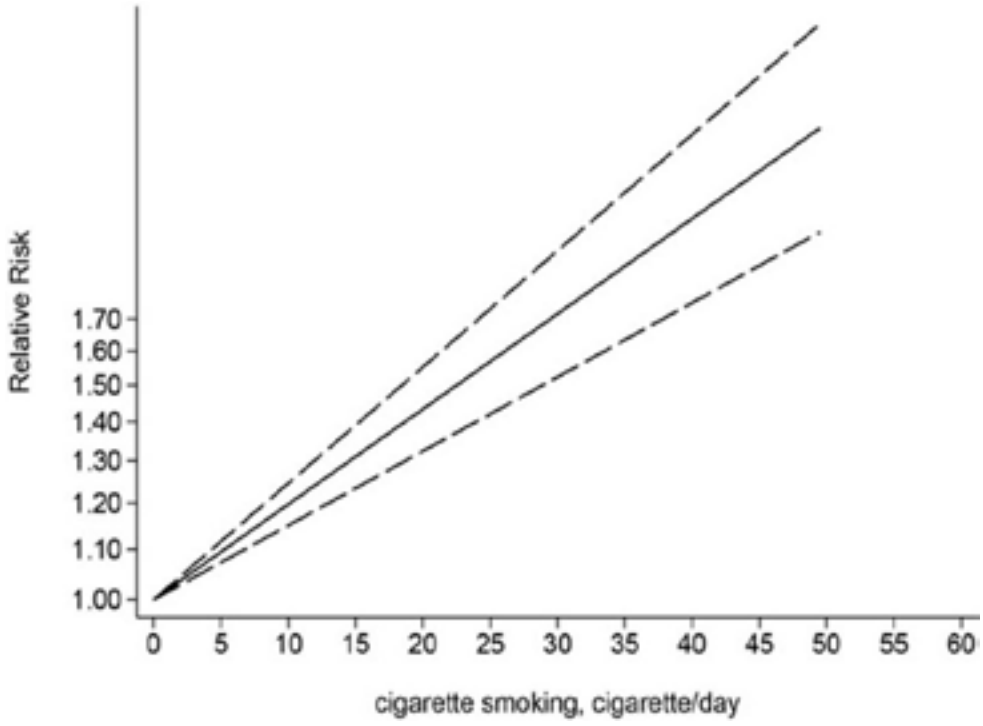
Quitting smoking and eliminating exposure to second-hand smoke significantly reduces the risk of stroke. A 12-year follow up study found a 34% significantly decreased risk of stroke for former smokers compared with current smokers. Former smokers who had stopped smoking for four years had an almost identical risk of stroke as that of lifetime non-smokers.

## ALCOHOL CONSUMPTION

Alcohol is consumed widely throughout the world. In Western countries, abstainers constitute the minority while most have a light to moderate alcohol intake (drinking a max of 2 and 3 drinks per day for women and men). Heavy alcohol intake is considered a risk

factor for stroke, whereas moderate or light drinking of alcohol is assumed to reduce the risk. However, more recent research has shown that even lower levels of alcohol Consumption may increase the risk of certain types of stroke risk.

A new study published in The Lancet links even low levels of alcohol consumption with increased blood pressure that can lead to strokes. The study comes after research published last year concluded there is no healthy level of drinking. In the new study, researchers found that one to two drinks a day increased stroke risk by 10% to 15% and that four drinks a day increased the risk of having a stroke by 35%. One drink was defined as either a small glass of wine, a bottle of beer or a single measure of spirits.




There are several ways in which alcohol may increase the risk of a stroke and may include:

Increased blood pressure — Alcohol abuse can lead to high blood pressure. High blood pressure is the number one contributing factor to a person's risk of stroke.

Increased Weight — being overweight and obesity can increase a person's risk of stroke. Regularly consuming alcohol can make it difficult to maintain a healthy weight.

Increased Risk of Diabetes — Alcohol can affect the way the body uses insulin, which can contribute to the development of type II diabetes. Diabetes doubles a person's risk of stroke.

Increased Risk of Liver Damage — Alcohol is a known cause of liver damage. Further, Alcohol consumption is associated with decrease platelet aggregation, increase fibrinolysis, and decrease plasma fibrinogen levels. This could increase the risk of hemorrhagic stroke.

Increased Risk of Atrial Fibrillation — Atrial fibrillation is a condition in which the heart beats irregularly. Excess alcohol consumption can trigger this condition, which can, in turn, increase a person's risk of stroke.

The more alcohol a person drinks, the higher the risk of stroke. Additionally, research has found that even moderate amounts of alcohol can increase stroke risk. According to some studies, there is no safe amount of alcohol to consume when it comes to lowering the risk of stroke.

## STRESS

Stress can be generally defined as undue, inappropriate or exaggerated response to a situation. Whereas anxiety about a situation could be positive, stress is always negative with attending adverse psychological and physiological changes leading to decreased productivity, disease and sometimes death. Stresses can be external (from the environment, psychological,




or social situations) or internal (illness, or from a medical procedure).

There are three categories of potential stressors environmental, organization, and personal.

Constant stress has been linked to higher activity in an area of the brain linked to processing emotions, and an increased likelihood of developing heart and circulatory disease.

Scientists have found that people who have heightened activity in the amygdala — the part of the brain commonly linked to stress — are more likely to develop cardiovascular disease. It contributes to aggravating health risks like hypertension, high cholesterol levels, diabetes, obesity, heart problems, etc. These health issues are categorized as 'high risk factors' for a stroke. It has been found in a study that stress apparently raises the risk of a Stroke or Transient Ischemic Attack (TIA) by 59%. Additionally, stress leads to developing harmful habits like smoking, which is again a well-established risk factor for a stroke.

Findings indicate that people can lower their stroke risk by attempting to reduce the stress in their life.

## NUTRITION

In the US, and probably also in other developed countries, the most prevalent risk factor is an unhealthy diet: The American Heart Association statistical report of 2015 indicated that only 0.1% of Americans consumed a healthy diet and only 8.3% consumed a moderately healthy diet. In China, the consumption of meat and eggs has increased markedly in recent years, while consumption of whole grains, vegetables and fruit has declined. This probably accounts for much of the increase in fasting cholesterol levels and coronary disease in China at the same time.

## NUTRIENT THAT AFFECT THE RISK OF STROKE

### Mediterranean Diet

The diet with the best evidence for stroke prevention is the Mediterranean diet from Crete. In the Seven Countries Study, it was discovered that coronary risk in Crete was 1/15th that in Finland and only 40% of that in Japan. In a retrospective article, Ancel Keys described it as 'a mainly vegetarian diet…favoring fruit for dessert'. The diet is not a low-fat diet; it is a low glycaemic/high-fat diet with 40% of calories from beneficial fats such as olive and canola oil, and high in whole grains, fruits, vegetables and legumes (peas, lentils, beans, nuts and so on) figure 2.

More recently, in high-risk primary prevention, a diet aiming for low fat intake was compared with two versions of a Mediterranean diet in a Spanish study. One version of the Mediterranean diet was supplemented with olive oil, the other supplemented with mixed nuts. In five years, the nut-supplemented diet reduced stroke by 46% and the olive oil supplemented version reduced stroke by 35%.

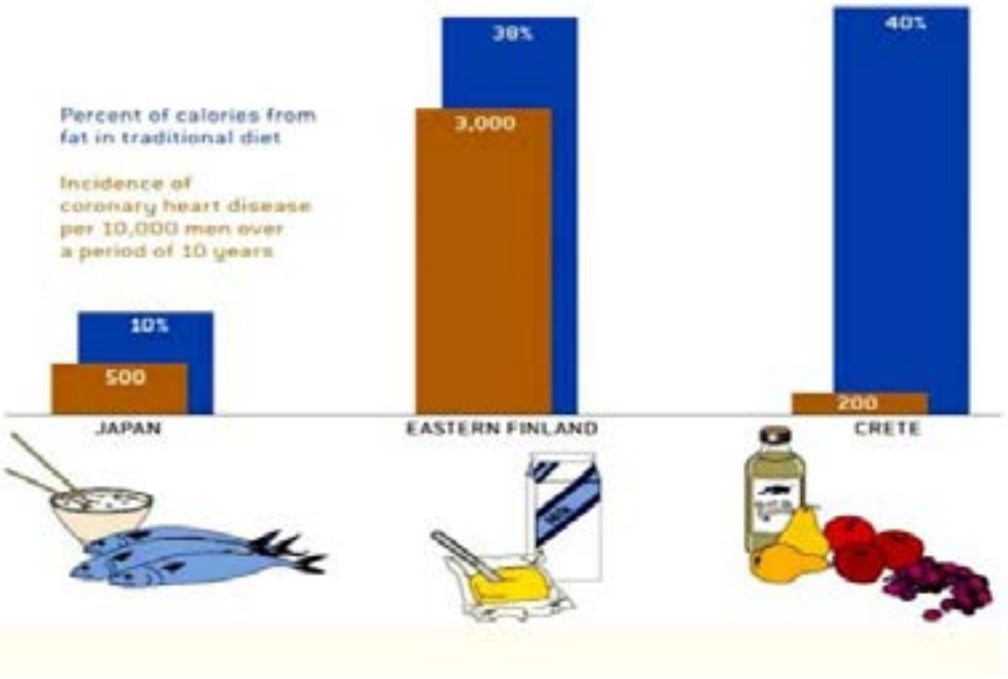



**Figure 2**
The Mediterranean diet. The Mediterranean diet is a high fat/low glycaemic diet with 40% of calories from fat; however, the fat is mainly beneficial oils such as olive and canola. Among men in the Seven Countries Study, the coronary risk in Crete was only 1/15th of that in Finland, where most of the fat was saturated fat (accompanied by cholesterol), and 40% of that in Japan, where the diet is a low-fat diet favouring fish. (Reproduced by permission of the artist, Cornelia Blik, from Willet and Stampfer [5]).

**Intestinal Microbiome**

Although dietary cholesterol does not increase fasting lipid levels by much, it clearly does increase coronary mortality. However, there is increasing recognition of the importance of the intestinal microbiome. While cholesterol content is essentially the same for any kind of animal flesh, red meat has more saturated fat, and has ~4 times as much carnitine as chicken or fish. Carnitine from red meat37 and phosphatidylcholine from egg yolk37 are converted by the intestinal bacteria to trimethylamine, in turn oxidised in the liver to trimethylamine n-oxide. TMAO causes atherosclerosis in animal models, 37 and in patients undergoing coronary angiograms plasma levels in the top quartile of TMAO levels after a test dose of two hard-boiled eggs predicted a 2.5-fold increase in the 3-year risk of stroke, myocardial infarction or vascular death.38 A 12-ounce Hardee's Monster Thickburger39 contains 265 mg of cholesterol and 320 mg of carnitine. The yolk of a 65g egg contains 237 mg of cholesterol and 250 mg of carnitine, so two egg yolks are worse than the 12-ounce burger. Patients at risk of stroke should avoid red meat and egg yolk.

**Carbohydrates**

Carbohydrate intake in quantities that exceed energy requirement (positive energy imbalance) is a major determinants of weight gain and adiposity, and the quality of carbohydrate intake affect metabolic health. Glycated proteins, and beverages and foods with high glycemic load, including added sugars, increase body weight. High carbohydrate intake from food with a high glycemic index, added sugars, and high glycemic load also leads to reduced intake of essential nutrients and has been associated with an increased of stroke mortality and coronary heart disease in observational studies.

**Protein**

Observational studies in Japan have showed that increase protein intake is associated with reduced risk of stroke.

**MINERAL**

Salt: It is clear that high salt intake increases the risk and severity of hypertension. High salt intake also leads to depletion of potassium and magnesium, as potassium and magnesium are excreted by the renal tubule in exchange for sodium. An optimal salt intake is ~2–3 g daily, and in most populations the average salt intake is much higher. Excess salt intake might increase cardiovascular and stroke risk by increasing blood pressure and causing fibrosis in the heart, kidneys, and arteries.

Potassium: potassium is an important nutrient in the diet, because it lessens the effects of salt, and studies have suggested it may help prevent high blood pressure, a major risk factor for heart disease and stroke. Eating food that high in potassium (sweet potato, leafy green, white beans) lower the risk of having a stroke.

Calcium: a recent observational study of 34 670 women reported that increasing supplemental calcium intake was not associated with ischemic stroke, but was associated with an increased risk of hemorrhagic stroke. Calcium supplementation without co-administered vitamin D, is associated with a significant increase In myocardial infarction and a trend towards an increase in stroke.

Co-administration of calcium plus vitamin D supplement was associated with an increased risk of stroke, myocardial infarction or the composite of both.

Coffee and tea: The role of regular consumption of tea or coffee on cardiovascular risk has not been tested in randomised controlled trials. There is moderately good evidence that consumption of coffee and tea reduces risk of myocardial infarction, but little mention of stroke in that body of literature. It seems likely that reduction of myocardial infarction would be associated with reduction of stroke risk. It seems likely that benefits of coffee is from bioflavonoids.

**OTHER VITAMINS AND SUPPLEMENTS INCLUDING OMEGA 3 FATTY ACIDS.**

The issue of supplements and vitamins for cardiovascular prevention was extensively reviewed, and meta-analysis performed, in 2018. Only folic acid and B vitamins reduced the risk of stroke. The authors commented that at the time of the meta-analysis, which found no benefit from omega-3 fatty acids, a study was under way with a higher dose of marine omega-3 fatty acids (2000 units). That trial, published in 2019, randomized 25,871 participants (men aged >50 and women aged ≥55) to omega 3 fatty acids vs. placebo in prevention of cardiovascular disease and cancer. There was no significant reduction of major cardiovascular events or cancer with omega-3 fatty acids.

**PHYSICAL ACTIVITY.**

Exercise and physical activity have an increasing evidence base in the primary and secondary prevention of stroke and in stroke rehabilitation. A recent research carried out and published in the American Journal of hypertension showed that:

Both systolic and diastolic BP at rest did not change in the nonexercising control group. All four exercise groups demonstrated significant reductions in both systolic and diastolic BP at rest. The magnitude of reductions in systolic BP was greater in the 61 to 90 min/wk group compared with the 30 to 60 min/wk group. There were no greater reductions in systolic BP with further increases in exercise volume. The magnitude of reductions in diastolic BP was not significantly different among four exercise groups.

Interface between physical activity and cerebrovascular disease is complex and of broad interest to clinicians, therapists, and epidemiologists. The importance of the relationship is becoming clearer: physical inactivity has been implicated by the INTERSTROKE study as one of the 5 key risk factors which account for more than 80% of the global burden of stroke. Physical fitness training is increasingly being recommended as a component of stroke rehabilitation programmes due to the emerging body of evidence surrounding the benefits in improving the function after stroke. The role of long-term physical activity in patients who have had a stroke in the prevention of further stroke is less clear.Exercise contributes to losing weight and lowering blood pressure, but it also stands on its own as an independent stroke reducer.

## CONCLUSION

Modifiable risk factors for recurrent stroke and cardiovascular disease are common in people with stroke. These risk factors often go unmanaged, which leads to increased risk for mortality and disability. Clinical guidelines recognize that lifestyle modification for risk factor management should be a priority following stroke. Improving diet, increasing physical activity, smoking cessation, and

reducing alcohol consumption, are specific health behaviors that should be targeted because of their effects on multiple cardiovascular risk factors. Health professionals therefore have a pivotal role to play in stroke prevention. The aim is to control modifiable risk factors to reduce the risk of stroke occurring in the first place (primary stroke prevention), and to reduce the risk of reoccurrence after a stroke or a transient ischaemic attack (TIA) (secondary stroke prevention).

## REFERENCES


1. Whelton, P. K. et al. 2017, Guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/ American Heart Association Task Force on Clinical Practice Guidelines. Hypertension (2018).

2. Iadecola, C. et al. Impact of hypertension on cognitive function: a scientific statement from the American Heart Association. Hypertension (2016).

3. Cai, A. & Calhoun, D. A. Resistant hypertension: an update of experimental and clinical findings. Hypertension (2017).

4. Calhoun, D. A. et al. Resistant hypertension: diagnosis, evaluation, and treatment. A scientific statement from the American Heart Association Professional Education Committee of the Council for High Blood Pressure Research. Hypertension (2008).

5. Norlander, A. E., Madhur, M. S. & Harrison, D. G. The immunology of hypertension. J. Exp.

6. Koroskenyi, K., Juba, F. & Vajda, G. Human vascular antigen complement consumption test of hypertensive patients (preliminary report). Experientia (1961).

7. Strazzullo P, D'Elia L, Cairella G, Garbagnati F, Cappuccio FP, Scalfi L. Excess body weight and incidence of stroke: meta-analysis of prospective studies with 2 million participants.Stroke. 2010

8. Flegal KM, Kruszon-Moran D, Carroll MD, Fryar CD, Ogden CL. Trends in obesity among adults in the united states, 2005 to 2014.JAMA. 2016

9. Scottish Intercollegiate Guidelines Network (SIGN) Management of Patients with Stroke or TIA: assessment, Investigation, Immediate Management and Secondary Prevention. SIGN 108. Edinburgh, UK: A National Clinical Guideline; 2008.

10. O'Donnell MJ, Denis X, Liu L, et al. Risk factors for ischaemic and intracerebral haemorrhagic stroke in 22 countries (the INTERSTROKE study): a case-control study. The Lancet. 2010

11. Gordon NF, Gulanick M, Costa F, et al. Physical activity and exercise recommendations for stroke survivors: an American Heart Association scientific statement from the Council on Clinical Cardiology, Subcommittee on Exercise, Cardiac Rehabilitation, and Prevention; the Council on Cardiovascular Nursing; the Council on Nutrition, Physical Activity, and Metabolism; and the Stroke Council. Stroke

12. Whelton, P. K. et al. 2017, Guideline for the prevention, detection, evaluation, and management of high

13. Hankey GJ: Nutrition and the risk of stroke. Lancet Neurol 2012

14. Shinton R, Beevers G. Meta-analysis of relation between cigarette smoking and stroke. BMJ 1989; 298:789±94.

15. Bonita R, Scragg R, Stewart A, et al. Cigarette smoking and risk of premature stroke in men and women. BMJ 1986; 293:6±8.

16. Bonita R, Duncan J, Truelsen T, Jackson RT, Beaglehole R. Passive smoking as well as active smoking increases the risk of acute stroke. Tob Control 1999; 8:156–160. [PubMed: 10478399]

17. Qureshi AI, Suri MF, Kirmani JF, Divani AA. Cigarette smoking among spouses: another risk factor for stroke in women. Stroke 2005; 36:e74–e76. [PubMed: 16081868]

18. NHMRC report. Effects of passive smoking on health. Med J Aust 1986; 145:404–405. No authors listed. [PubMed: 3762477]

19. CDC. 1986 Surgeon General's report: the health consequences of involuntary smoking. MMWR Morb Mortal Wkly Rep 1986; 35:769–770. [PubMed: 3097495]

20. Health effects of exposure to environmental tobacco smoke. California Environmental Protection Agency. Tob Control 1997; 6:346–353. No authors listed. [PubMed: 9583639]

21. AbramsDB, GlasserAM, PearsonJL, VillantiAC, Collins LK, Niaura RS. Harmminimizationandtobaccocontrol: reframing societal views of nicotine use to rapidly save lives. Annu Rev Public Health. 2018; 39(1):193-213. Doi:

10. 1146/annurev-publhealth-040617-013849

22. GlantzSA, BarehamDW. E-cigarettes: use, effects on smoking, risks, and policy implications. Annu Rev Public Health. 2018; 39:215-235. Doi: 10.1146/annurev-publhealth-040617-013757

23. HuangJ, FengB, WeaverSR, PechacekTF, SlovicP, EriksenMP. Changing perceptions of harmofe-cigarettevs cigarette use among adults in 2 US national surveys from 2012 to 2017. JAMA Netw Open. 2019; 2(3):e191047. Doi: 10. 1001/jamanetworkopen.2019.1047

24. National Academies of Sciences, Engineering, and Medicine. Public Health Consequences of e-Cigarettes. Washington, DC: National AcademiesPress;2018.doi:10.17226/24952

25. Alzahrani T, Pena I, TemesgenN, GlantzSA. Association between electronic cigarette use and myocardial infarction. Am J Prev Med.2018; 55(4):455-461. doi:10.1016/j.amepre.2018.05.004

26. Hackam DG, Spence JD. Combining multiple approaches for the secondary prevention of vascular events after stroke: a quantitative modeling study. Stroke 2007; 38:1881–5.

27. Chiuve SE, Rexrode KM, Spiegelman D, et al. Primary prevention of stroke by healthy lifestyle. Circulation 2008; 118:947–54.

28. Larsson SC, Akesson A, Wolk A. Healthy diet and lifestyle and risk of stroke in a prospective cohort of women. Neurology 2014; 83:1699–704.

29. Akesson A, Larsson SC, Discacciati A, et al. Low-risk diet and lifestyle habits in the primary prevention of myocardial infarction in men: a population-based prospective cohort study. J Am Coll Cardiol 2014; 64:1299–306.

30. Mozaffarian D, Benjamin EJ, Go AS, et al. Heart disease and stroke statistics--2015 update: a report from the American Heart Association. Circulation 2015; 131:e29–322.

31. Weiwei C, Runlin G, Lisheng L, et al. Outline of the report on cardiovascular diseases in China, 2014. Eur Heart J Suppl 2016; 18(Suppl F):F2–11.

32. Reddy KS. Cardiovascular disease in non-Western countries. N Engl J Med 2004; 350:2438–40.

33. Critchley JA, Capewell S. Mortality risk reduction associated with smoking cessation in patients with coronary heart disease: a systematic review. JAMA 2003; 290:86–97.

34. Go AS, Mozaffarian D, Roger VL, et al.Heart disease and stroke statistics—2014 update: a report from the American Heart Association. Circulation. 2014; 129:e28-e292. [PMC free article] [PubMed] [Google Scholar]

35. Burn J, Dennis M, Bamford J, Sandercock P, Wade D, Warlow C. Long-term risk of recurrent stroke after a first-ever stroke. The Oxfordshire Community Stroke Project. Stroke. 1994; 25:333-337. [PubMed] [Google Scholar]

36. Hardie K. Ten-year risk of first recurrent stroke and disability after first-ever stroke in the Perth Community Stroke Study. Stroke. 2004; 35:731-735. [PubMed] [Google Scholar]

37. 37 https://www.verywellmind.com/the-connection-between-stress-and-stroke-risk-3144489

38. Global Health Estimates. Geneva: World Health Organization; 2012. Available from: http://www.who.int/healthinfo/global_burden_disease/en/ [cited 2016 June 1].

39. Owolabi MO, Akarolo-Anthony S, Akinyemi R, Arnett D, Gebregziabher M, Jenkins C, et al.; Members of the H3Africa Consortium. The burden of stroke in Africa: a glance at the present and a glimpse into the future. Cardiovasc J Afr. 2015 Mar-Apr; 26(2) Suppl 1:S27–38.http://dx.doi.org/10.5830/CVJA-2015-038 pmid: 25962945

40. Feigin VL, Lawes CMM, Bennett DA, Barker-Collo SL, Parag V. Worldwide stroke incidence and early case fatality reported in 56 population-based studies: a systematic review. Lancet Neurol. 2009 Apr;8(4):355–69. http://dx.doi.org/10.1016/S1474-4422(09)70025-0pmid: 19233729

41. 41.. Strong K, Mathers C, Bonita R. Preventing stroke: saving lives around the world. Lancet Neurol. 2007 Feb;6(2):182–7. http://dx.doi.org/10.1016/S1474-4422(07)70031-5 pmid: 17239805

42. 42[Guideline] Whelton PK, Carey RM, Aronow WS, et al. Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Hypertension. 2018 Jun.

43. Benjamin EJ, Blaha MJ, Chiuve SE, et al, for the American Heart Association Statistics Committee

and Stroke Statistics Subcommittee. Heart disease and stroke statistics-2017 update: a report from the American Heart Association. Circulation. 2017 Mar 7.

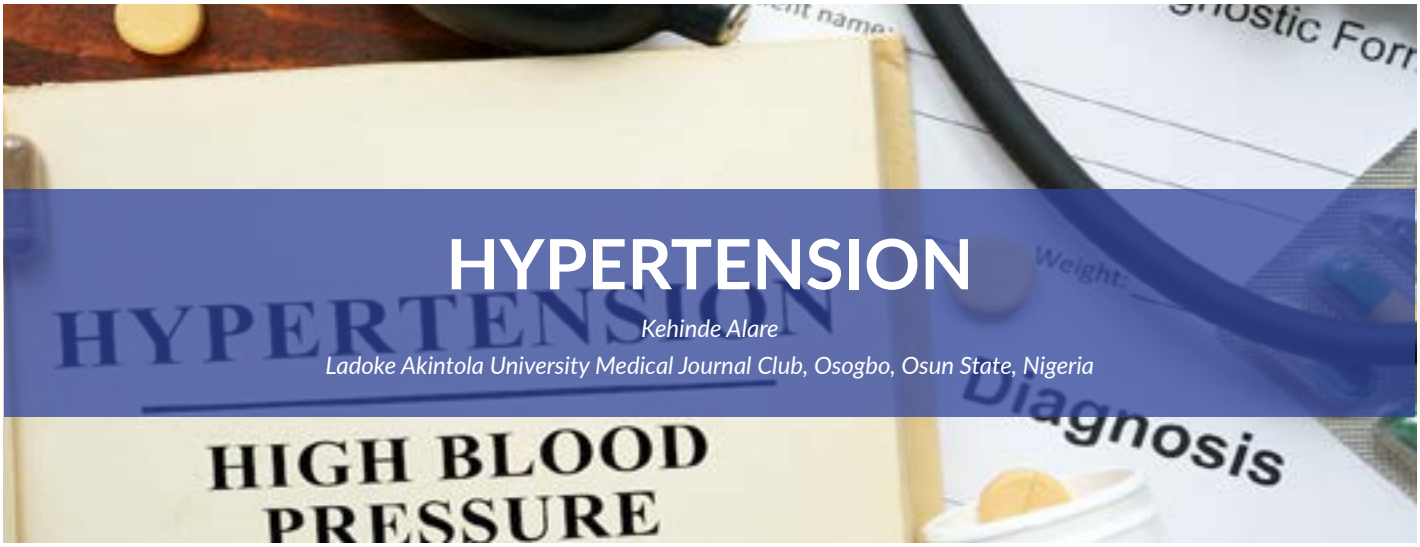
44. American Heart Association. Heart Disease and Stroke Statistics—2007 Update. Dallas, TX: AHA,2007.

45. 45 He J, Whelton PK. Elevated systolic blood pressure and risk of cardiovascular and renal disease: overview of evidence from observational epidemiologic studies and randomized controlled trials. Am Heart J. 1999

46. American Journal of Hypertension, Volume 29, Issue 8, August 2016, Pages 900–903, https://doi.org/10.1093/ajh/hpw054

47. Rhoads GG, Kagan A. The relation of coronary disease, stroke and mortality to weight in your hand in middle ages. Lancet 1983;1:492-495

48. Laver RM, clarke WR. Childhood risk factors for high adult blood pressure




# HYPERTENSION

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SUMMARY

*In this article, hypertension will be discussed under these headings : Epidemiology, Diagnosis, Causes, Stages, Management and Complications of uncontrolled hypertension. With this, the reader will be enlighten on the topic and how it's related to our health and lifestyle.*

*Key Facts:*

- *In 2015, 1 in 4 men and 1 in 5 women lived with hypertension*
- *Fewer than 1 in 5 people with hypertension have it controlled.*
- *Hypertension is a major cause of premature death world wide*
- *About 1.13 billion individuals of the world's population have hypertension*
- *Hypertension accounts for about 7.1 million death each year globally*

## INTRODUCTION

Hypertension can be defined as sustained systolic blood pressure of equal and above 140mmHg against the diastolic blood pressure of equal and above 90mmHg (WHO, 2009). However, there are various conditions that can account for raised blood pressure other than hypertension such as anxiety, heavy exercise, fear(as in white coat hypertension), hypertension in pregnancy, etc.

Blood pressure is the force exerted by the circulating plasma (blood) on the vascular walls especially the arterial walls. The measure of this is very crucial in the diagnosis and management of hypertension. Increased force exerted on these vascular walls beyond normal has effect on the integrity of the walls which could lead to wearing and tearing of the wall which can cause injuries to the blood vessels. Exposure of blood to an injured surface of the blood vessels causes coagulation of blood around the injured surface forming a thrombus which narrows the blood vessel and obstructing blood flow. With time, the thrombus formed will be slogged off as embolus circulating in the blood and may occlude some tiny vessels leading to reduction in blood supply to the affect area.

Hypertension has no known cure but can be properly managed to avoid complications. Hypertension has been proven to be associated with physical inactiveness. Intake of fatty diet can be managed by lifestyle changes. Some drugs has been helpful in the management of hypertension.

There is prevalence of hypertension in the non-hispanic communities which African communities are included with more males been affected.

The cause of hypertension may be secondary to some disease conditions or may be idiopathic and this is known as essentially hypertension.

## EPIDEMIOLOGY

The world prevalence estimation for hypertension is about 1 billio and about 7.1 million death per year (Gebreselassia, 2015).

There is prevalence of hypertension among the non-

hispanic blacks population than in Hispanic and non-hispanic whites combined and there has been increased in the incidence of hypertension in Africa which is sequential with the adoption of western lifestyle.

There is prevalence of hypertension in men than

women. It is common in adults above 40 years of age this may be due to reduction in the elasticity of vascular walls in adult.

In Nigeria, about 1.5 million cases of hypertension are reported each year. The prevalence is about 28.9% with prevalence of about 29.5% among men and 25.0% among women (Adeloye, 2015).

**DIAGNOSIS**

Diagnosis of hypertension can be made in the clinic, at home or any other place. However, it is advisable that a clinician makes the diagnosis because not all rise in blood pressure could be accounted for by hypertension and there are principles to blood pressure measurement such as positioning and all these can lead to misdiagnosis.

**PRINCIPLES OF PROPER BLOOD PRESSURE MEASUREMENT**

In order to avoid orthostatic hypotension or other factors influencing the result gotten while taking blood pressure readings, some measures are set in regards to the positioning and conditions in which the patients should be.

These principles are meant to put the patient in good condition for proper blood pressure investigation and they're highlighted briefly as follows:

- Blood pressure should be taken in a cool and quiet room.
- Blood pressure should be taken with the patient relax, not in any distress and also the patient should comfortable to reduce.
- The patient should be seated with an outstretched and supported arm. The measurements should be taken at least five minutes after the patient have been positioned.
- An automated machine with appropriate cuff size is recommended which should be placed at the same level with the heart of the patient. The cuff should be deflated at a rate of 2mm/s and the blood pressure should be measured to the nearest 2mmHg.

**CONFIRMATION OF DIAGNOSIS**

The World Health Organization recommends that the blood pressure should be measured with systolic blood pressure greater than or equal to 140mmHg and diastolic blood pressure greater than or equals 90mmHg at two different consistent instances before the diagnosis of hypertension can be ascertained.

When assessing cardiovascular risk the average blood pressure at separate visits is more accurate than the blood pressure taken at a single visit.

**CAUSES**

Hypertension could be due to an underlying disease known as secondary hypertension. Most times, hypertension is of unknown cause and it is referred to as essential hypertension.

**ESSENTIAL HYPERTENSION**

The etiology of essential hypertension is multifactorial some of are highlighted below:

- Genetic factors: there is familial concordance of hypertension in which children of hypertensive parents or anyone having first degree relative with hypertension tends to be hypertensive later in life too.
- Fetal factors: hypertension has been linked to low birth weight , many fetus born with low weight tends to have hypertension later in life.
- Others: obesity, excessive sodium intake, excessive alcohol intake and stress.

**SECONDARY HYPERTENSION**

Secondary hypertension is elevation in blood pressure as a result of underlying diseases which may be curable, e.g., diabetes mellitus, renal pathology , cardiovascular disease and others.

**RISK FACTORS**

The risk factors of hypertension can classified into:

- Modifiable factors
- Non-modifiable factors

Modifiable factors can be regulated by therapeutic moderations. They include diet, physical inactiveness, obesity, smoking , alcohol consumption.

Non-modifiable risk factors can be regulated or moderated. They include age, family history, race, sex, co-existing disease like diabetes and renal disease.

**STAGES**

Hypertension can be classified into stages on the basis of severity or readings gotten from the blood pressure measurement:

1. Stage I - Systolic blood pressure ranges between 140 - 159mmHg and the diastolic blood pressure ranges between 90 - 99mmHg.

2. Stage II - Systolic blood pressure ranges between 160 - 179mmHg and the diastolic blood pressure ranges between 100 - 109mmHg.

3. Stage III - Systolic blood pressure is greater than or equal to 180mmHg and the diastolic blood pressure is greater than or equal to 110mmHg.

**MANAGEMENT**

The management of diabetes can be classified under two broad categories:

1. Lifestyle management
2. Pharmacological therapy

**LIFESTYLE MANAGEMENT**

Some changes in lifestyle have proven to be of good therapeutic importance in the management of hypertension.

- o Diet : increase in consumption of low fat food, vegetables and fruits are encouraged in patient with hypertension and also reduction in dietary sodium intake.
- o Regular Physical Exercise: regular exercise has proven to be helpful in the management of early stage hypertension.
- o Reduction in alcohol consumption
- o Smoking cessation
- o Weight loss

**PHARMACOLOGICAL THERAPY**

Classes of drugs used in the management include:

- Angiotensin Converting Enzyme (ACE) Inhibitors e.g Lisinopril, Enalopril, Benazepril, Fosinopril...
- Angiotensin II Receptor Blockers (ARB) e.g Losartan, Valsartan, Olmesartan
- Renin Inhibitors e.g Aliskiren
- Diurectics e.g Hydrochlorothiazide, Furosemide, Amiloride, Torsemide, Spirolactone
- Calcium channel blockers e.g Amlodipine, Felodipine, Nifedipine, Verapamil
- Beta blockers e.g Atenolol, Propanolol, Carvedilol
- Alpha blockers e.g Doxazosin, Prazosin, Terazosin

The pharmacological management of hypertension is done step wisely with aim of restoring the blood pressure back to about 140/90mmHg and to around 150/90 mmHg in patient above 80 years of age.

This step wisely administration of drugs are done to avoid polypharmacy and the steps include:

Step 1: The first line of drugs in the management of hypertension is ACE inhibitors or ARB in patient under the age of 55years. In African population, resistance to ACE inhibitors and ARB has been observed. Calcium channel blockers are the first line drugs in the management of hypertension in Africans. Also, thiazide-like diuretics has been indicated as the first line of drugs in the management of hypertension in patient with heart failure.

Step 2: If optimal blood pressure has not been reached with step 1 then another class of drugs can be added . A combination of ACE inhibitor or ARB with calcium channel blocker is recommended or with a thiazide-diuretic. At this stage, two classes of anti-hypertensive drugs are used.

Step 3: After step 2, if an optimal blood pressure hasn't be achieved; then another class of anti-hypertensive can be added to make it a triple therapy. This is highest stage as polypharmacy is highly discouraged in the management of hypertension. The triple therapy of ACE inhibitor or ARB, calcium channel blocker and thiazide-diuretic can be used.

Step 4: After triple therapy has been administered to

optimal dosage and the targeted blood pressure of 140/90mmHg has not been reached then the patient can be referred to a specialist.

**COMPLICATIONS**

Complications of uncontrolled hypertension result in organ damage:

- Blood Vessels: Hypertension can lead atherosclerosis, vascular aneurysms and aortic dissection which all could be fatal.
- Kidney: Hypertension may cause chronic renal disease which is manifested as heamaturia , uremia, proteinemia.
- Heart : Hypertension can lead to cardiac failure which is evident as pulmonary oedema, myocardial infarction and left ventricular hypertrophy.
- Brain : Hypertension can lead to haemorrhagic and ischemic stroke, transient ischemic attack, seizure and vascular dementia
- Eye ; Hypertension can result in retinal damage.

When not properly managed, hypertension reduces life expectancy.

## REFERENCES

1. Kumar P , Clark M; Kumar & Clark's Clinical Medicine. Elsevier 9th Ed. 2015 pg1036-1038.

2. Whalen K. Pharm.D; LIPPINCOTT REVIEWS: PHARMACOLOGY. Wolters Kluwer 6th Ed. 2015 pg225- 227.

3. Kirubel Zemedkun Gebreselassie, Mojgan Padyab, "Epidemiology of Hypertension Stages in Two Countries in Sub-Sahara Africa: Factors Associated with Hypertension Stages", International Journal of Hypertension, vol. 2015, Article ID 959256, 12 pages, 2015. https://doi.org/10.1155/2015/959256

4. World Health Organization (WHO). (2019). Hypertension https://www.who.int/news-room/fact-sheets/detail/hypertension

# ROLE OF NUTRITION IN CHILDREN'S NEUROCOGNITIVE DEVELOPMENT FROM PREGNANCY THROUGH CHILDHOOD.

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## INTRODUCTION

Nutrition is one of the many factors that affect the development of the brain and cognitive ability of children. Cognition represents a complex set of higher mental functions sub-served by the brain. It includes attention, memory, thinking, learning and perception (Bhatnagar and Taneja, 2001). Cognitive development in pre- scholars is predictive of later school achievement (Tramontana et al., 1988; Clark et al., 2010; Engle, 2010). Academic achievement is important for future personal health and well- being. Schooling builds human capital-skills, abilities, resources which in turn shape an individual's health and well- being (Ross and Mirowsky, 1999). Indeed, more education has been linked to better jobs, higher income, higher socio-economic status, better health care access and housing, better lifestyle, nutrition and physical activity which are all well-known health determinants.

## DIETARY INFLUENCES ON COGNITIVE DEVELOPMENT

### Carbohydrates:

Carbohydrates are the main source of energy for the brain. The dietary reference intake for carbohydrates has been set at 130 grams per day for children. This value is based on the average minimum amount of glucose used by the brain. Carbohydrates aid in production of serotonin that helps a child's brain.

### Protein

Protein plays a role in cell growth and replication. Meat, milk, fish and cheese are good source of protein. These play a role in production of brain neurotransmitter - along with other nutrients - myelination, and development of dendritic arborisation of the developing brain. Protein and energy deprivation during brain development has been linked to reduction in brain cells, reduced brain size and poor cognitive performance. Lack of protein, along with calorie deficit in the diet, affects brain maturation which results in cognitive functional delays and long-lasting brain damage.

### Iodine

Iodine deficiency is a significant public health issue especially in children and pregnant women (World Health Organization, 2004). Iodine deficiency in many countries has led to food fortification which has led to use of iodized salt (World Health Organization, 2004). The relationship between iodine and cognitive development is extensively researched. Iodine helps in production of thyroid hormone and promotes growth. Deficiency during early years of life causes mental retardation and loss of IQ. In school-age children, iodine deficiency is linked with poor cognitive performance. Rich sources of iodine are fish, seafood, and algae. Although with iodization of salt, iodine deficiency in children has been controlled to a certain extent it is still present in some regions of the world. Psychomotor effects of iodine deficiency in children are more visible after the age of 2.5 years.

### Iron

This is one of the commonest nutritional deficiencies in both developing and developed countries. Iron deficiency anemia affects 30% of the world's population. Iron is an important brain nutrient and is found in varying concentrations in different parts of the brain. Iron requirements are higher during infancy and early childhood compared to adulthood. Iron intake is important for brain development, synthesis of neurotransmitters such as dopamine and

serotonin, and neuronal metabolism.

**Zinc**

Zinc deficiency is a major problem worldwide. It affects 40% of the global population (Maret and Sandstead, 2006). It is believed that zinc is a vital nutrient for the brain, with important structural and functional roles (Bhatnagar and Taneja, 2001; Black, 2003a; Bryan et al., 2004; Shah and Sachdev, 2006; Georgieff, 2007). Zinc is known to play an important molecular role in processes of gene replication, activation, repression, and synthesis. Zinc plays a role in neurogenesis, maturation, migration of neurons and in synapse formation.

Indirectly, zinc deficiency may affect cognitive performance through its interactions with other nutrients. Because iron and zinc are most bioavailable from many of the same foods and their absorption is inhibited by many of the same dietary substances, iron and zinc deficiencies often occur simultaneously. Negative effects of iron supplementation on zinc and copper status have been reported especially when the two nutrients were given in an aqueous solution, a situation similar to the intake of supplemental iron. No or fewer negative effects were observed when the two nutrients were given in a meal. Zinc is also known to interact with vitamin A metabolism. It contributes to the consequences of vitamin A deficiency.

**Vitamin B12**

Vitamin B12 impacts the rate of nerve conduction and helps in conversion of folic acid into its active form. It has been found that teenagers who consume macrobiotic diet till the age of 6 years had lower intelligence levels, spatial ability and short-term memory compared to teenagers who consume omnivorous diet. Deficiency of vitamin B12 leads to poor brain function, irritability, brain atrophy, and retarded myelination. Children with low consumption of animal foods such as milk, liver, meat, and eggs are at risk of vitamin B12 deficiency.

**Choline**

Choline is the precursor of neurotransmitter acetylcholine. It is found in brain membranes. It has been found to influence changes in brain structure and function. Choline deficiency causes memory and learning disorders in young children. Legumes, eggs, peanuts, salmon, and beef liver are good source of choline.

**Folic acid**

Folic acid affects neural stem cell proliferation and reduces cell death. Vegetables, legumes, and milk are good source of folic acid. Maternal folic acid intake has long-term effects on the child's neurocognitive development as it facilitates closure of fetal neural tube. Deficiency is associated with neurocognitive defects, language delay, autism and schizophrenia.

**Omega-3 Polyunsaturated Fatty Acids**

In recent years, there has been an increasing interest in the effect of essential fatty acids, particularly long chain polyunsaturated fatty acids (LCPUFA), on brain development. Of the human brain's dry weight, 60% is comprised of lipids of which 20% are decasahexaenoic acid (DHA; which is an omega- 3 fatty acid) and arachidonic acid (AA; an omega-6 fatty acid). This represents the two core fatty acids found in gray matter (Benton, 2010b; De Souza et al., 2011). Essential fatty acids play a central functional role in brain tissue. They are not only the basic components of neuronal membranes, but they modulate membrane fluidity and volume and thereby influence receptor and enzyme activities in addition to affecting ion channels. Essential fatty acids are also precursors for active mediators that play a key role in inflammation and immune reaction. They promote neuronal and dendritic spine growth and synaptic membrane synthesis thereby influencing signal processing and neural transmission. In addition, essential fatty acids regulate gene expression in the brain (McCann and Ames, 2005; Eilander et al., 2007; Innis, 2007; Cetina, 2008; Wurtman, 2008; Ramakrishnan et al., 2009; Ryan et al., 2010; Schuchardt et al., 2010; De Souza et al., 2011).

Most of the research on the relationship between omega-3 intake and cognitive development has focused on pre-term infants or those with low birth weight. Studies have shown that visual acuity, visual recognition, memory, and enhanced performance on measures estimating cognitive performance in early infancy are significantly affected by omega-3 deficiency in human preterm infants as well as by long-chain PUFA supplementation. There is evidence that fatty acid metabolism may be implicated in a

cluster of neurodevelopmental disorders including attention deficit-hyperactivity disorder (ADHD), dyslexia, dyspraxia, and the autistic spectrum. These neurodevelopmental disorders have similar feature of low level of PUFAs in red cell membranes and plasma. The mechanism is thought to be related to inefficiency in the synthesis of long-chain PUFAs from their precursors.

## EFFECT OF NUTRITION ON BRAIN DEVELOPMENT

Brain development is a temporally extended and complex process. Nutrition plays crucial roles in brain development at the macrostructure level and (e.g., development of brain areas such as the hippocampus) microstructure level (e.g., myelination of neurons) which have impacts on cognitive development. The timing of nutritional influences on the brain is also important. Nutritionists believe that there are critical periods of rapid brain growth and development; from gestation until the first 2 years of life during which brain development is sensitive to insult and deficiencies in diet have greater impacts. Thus, most research investigating the impact of nutrition on cognitive development has focused on effects in children under the age of 2 years.

This view does not consider that certain brain areas are not yet developed by the age of 2 years. For example, the slowest area to be completely myelinated is the frontal lobe which begin myelination at approximately 6 months of age and continues throughout childhood, adolescence, and adulthood. Spurts of development of the frontal lobes occur from birth to 2 years, 7 to 9 years, and in the mid-teenage years. The frontal lobes are thought to be responsible for executive, "higher order" cognitive activities such as planning, developing strategies, testing hypotheses when problem solving, focusing attention, inhibiting irrelevant stimulation, and collating memories. Therefore, poor nutrition or nutritional interventions may have effects on the developing functions of the frontal lobes throughout childhood.

Folic acid deficiency between 21 and 28 days after conception (when neural tube closes) predisposes the fetus to a congenital malformation called a "neural tube defect". Individual brain development follows a genetic program which is influenced by environmental factors including nutrition (Bryan et al., 2004; Toga et al., 2006; Giedd et al., 2010). Environmental influences may modify gene expression through epigenetic mechanisms; gene function is altered through the process of DNA methylation, histones modification and the modulating effect of non- coding RNAs without the alternation of the gene sequence per se. These epigenetic factors can cause long lasting or even heritable changes in biological programs ( Levi and Sanderson, 2004; Rosales et al., 2009; Murgatroyd and Spengler, 2011; Lillycrop and Burdge, 2012).

### Effect of breast feeding on cognitive development

Breast feeding plays a vital role in the cognitive development of infants. Breast milk provides optimal nutrients for babies. It has the right amount of nutrients needed for infants. Breastfeeding provides nutritional and immunological benefits to infants and also prevents infections. Breastfeeding improves cognitive development and emotional interactions between the mother and infant. A previous meta-analysis of 20 studies that compared breastfeeding with formula-feeding reported that breastfeeding is associated with significantly improved cognitive development from infancy to adolescence (www .ncbi.nlm.nih.gomc) .

In this cohort study of 50 mothers and their babies, investigators analyzed breast milk composition and frequency of feeding at 1 and 6 months of age. Cognitive development was measured at 24 months using the Bayley-III scale, a standardized test of infant and toddler development. The study showed that the amount of 2'FL in breast milk in the first month of feeding was related to significantly higher cognitive development scores in babies by the age of 2. The amount of 2'FL in breast milk at 6 months of feeding was not related to cognitive outcomes indicating that early exposure to breast milk may be more beneficial. Using a statistical technique, mediation analysis, the investigators were able to independently evaluate the effects of breastfeeding in general and the effects of the oligosaccharide 2'FL. These observations allowed the team to conclude that the increased neurodevelopment provided by breastfeeding was due primarily to mothers who were producing more 2'FL for the baby to consume. While the investigators observed neuro-enhancement explained by higher 2'FL during the first month of a baby's life, this effect was not observed when looking at 2'FL content of breast milk at the six-month time-point. Being able to identify factors critical to early neurodevelopment offers the possibility for supplementing women's breast

milk in individuals who produce lower quantities of this important substance.

## Effect of breakfast on cognitive development

The metabolism of glucose in children's brain increases and almost if not twice of that of the adult's metabolic rate, children and adolescent who habitually consumes breakfast are more likely to have better micronutrient and macronutrient. Children have a higher brain glucose metabolism than adults.

Fourteen studies reported equivocal findings on attention outcomes, of which 4 of 14 studies (29%) were in well-nourished adolescents ,7 of 14 studies (50%) were in younger well-nourished children and 3 of 14 studies (21%) were in undernourished children. However, one study observed a ceiling level of performance on the attention task, which may account for the null effects . Moreover, one study assigned participants to a condition based on habitual breakfast intake, which is an obvious bias.(Nutritional effect of behavior in children),( J psychiatr Res 1982). In summary, effects of breakfast on cognitive development are thus; breakfast supplies the required glucose to the brain for its high metabolism, children are said to be more prone to the adverse effect of overnight fasting, breakfast is important to fuel the brain in the morning, breakfast influences the volume of gray and white matter in children, the type of breakfast children eat can potentially have a long term influence on cognitive development. In malnourished children, systematic review concluded that having breakfast is beneficial for cognitive function and development.

## Effect of malnutrition on cognitive development

Malnutrition is a lack of adequate nutrition. This can be under-nutrition or over-nutrition. Malnutrition is the consequence of a combination of inadequate intake of protein, carbohydrates, micronutrients and frequent infections. Malnutrition is associated with both structural and functional pathology of the brain. Structurally, malnutrition results in tissue damage, growth retardation, disorderly differentiation, reduction in synapses and synaptic neurotransmitters, delayed myelination and reduced overall development of dendritic arborization of the developing brain.

Nutrients like iron, zinc, folic acid, iodine should not be lacking in a child's diet as deficiency in these can be fatal on cognitive development. Malnutrition causes children to have less energy and interest for learning. Research has shown that malnutrition is responsible for stunted growth in children. Also, recent reviews concluded that even mild but persistent malnutrition in the first two years of life negatively influences reasoning, visuospatial functions, IQ, language development, attention and learning.

## Neuro developmental disorders associated to poor nutrition

**Attention deficit hyperactivity disorder (ADHD):** is a neuro developmental disorder characterized by inattention or excessive activity and impulsivity which are otherwise not appropriate for a person's age. Individuals with ADHD display difficulty in regulating emotions or problems with executive function. Symptoms appear before age 12 and present for more than 6 months.

**Dyslexia:** is a learning disorder that involves difficulty in reading due to problems with identification of speech sound and learning how to relate with letters and words. It affects areas of the brain that process language.

**Dyspraxia:** is a chronic neurological disorder beginning in childhood affecting movement and co-ordination as a result of inaccurate transmission of neural messages to the body systems.

## CONCLUSION

Early maternal nutrition during pregnancy is important, as well as early childhood nutrition, to achieve future personal health and well-being. It is also important in building a good economy in developing and under-developed countries by grooming able and fit generation of leaders.

## REFERENCES

1. Nyaradi A, Li J, Hickling S, Foster J and Oddy WH (2013). The role of nutrition in children's neurocognitive development, from pregnancy through childhood.

2. Janet Bryan, Ph.D., Saskia Osendarp, Ph.D., Donna Hughes, MPsych, Eva Calvaresi, MPsych,

Katrine Baghurst, Ph.D., and Jan-Willem van Klinken, Ph.D. Nutrients for Cognitive Development in School-aged Children.

3. Cristina Campoy, MD, phD. Impact of Pregnancy Nutrition on Cognition.

4. Paige K. Berger, Jasmine F. Plows, Roshonda B. Jones, Tanya L. Alderete, Chloe Yonemitsu, Marie Poulsen, Ji Hoon Ryoo, Bradley S. Peterson, Lars Bode, Michael I. Goran. Human milk oligosaccharide 2'-fucosyllactose links feedings at 1 month to cognitive development at 24 months in infants of normal and overweight mothers. PLOS ONE, 2020; 15 (2): e0228323 DOI: 10.1371/journal.pone.0228323

# CLAIM OF ADVERSE DRUG REACTIONS MASQUERADING AS MYIASIS IN A 13 - YEAR OLD NIGERIAN TEENAGER - A CASE REPORT AND REVIEW OF LITERATURE

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## ABSTRACT

*Myiasis is the infestation of the skin and subcutaneous tissue by the larvae of obligatory dipterous insects. It affects under – developed countries especially in the tropics and some parts of the temperate countries. It is however under – reported because of wrong diagnosis and fear of stigmatization even when the diagnosis becomes evident. This case report and literature review highlights how a coincidental diagnosis of malaria in a boarding house student and the development of multiple nodular swellings following commencement of prescribed antimalarial drugs in the same subject, raised the claim of idiosyncratic drug reactions by the parents with the threats of initiating legal proceedings against the facility of care. This calls for more awareness and education for the first on call clinicians and the general population.*

## INTRODUCTION

Myiasis is a distressing illness due to burrowing of larva from the eggs deposited by some common flies (1) and may appear frightening to the patient and parents; it may be challenging for the attending clinician especially if unfamiliar with this disease condition. This case is reported because of adverse drug reactions' claim by the parents and the unusual presentation.

Myiasis is endemic in under – developed and tropical countries like Sub-sharan Africa, Asia. It has also been reported in South East Mexico, South America and in some temperate regions of the world. Cases of myiasis though established in Southern part of Nigeria have been under – reported and under – published for reasons varying from stigmatization, delay presentation and wrong diagnosis. (2,3)

Families of Myiatic flies like Calliphoridae, Sarcophagidae, Muscidae and Phoridae often create infestation when they lay eggs that hatch into larva on open wounds and lesions. (1 - 3) However, myiasis has been reported rarely even on an unbroken skin. (2), (3). This is especially possible in handicapped, physically challenged subjects, abandoned neonates or neglected younger infants who cannot ward off flies on their own. More so, myiatic flies can lay their eggs on washed clothing spread out on the grasses or on the lawn for drying and the larva then burrow into the body when subjects wear such clothes un-ironed. More importantly risk factors for myiasis include poor personal and environmental hygiene, extremes of age, low socio – economic status, Parental deprivation or neglected children, immunosuppression including malnourished children and infection with Human immunodeficiency virus. (1), (2)

We report a 13 – year old male boarding house student who was re - presented by parents with swelling of the lower lip and armpit coincidental with commencement of antimalarial and analgesics for a febrile illness that started 3 days to onset of the swellings.

Parents were on the verge of suing the hospital due to alleged drug reaction when a maggot was noticed at the opening of the wound which was subsequently recovered from the lips and the left armpit.



*Figure 1a*



*Figure 1b*



*Figure II. Credit: The Natural History Museum*

## CASE REPORT

Master C. A is a 13 - yr J.S S 3 boarding house male student who presented at a Tertiary private facility in Port Harcourt – Meridian Hospital, Rivers state, with fever and malaise of 3 days prior to his first presentation. Developed swelling over the lower lip and the left armpit 24hours after commencement of antimalarial and anti pyretics.

Fever was described as not very hot and relieved temporarily with the use of oral paracetamol. There was feeling of unwell (malaise) from the onset of fever. Parents brought in patient for a review when symptoms were not resolving. Being a boarding house student, history suggestive of meningitis (neck stiffness, headache and photophobia) was sought but reported in the negative. His past medical history was however adversely eventful. Patient had previous history of fractured ulnar and radial bones sustained from a fall while playing at the hostel 4 weeks prior to the latest presentation. Master C. A had been placed on Plaster of Paris (POP) for immobilization by the hospital Orthopaedic surgeon about 4 weeks prior to this illness. Parents live in a self – owned duplex in an estate. No bushes or stagnant water around. Father is a Petroleum Engineer who works in an oil firm and Mother is a Legal practitioner. Both are of class 1 Socio – economic classification. Nevertheless, child is a J.S.S 3 class student who attends a boarding house school where there are bushes around. He also launders his clothes and spread them on the grasses in the school. He does not iron his washed clothes because there is no electric iron in his hostel.

His examination findings were those of an ill looking teenager, with an above elbow POP cast on his right forearm. He was febrile with temperature of 38.2 degrees Celsius. The chest was clear clinically and the Blood pressure and the heart sounds were normal. No signs of meningeal irritation. Anthropometry include a premorbid weight of 42kg that corresponds to the 30th percentile, (weight with POP cast was higher and unreliable); height of 155cm at 25th percentile and Body Mass index (BMI) of 17.5Kg/metre square (35th percentile). Other systemic examinations were essentially normal.

His malaria parasite result was scanty. His full blood count showed a total count of 9900 cell/mm3 with neutrophilia of 72% considered appropriate for age. He was subsequently started on tabs ACT 20/120 IV twice daily for 3 days, paracetamol tablet II thrice daily for 2 days.

About 24 hours into commencement of ACT, patient developed lip swelling and left armpit nodular swelling which necessitated re – presentation in the same facility. There was crawling sensation on the lips which warranted patient intermittently requesting his parents to clean his lips with soft clothing. There was poor sleep due to the pains and the crawling sensation. The swellings were progressively increasing with associated lip pain and painful left arm pit. The parents were worried and threatened legal action on the ground of drug reaction.

On further review, there was redness and swelling of the lower lip with a pale centre discharging scanty serous fluid. The lower lip was also tender. No lymph nodes enlargement noted. The left armpit also showed similar lesion – 2 nodular swellings. A presumptive diagnosis of furunculosis to keep in view suspected idiosyncratic adverse drug reaction to arthemeter - lumenfantrine was made. Parents were counseled with regards to non -peculiarity of ACT to such reactions. The meaning of idiosyncratic reaction was also explained to the parents with emphasis on possible patient specific reaction that is not peculiar to this specific ACT. Nevertheless, the ACT in use arthemeter – lumenfantrine - was discontinued while patient was admitted for close monitoring. The repeat full blood count showed relative lymphocytosis of 79.5% at this time with 17.5% neutrophils and 3% eosinophils. Oral camoquine was administered with oral artesunate to complete Malaria treatment already started with arthemeter – lumenfantrine; while monitoring of patient closely continued. Intravenous (IV) Antibiotics [IV cefuroxime and IV metronidazole] were also given in view of the presumptive diagnosis of furunculosis.

The swellings became more intense 3rd day into admission when patient persistently called the parents attention to the crawling sensation on his lips. There was application of petroleum jelly on the patient's lips by the parents in an effort to soothe the patient's feeling of crawling sensation. Yet the parents did not see anything crawling until the head of a maggot (suspected fly larva) became conspicuous on the lower lip. This was frightening to both the patient and the parents. Fears were allayed while the attending paediatrician extracted the maggot with a pair of forceps under parenteral analgesia. Two similar maggots were rcovered from the swelling at left arm pit. A total of 3 larvae in all [See figure 1a and 1b1] and they were sent to Microbiology arm of the facility for confirmation as larvae of Tumbu fly (Condylobia anthropophaga). Patient's retroviral screening requested following the diagnosis of myiasis was negative.

The residual ulcers after the removal of the larvae were thoroughly cleaned with diluted hydrogen peroxide, followed by normal saline. IV Antibiotics [cefuroxime and metronidazole] were continued for 72 hours further before the child was discharged. The lip swelling, fever and left arm pit swelling resolved within few days after the extraction. The child was also given Tetanus toxoid booster dose at this time. Ivermectin was not available as anti – parasitic agent as at the time of patient's admission. Nevertheless, patient had Tetanus toxoid Booster dose, was seen at follow up clinic in the following weeks and the lesions had healed well.

## DISCUSSION

Myiasis is the parasitic infestation of the body of a live mammal by fly larvae (maggots) that grow inside the host while feeding on its tissue.(1) Myiasis therefore occurs when fly larva grows and feeds in the body of a mammal. (2), (3). In human, certain flies need human host for larval development. Examples of such flies are Dermatobia hominis (human botfly), Cordylobia anthropophaga (tumbu fly) See figure II; and Oestrus ovis (Sheep botfly). Whereas Dermatobia hominis is found in Central and South America, C. anthrophaga is common in the tropical Africa. (2), (3) (4)

Although most frequently observed in under – developed and tropical countries, human myiasis

has been observed throughout the world including the temperate regions. Cases of human myiasis in Nigeria and most African countries are most probably under – reported because some remain undiagnosed, unidentified or unpublished. In the earliest 20s or 30s, people with myiasis were dumped in evil forest as abominators. (2)

Flies are generally most commonly attracted to open wounds, urine or faeces but myiatic flies can create infestation even on unbroken skin or enter the body through the nose or ears. Some larva can burrow onto the lips or the skin if their eggs are deposited on the lips or on un-ironed laundered clothing. Eggs can hatch into larva as early as 1 – 3 days when conditions are favourable to 15 days when conditions are not so favourable.(2) (3).

There are various forms of myiasis depending on the site of manifestation. For example, cutaneous myiasis involves the skin leading to painful, slow – developing ulcers or furuncle – (boil-) like sores that can last for a prolonged period. When furuncle or boil – like lesions occur due to infestation by fly – larva, it is called Furuncular myiasis. Cutaneous myiasis can therefore present as furuncular, migratory and wound myiasis depending on the type of infesting larvae. It is however, a migratory myiasis when there is exposure to infested cattle or horses such that the fly larva forms an apparent tortuous thread – like red line under the skin that ends in a terminal vesicle. The larva lies ahead of the vesicle in apparently normal skin. (1) (2)

Our patient has furuncular myiasis based on the clinical presentation of boil – like lesions preceeding the extraction of the maggot. The initial crawling sensation noted by patient is suggestive of initial migratory form of cutaneous myiasis before the larva exited the body. Common sites of burrowing and exit include the axillae, the limbs, umbilical cords, vulva, the face and rarely lips. (1) (2). Our patient's furuncular myiasis involved the left axilla and the lower lip. The involvement of the lip in our patient was a rare finding in most of the reports encountered (1), (2), (5) (6) and this made the suscipicion of a drug action more plausible. Notwithstanding, myiatic flies can patch on the lips of subjects who are unable to ward off flies as earlier mentioned. Our patient might have been incapacitated to ward off wandering flies because of his immobilized right forearm and being a right handed person hitherto to having a fractured right upper limb.

In addition, application of petroleum jelly by the patient's parents as adjuvant to their child's lip swelling was significantly (albeit unknowingly) helpful in forcing the larva out. Use of palm oil, petroleum jelly and cholesterol free oil – laser oil have been documented to asphyxiate the larva by blocking their respiratory sinuses thus forcing them to exit the subcutaneous tissue. (2), (3), (7) Also, ivermectin and use of appropriate antibiotics were part of the treatment plan in myiasis.(8) Our patient only had broad spectrum antibiotics with anti – staphylococcal and anti – anaerobic properties on the suscipicion of furunculosis as ivermectin was not available. In fact, Oguugua et al use ivermectin solution and ointment to wash and seal the pores created after extracting the larva with promising result.(2) Ivermectin is thus a recognised anti – parasitic agent for myiasis. (2), (3). Nonetheless, secondary bacterial infections are complications of myiasis and this was covered by the use of those broad spectrum antibiotics. Other measures taken after extracting the larva of the myiatic flies is the use of the tetanus prophylaxis because myiasis can be a portal of entry for Clostridium tetani; therefore, vaccination should be considered in affected individuals. (8).

Of particular interest is the occurrence of myiasis in children of higher socio – economic class: a non – risk factor for myiasis. The historically established risk factor of myiasis in this child was the spread of his washed clothing on the grasses in his school hostel and his wearing of un -ironed clothing after laundry. Kabir et al (3) reported similar cases of Furuncular myiasis in 2 Nigerian Siblings of higher Socio – economic class parents. Both siblings had history of wearing un – ironed laundered clothes hitherto spread on the lawn in their compounds. This calls for serious concerns regarding care of children's clothing and scaling up of environmental hygiene practices even in the homes of higher socio – economic class parents.

Going further, Myasis can occur in physically challenged subjects or neglected neonates and or young infants when myiatic flies settle and lay their eggs on the body surfaces or mucosal membranes of the physically challenged or physically helpless subjects. Our patient had fractured right upper limb on immobilization and that put him at risk of being incapacitated albeit temporarily. Also, the history of fever and general unwellness in our patient made his first diagnosis of having malaria appropriate and confounded the diagnosis of Myiasis ab initio. Nigeria being in the endemic region suffers the world's greatest malaria burden, with approximately

51 million cases and 207,000 deaths reported annually (approximately 30 % of the total malaria burden in Africa). (9) Although, this should not be taken for granted, clinicians on first contacts with the Paediatric age group must strive to look for other aetiologies of febrile illnesses because not all fever may be due to Malaria parasites in the blood stream. Nevertheless, some forms of myasis may elicit systemic response similar to a febrile illness like malaria or sepsis.(10)

## CONCLUSION

Myiasis is an under – reported distressing illness in Nigeria despite the establishment of its occurrence especially in Southern Nigeria. Further awareness, patients and parental education on environmental and personal hygiene practices with appropriate treatments are very crucial to reducing the burden of the disease.

## REFERENCES

1. Fabio F. Omar L. Myiasis, Clin Microbiol Rev 2012;25:79 105.

2. Ogugua KO, Ted GA, Eme EO, Dorcas SB, Chika HA, Lekia K, et al. Human Myiasis in neonates and children of Niger Delta wetlands and South – East Nigeria. JCDSA 2011;1:171-6

3. Kabir H, Abdallah RJ, Lawal NO, Farouk ZL. Furuncular myiasis in two Nigerian siblings – case report and review of literature. Niger J. Basic Clin Sci 2019; 16:145-8.

4. Edugbola LD. Cutaneous myiasis due to Tumbu fly, Cordylobia arthropophaga in Ilorin, Kwara State, Nigeria, Acta Trooical. 1982; 39: 355 – 62.

5. O. K. Ogbalu, T. G. O. Achufusi and C. Adibe, "Incidence of Multiple Myiasis in Breasts of Rural Women and Oral Infection in Infants from the Human Warble Fly Larvae in the Humid Tropic: Niger Delta," International Journal Dermatol. Vol. 45, No. 9, 2006, pp. 1089- 1070. doi:10.1111 /j.1365-4632.2006.02983.

6. R. Jelinek, H. D. Nothdurft, N. Rieder and T. Löscher, "Cutaneous Myiasis: Review of 13 Cases in Travellers Returning from Tropical Countries," International Journal of Dermatology, Vol. 34, No. 9, 1995, pp. 624-626.

7. T. A. McGraw and G. W. Turiansky, "Cutaneous Myiasis," Journal of the American Academy of Dermatol Vol. 58, No. 6, 2008, pp. 907-926.

8. Bolognia JL, Jorizzo JL, Rapini R. Cutaneous myiasis. Dermatology. 2nd ed. Mosby Elsevier; 2008. Vol 1: 1300-01.

9. WHO . World malaria report 2014. Geneva: World Health Organization; 2014.

10. Hartley M. Cutaneous myiasis. @https://dermnetnz.org/topics/cutaneous-myiasis/#:~

# IMPLICATIONS OF COVID-19 ON CLINICAL CLERKSHIP AND HOSPITAL POSTINGS - A NIGERIAN MEDICAL STUDENT'S PERSPECTIVE

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**ABSTRACT**

*The corona virus pandemic hit the world, sweeping through all its parts; crumbling economies and overwhelming health systems. Undergraduate medical education has been disrupted and put to a halt. As Nigeria braces to open up medical schools and clinical rotation about to begin, the palpable panic that accompanies patient-student spread of the virus will definitely affect undergraduate medical education in Nigeria. Presently, little is known in Nigeria about the long term effects of COVID-19 on clinical clerkship and undergraduate medical education and adaptations to juxtapose future pandemics with medical education. This review accesses risks, plans and policies in place and how they will affect undergraduate medical education.*

**Keywords:**
*COVID-19, Clinical Clerkship, Undergraduate Medical Education, Clinical Rotations, SARS-CoV2*

## INTRODUCTION

The Coronavirus disease (COVID-19) is an infectious disease caused by the novel Coronavirus Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) 1, 2. COVID-19 was first discovered in China in December 2019 and the first case confirmed in Nigeria announced in February 2020 2, 3.

The rapid spread of the virus and its declaration as a Public Health Emergency of International Concern by the World Health Organisation (WHO)1 has caused countries to fold up businesses, schools, markets, places of worship and all forms of public gatherings. Among the various economic4 and social impacts of COVID-19, medical schools have been shut down and education put to halt around the world5.

This review aims to put side by side the impact of SARS-CoV2 on the fate of medical students and how it might affect hands-on practical training and clinical clerkship.

## REVIEW

In Nigeria, undergraduate medical education is divided into two parts: a pre-clinical part and a clinical part comprising of various levels of professional exams 6. Undergraduate medical students spend the first three years of their education in the pre-clinical school and the last three years in clinical school. In clinical school, students begin hospital rotations gaining exposure to clinical practice and methodology 7. Undergraduate medical education in Nigeria involves series of didactic lectures, tutorials, seminars, presentations, small group discussions and clinical postings. The major part of learning for medical students takes place in the hospital; here students get instructional teaching and guided learning.

At the wake of COVID-19 and attempts to curb its spreads among students, the National Universities Commission initially ordered the closure of Nigeria universities for one month beginning from 23 March 2020 but this has since been extended to several months 8.

While clinical clerkship has been cancelled or suspended in some countries around the world 9. No news about cancellation/suspension of hospital rotations has been announced in Nigeria yet but there are debates and concerns about whether adequate measures are in place to protect medical students against the virus.

Student-Patient contact will be affected drastically in an attempt to curtail the spread of Coronavirus. Participation of students in patient healthcare might

be affected and invariably learning impeded. Clinical exposure is a crucial part of medical learning and COVID-19 might reduce access to this exposure. Teachings in wards, accident and emergency units and clinics might be eliminated in many medical schools with high cases of SARS-CoV2. The fear of the virus might also affect the composure and willingness of medical students to participate in patient-care.

Recent papers on the effect of COVID-19 on the mental health of students have also shown that the virus has caused an increase in the academic stressors on students 10. The widespread fear and uncertainty of the virus is likely to increase the burden on students making learning uncomfortable. These suggest that the COVID-19 pandemic may have continuing implications on the academic performance, careers and competence of many medical students. Many students have been out of school for a long time and this is also likely to have effect on their academic performance.

The economic impact of COVID-19 is also a tale to tell. Many countries are at the brink of recession and inflation rates skyrocketing (11). It is similarly likely that this might cause a financial and mental health burden on students, affecting their academic performance.

As medical education requires physical hands-on training, the option of embracing technology and switching to online teachings might be a far call.

## CONCLUSION

As medical schools prepare to open up, clinical clerkship and hospital rotation would be hit greatly. The Nigerian University Commission and Medical and Dental Council of Nigeria (MDCN) the governing councils of medical education in Nigeria has not put forth any guideline outlining the fate of medical students in the face of this pandemic.

COVID-19 has the potential to sub-acutely affect medical education in Nigeria if timely intervention and policies are not put in place.

## RECOMMENDATION

1. Provision of free or subsidized Personal Protective Equipment for students.
2. A review of undergraduate medical education policies in the situations of global public health challenges.

## LIMITATION TO THE STUDY

There is little information on guidelines for medical schools in Nigeria in the post COVID-19 era.

## CONFLICT OF INTEREST

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## REFERENCES

1. WHO. Coronavirus. https://www.who.int/health-topics/coronavirus#tab=tab_1. Accessed 5th September, 2020.

2. WHO. Coronavirus disease (COVID-19) pandemic. https://www.who.int/emergencies/diseases/novel-coronavirus-2019. Accessed 5th September, 2020.

3. First Case Of Corona Virus Disease Confirmed In Nigeria https://ncdc.gov.ng/news/227/first-case-of-corona-virus-disease-confirmed-in-nigeria . Access 5th September, 2020.

4. Anoushiravani AA, O'Connor CM, DiCaprio MR, Iorio R. Economic Impacts of the COVID-19 Crisis: An Orthopaedic Perspective. J Bone Joint Surg Am. 2020;102(11):937-941. doi:10.2106/JBJS.20.00557

5. Clark J. Fear of SARS thwarts medical education in Toronto. BMJ. 2003;326(7393):784. doi:10.1136/bmj.326.7393.784/c

6. Ibrahim M. Medical education in Nigeria. Med Teach. 2007;29(9):901-905. doi:10.1080/01421590701832130

7. Marjo Wijnen-Meijer, William Burdick, Lonneke Alofs, Chantalle Burgers, Olle ten Cate. (2013) Stages and transitions in medical education

around the world: Clarifying structures and terminology. Medical Teacher 35:4, pages 301-307.

8. Azeezat. Adedigba. Coronavirus: NUC announces closure of Nigerian universities. from https://www.premiumtimesng.com/news/headlines/382880-coronavirus-nuc-announces-closure-of-nigerian-universities.html; Accessed 5th September, 2020

9. Akers A, Blough C, Iyer MS. COVID-19 Implications on Clinical Clerkships and the Residency Application Process for Medical Students. Cureus. 2020;12(4):e7800. Published 2020 Apr 23. doi:10.7759/cureus.7800

10. Nicholas Grubic, Shaylea Badovinac, Amer M Johri. Student mental health in the midst of the COVID-19 pandemic: A call for further research and immediate solutions. First Published May 2, 2020 Letter. https://doi.org/10.1177/0020764020925108

11. UNDP COVID-19 Socio-economic impact. https://www.undp.org/content/undp/en/home/coronavirus/socio-economic-impact-of-covid-19.html Accessed 5th September, 2020

# COVID-19: CAN SURGICAL MASKS OR RESPIRATORS PREVENT CORONA VIRUS?

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The world is suffering from the deadly infection of Coronavirus. The World Health Organization (WHO) has declared the infection as 'pandemic'. The early symptoms of this deadly infection are high fever, shortness of breath, tiredness and dry cough. So, the entire world is under lockdown to break the chain of COVID-19 with certain precautionary steps being ensured. The contemporary ways to sanitize oneself is the 20 seconds formulae of hand wash or using disinfectant containing at least 60% alcohol, avoid touching surfaces and handshakes and unnecessary contacts. The use of mask and covering face using napkin or elbow while coughing and disposing the napkin would add an added advantage.

We all know that sanitizer containing good quality alcohol can kill most of the germs, bacteria and viruses including the fatal Coronavirus. In a research, it was found that the Coronavirus can survive for 3 days (72 hours approx.) on surfaces like steel or plastic.

Now, the question arises whether any face mask or covering of the mouth with any cloth helps to prevent Coronavirus from invading our body?

Firstly, I want to let you know that no mask can provide 100% safety from Corona Virus. There are different types of masks available in market and most common of them are surgical masks and Respirators.

They act as a barrier to prevent the foreign body from entering our body, but the main difference between them is that Surgical masks are used only to prevent dust particles or large particle droplets that enter through our nose or mouth while Respirators can prevent vapours, airborne microorganisms, fumes or particulate matter that enter through nose or mouth . But as we know that the virus can enter our body through the eyes so the person wearing spectacles or googles can prevent Covid19 from entering the body to some extent. As spectacles or googles are left open from its sides so virus can reach our eyes easily. Hence, it is advised to stay at home and take precautionary steps while stepping out of home. The frontline workers including police, media persons, essential service and sweepers wear face shield along with N95 masks to safeguard themselves.

Surgical masks are mostly used by people who are not into any essential services as these are not designed to protect the wearer from breathing in airborne bacteria or viruses. Coronavirus is smaller than the PM 2.5 cut-off but bigger than some regular dust particles and gases. These masks can be used by a person with weak immune system or suffering from chronic health diseases. Persons dealing with severe respiratory or breathing issues are more prone to get infected from coronavirus, hence it is advisable for the infected person to use face mask as well in order to protect others from getting infected. The use of masks is also recommended for healthy people to prevent infection. Surgical masks are of three types-: 2-layered surgical mask, 3-layered surgical mask and 6-layered surgical mask

The protection also depends on the number of layers, more number of layers contribute more protection. A 3-layered mask is better than 2-layered masks, the cost of 3-layered mask is also higher than 2-layered mask. While the 6-layered masks are best for now as it can protect us from some of the viruses but cannot protect from SARS CoV-2 to enter the body. This type of mask is loose-fitting and virus can easily enter through the opening sides. And one more drawback is that it is not having any particular side from which a person have to wear it. So as a human it's common to make mistake and use different side to wear while reusing, keeping it anywhere after using can spread risk of virus at that particular place and without knowledge of its proper way of wearing help the virus to enter

the body. Also, the outer layer of the mask consists of virus and accidental touching can lead to the novel coronavirus.

As mostly surgical masks tend to be disposable, it get wet due to water vapours generated during respiration, that means they need to be replaced with new mask or washable 6-layered mask, washing it with 60°C hot water with a disinfectant to kill the germs can help in reusing them.

Respirators are used by health professionals as they are having contact with infected person quite often. These are at a risk of getting contaminated by infected person that's why they use it along with face shield to be more secure from the Coronavirus. N95 mask is most common respirator consisting of filters that can prevent 95% i.e. 0.3 micron airborne particles. It is a type of mechanical filter respirator and is better than any type of surgical mask. Respirator is tight fitting and have no opening, it protects in a better way. It protects us from most of the viruses which enter from mouth or nose openings but still protects against SARS CoV-2 to a small extent. When someone around us coughs or sneezes then the germs or droplets exhaled can stick on a person's clothes or long hair especially in women, so the hair must be covered with a cloth while going out and changing clothes after coming back to home is recommended. For men too keeping a long bushy beard and facial hair are also make one prone to Corona virus so the best way is to avoid going out or to keep short or no beard on face or clean it after every outside tour. It contains valve which releases unfiltered air when the wearer breathes out, hence the spread cannot be stopped just by using respirator.

So along with respirator, using a sanitizer is a must, to clean our hands from time to time. A respirator cannot be reused and after a particular time period of usage it becomes difficult for us to breathe through it and it indicates the time to replace or dispose after use. Most of the doctors and nurses who are curing the Corona patients use surgical masks, face shield, Personal Protective Equipment (PPE) kits and respirator to get best protection. No doubt, for full day wearing these causes skin irritation and problems especially in the summer season, discomfort due to sweat on the human body, but still to save lives and to contribute the nation, the Corona Warriors are doing their best work to keep us safe during this crucial period.

Therefore, for the common people, the best way is isolation and social distancing. Mask is better than no mask. A common man must keep physical distance from everyone and use sanitizers to kill the microorganisms on hands and wrist. In fact if we apply some sanitizer over the mask before using then the germs which stick will be killed at the moment and the mask will be more risk free. And if a person is having any symptoms of COVID-19, then he much contact the health care team or can go to a hospital for a check up. As prevention is better than cure, all things must be kept in mind to fight this virus. If tested positive than one can get cured. Either the person has good immunity or should eat healthy to boost the immune system, for example, Citrus fruits, broccoli, garlic, ginger, spinach, curd and almonds are best immunity boosters, drinking plenty of normal water everyday also help in keeping our body fit. Immunity system helps in recovery.

So we must cooperate with the health professionals' advice and must take all the precautions seriously and not lightly. This is the only way to survive during this period of time. Everyone is using masks nowadays but the people who are able to dispose or reuse it with care are only able to defend themselves and their families from viral infection.

All the small kids and youth must understand it necessary to stay at home and avoiding any kind of meeting with friends or relatives.

This document provides advice on the use of masks in communities, during home care, and in health care settings in areas that have reported cases of COVID-19. It is intended for individuals in the community, public health and infection prevention and control (IPC) professionals, health care managers, health care workers (HCWs), and community health workers. This updated version includes a section on Advice to decision makers on the use of masks for healthy people in community settings.

In some countries masks are worn in accordance with local customs or in accordance with advice by national authorities in the context of COVID-19. In these situations, best practices should be followed about how to wear, remove, and dispose of them, and for hand hygiene after removal.

Advice to decision makers on the use of masks for

healthy people in community settings as described above, the wide use of masks by healthy people in the community setting is not supported by current evidence and carries uncertainties and critical risks.

- Public health experts say you need to wear masks outside, especially when near other people.
- Available research shows wearing masks and social distancing appear to limit the spread of the virus.
- Disposable surgical masks or reusable cloth masks are ideal for everyday use. N95 masks, and other medical-grade respirators, should be reserved for health-care professionals.
- Masks should completely cover your mouth and nose with a snug (but not tight) fit.

If we, common people maintain social distancing and use just any cloth mask or scarf or just simple 3-layered or 6-layered, it is enough with respect to follow all precautionary measures then there will be no problem as such balefully.

That's why Stay Home, Stay Safe, Stay Healthy & Stay Fit and the world will soon defeat this Coronavirus.



# DISCRIMINATION AND RACISM IN HEALTHCARE

*Dr Abimbola Olanrewaju*

The issues surrounding racism and discrimination in the world are far reaching and cannot be fully discussed. These issues have either been ignored or very little has been done to deal with them, leading to a system or culture of acceptance of the perceived "norm". Unfortunately, they are found in every fiber of the society, family, peer groups, and organizations, including healthcare.

For example, data from the United States, show that over the past 50 years, there has been an increase in life expectancy and declines in infant and adult mortality as the health of both black and white persons has improved (U.S National Center for Health Statistics, 1998). But indicators still show more Black and Hispanic morbidity and mortality as compared to whites, in a country where non-whites (minorities) make up just about 27% of the total population.

Healthcare is multi-faceted and it begins from education to prevention to diagnosis, care, treatment and follow up. Racism and discrimination can happen in any of these facets and can have far reaching consequences for the individual who is the patient and also the society at large.

According to Wikipedia, racism can be defined as "prejudice, discrimination or antagonism directed against a person or people on basis of their membership of a particular racial or ethnic group, typically on that is a minority or marginalized". Racism can also be defined as "the belief that different races possess distinct characteristics, abilities, or qualities, especially so as to distinguish them as inferior or superior to one another.

Discrimination is defined by Wikipedia as "the unjust or prejudicial treatment of different categories of people, especially on grounds of race, age or sex.

## RACISM AND DISCRIMINATION

Racism and discrimination are quite different but their effects are somewhat the same. It is not easy to effectively distinguish between the two as they share common characteristics and may get mixed up in context or meaning. While racism refers to a particular 'people or peoples', discrimination can refer to a lot of characteristics like race, sex, age, social class, ideology, religion, tribe, sexual orientation, health and others.

The context in which these two happen in healthcare is almost always the same as they occur in any other institution or organization. The only difference is the fact that in healthcare, far reaching consequences, morbidity and ultimately fatalities can occur.

For example, between 1932 and 1972, hundreds of black men in Tuskegee Alabama, USA were infected with Syphilis and technically denied access to treatment for syphilis so scientists could study how the disease progresses with time. It was devastating for the local black population as a lot of the infected men died and this caused a deep distrust of the health system by black people.

This is one of the reasons why black people in the US, are skeptical about donating their organs to persons in need when they die as they feel their bodies will be "mutilated or experimented with", instead of being given to a recipient.

Healthcare racism can also occur when patient is denied full and comprehensive care because of their race. This could arise when a patient is wrongfully diagnosed either wholly or in part due to assumptions of their health status or actual prejudice. For example, there is an established racial bias on pain management due to the slavery-era myth that "blacks have a higher pain threshold than whites". This has caused under-prescription for a lot of black patients with chronic pain and led to so much morbidity and suffering.

Discrimination on basis of medical condition is still rife in many parts of the world. Medical ethics are against any form of discrimination or racism and most of the ethics and legal committees of many health institutions talk tough over the issue but little is always done to correct the ugly trend. A lot of people are still

being denied adequate care based on their medical condition. Some medical conditions for example, HIV/AIDS are termed 'shameful' or 'embarrassing' and patients suffering from these conditions are sometimes subjected to prejudice. The situation with the above is better now, compared to the 80s and 90s, due to education and advocacy, but a lot of work still needs to be done.

There are situations where the healthcare giver is at the receiving end of discrimination, as there are reported cases where a doctor or a care nurse is racially insulted or refused the opportunity to treat by a patient just because of their skin color or ancestry. For example, in 2016, The Washington Post reported a story about a young black doctor, Tamika Cross who was dismissed from attending to a passenger in need of medical help aboard a Delta Air flight, on basis that she wasn't a real doctor.

The above situation is also in addition to the fact that healthcare workers from black and other minority ethnic groups in the UK, are likely to be victims of lopsidedness in the recruitment process. According to a study done by Roger Kline for the Care Quality Commission in 2018, white staff are about one and a half more likely to be appointed once shortlisted for a job compared to black and other minority ethnic groups.

## CONCLUSION

Public perception of race, is one the factors fueling this crisis and people would rather be influenced by public knowledge than actual facts. The consequences of racism and discrimination in the society are far reaching, but even more devastating in healthcare as lives are being permanently affected. Governments, societies, organizations, healthcare workers, patients and even the general population should take the case of medical racism and discrimination more seriously with a view to stamping it out of healthcare, and by extension, the world at large.

**Dr Abimbola Olanrewaju writes from Lagos, Nigeria.**

## SOURCES

1. National Center for Health Statistics (US). Health, United States, 2015: With Special Feature on Racial and Ethnic Health Disparities. Hyattsville, MD: National Center for Health Statistics (US); 2016 May. Report No.: 2016-1232.

2. The Department of Health and Human Services, United States (HHS). National partnership for action to end health disparities: Offices of Minority Health. Washington, DC, 2011.

3. Rachel R. Hardeman, Ph.D., M.P.H., Eduardo M. Medina, M.D., M.P.H., and Katy B. Kozhimannil, Ph.D., M.P.A. Structural Racism and Supporting Black Lives — The Role of Health Professionals. New England Journal of Medicine, October 12, 2016.

4. David R. Williams, Ph.D., M.P.H., and Toni D. Rucker, Ph.D., (2000). Understanding and Addressing Racial Disparities in Healthcare, Health Care Financing Review Volume 21, Number 4.

5. Inequality in recruitment outcomes for Black and Minority Ethnic Staff within the Care Quality Commission – A report for the CQC on their causes and steps to be considered to remedy them by Roger Kline. April, 2018.

6. Tuskegee Syphilis Study – Wikipedia.



# COUNSEL TO NEW HIPPOCRATES

Dr. Olaewe opeyemi

The oath on your lips today gives you a new surname: Hippocrates. On behalf of our elders, welcome to the Hippocratic clan– brethren of the stethoscope and scalpel. This is still the noble family, we are not just learned, we are noble, the situation in our land regardless.

Saving lives remains our priority, far more than the charm of gold, more than the shine of silver. Let this passion fuel your drive and you'll be seldomly disappointed. Practice your craft

with excellence, do your business with integrity, hold your head high, work to earn the respect of your colleagues: they are your greatest protection, then you will undoubtedly earn that of your patient and society.

From now on, many people will not call you by your name anymore even the people who gave you the name and were present at your christening will now call

you by your new name: Doctor. As often as you hear it called, let it remind you of the weight of responsibility on your neck, there is the most minimum standard, higher than anyone in your society: a wall you must fall from. Let your hands convey warmth and comfort as you palpate and percuss, let your stethoscope relief men as you listen to their deepest fears and intimate thoughts. Let your scalpel bring hope to the hopeless : everything is not drugs, let your presence, your bright smile and words of grace be your first

and ultimate therapy. There are days, you'll be happy you came through this path; there are days you wish you hadn't: you will be present at the gateway of life (and death) ; than any other, you'll realize the frailty of life. You'll go to the grave

with many secrets which you must keep on trust. If you desire wealth, invest your skill like a spider. Respect all your coworkers: your brethren of this craft and others, it'll make your life easier. You cannot divorce your books – they are your lifelong companions.Remember to choose a field that makes you cheerful and makes your work worthwhile.

Do not neglect men of your immediate family, they'll be all you have at some point in your life



# Experts Say in this Bloodless War, Withdrawal is the Weapon

Dr. Ajala Adeshina

My tutors tell me that psychiatry is a faucet
where things get stuck unless they become formless like fluids.
That sanity takes the body of a chameleon; an entry & exit
for neon of colours
& time could crop the portrait of ab//norm//ality into the norm.

The world clocks into a new time when the fulcrum on which survival
balances is no longer in community. No longer in the touch of fellowship. Of togetherness & communality.
An abnormality is being framed into our psyche.
Every man in the confines of his tent, because there's a death
that lurks in the web of humans.

They say our world writhes in the wild fists of a warrior, who sings war songs from Wuhan.
A health worker kits for war, zips Hamzat suit, face shield,
N95 mask & gloves.
& experts say in this bloodless war, withdrawal is the weapon.
You come into this battle at a range of 2 metres because you
don't know who carries the shrapnel of death in his breath.
You arm yourself with bayonet of facemask, palms wet with water & spirit.
You pray evil away from your way, from your family, from your loved ones.

# Baami

*OmoobaJesu Adetunji*

Funmike is Baba's favourite because she is the last child from his first wife whom the others claim he loves the most. But if he really did love Yejide as much as they say he does, why did he choose seven more after her? No one knew for sure. Some people said it was because money had come and when money comes, it is only right that your wives increase. Some said he wanted a boy and Iya Kemi did not seem like she would produce a boy after already birthing him five girls. Funmike was the sixth. The child couples usually describe as a mistake; a product of rash careless passion. Her father had named her himself. Olufunmike meant 'God has given me this one to pamper'. Their bond was evident from birth. Funmike adored him as much as he did her. Yejide watched them dote over one another with scepticism. She knew her husband, love was definitely not a thing that lasted with him.

Baba worked in Ife and they lived in Ilesha. He left every Sunday and came back on Fridays to the big family house that stood robust at the very mouth of their street. It was painted a stunning white and repainted every three months for renewed resplendence. One could easily mistake it for a giant star from a distance; it shone and stood out completely from the surrounding battered brown roofed houses it was flanked by.

***

When the long vacation rolled by in July 2005 and Funmike, just finishing primary four, started to sing the song of wanting to go with her father on his trips into her mother's ears, Yejide snapped her fingers hurriedly behind her head and said it would be over her dead body. But it would only take many nights of refusing food and crying to sleep to change Iya Kemi's verdict. She pleaded with Funmike not to go, she told her that she would suffer in the hands of her father's other concubine but Funmike was adamant. Baba loved her the most, he would protect and spoil her, there would be no other child to share his love with in Ile-Ife. This was her reasoning. By the time Baba came home that weekend it was settled. Fumike would leave with him on Sunday.

***

In Ile-Ife, Funmike did not see her beloved father often. He left for work very early and he came back when she had already collapsed into sleep. Collapsed because Sister Anike, in whose house they lived in Ife used her like a horse in the day.

Funmike had cried and cried the first day and asked for her father, refusing to wash the pile of clothes before her and Sister Anike had come to twist her ears and whip her with her dunlop slippers. The way the rubber slapped against her body sent painful shivers down her spine and she determined to do as she was told from then on. She also did not get to eat that day as punishment for being stubborn. Little did she know that she would not be eating properly here regardless of her behaviour. By the third day, after going hungry to the point of blurry vision and bending-over stomach aches, she found her way out of the house to a roadside buka* and bought a plate of hot beans with some money she found in her father's trouser pocket when she washed his clothes the day before.

The beans satisfied her beyond words and so she made the place her regular spot and this was what she would have for the next two days she remained there. She was relived she would be spending only a week in this miserable place. Sometimes she would remember the words of her mother and her heart would break but she would remind herself that she will soon be back home. She also did not understand why her father did not acknowledge her as though he knew her here. She did not want to dislike him, but this, this sudden animosity, would be difficult to forgive.

***

When Yejide saw her daughter alight from the white regular Benz she knew that something had happened. Even if Fumike had not run to embrace her such that she could feel the hollow spaces in her dress and the flatness of her cheeks, she could see clearly that her daughter had been malnourished.

"Omo mi**" She exhaled pulling her daughter close once more. She wished she could carry this eight year old body all the way inside but even with the weight lost she could but only lift her a few inches above the ground for a short while.

It was about a week when Funmike could no longer do her chores. She complained of headaches and frequent watery stool. Iya Kemi gathered some herbs and asked the other children to help her prepare it as she cooked Fumike's favourite dish; pounded yam with egusi soup and fried fish.

Funmike struggled to eat, but she did eventually and took the herbs. By the next morning the fever had broke only to return by evening even worse. Baba Kemi was not around, it was a weekday and the other wives kept to themselves. They all tried to stay out of each others way. Who will I run to? Yejide thought. She made her daughter more herbs and knew that God will have mercy on her and make this child well.

When Baba came during the weekend and saw Funmike, he rushed her to the hospital. They said she had typhoid. On hearing this Iya Kemi pulled the neck of Baba's buba, pulling him back and forth vigorously. Shouting that it was his fault. That he wanted to kill her child in Ife. The doctors and nurses threatened to walk them out of the ward so Iya Kemi calmed a little and asked to see her daughter. They said they would need money to treat Fumike immediately. Iya Kemi looked to Baba with pleading eyes. He shook his head. Iya Kemi dropped to the floor like a bag of rice. A nurse ran to her to console her and advise her to find a way to get the money. When Yejide looked up, her husband was nowhere to be found. He had left.

***

Kemi had just left for the university the previous week. Yejide had given her all her savings. She had sold every precious thing she owned to gather that money, now there was nothing more to sell. Who will help her now? Baba did not come home for two weekends after that. She tried to borrow money and always got less than what she needed, only enough to keep Funmike in the hospital but not treat her well enough.

By the third weekend when Baba finally showed up, Funmike was a different being. There was no colour in her eyes. Her stomach was as big as a soccer ball and she had lost even more weight. He felt bad. They were asking for blood now. She needed blood. He called Yejide aside and shoved naira notes into her hand. He was turning to leave when the doctor called him back, asking him politely to get tested. Perhaps his blood would match hers. Baba wanted to explain that he had given his wife money, that everything would be fine but the doctor insisted so Baba agreed.

It so happened that he was a match to her O-type blood. Funmike had woken up by now and was told the good news. She smiled weakly at her father, happy that he was there. It had been so long she had seen him. The doctor asked Baba to come with him to the blood bank to donate the blood.

As they walked past the hallway, outside the ward and made steady strides toward the lab complex a few feet away Baba suddenly exclaimed with his arms thrown up. Ah he had forgotten something important in the ward, he said. He needed to get it urgently. The doctor nodded and pointed ahead to the lab complex, telling him to meet him there. Baba nodded emphatically and assured him he would be there in no time.

An hour had passed when the doctor came in Funmike's room looking for Baba. That was how they all found out he was long gone. Funmike's heart cracked some more.

Kemi arrived from school that evening to see her baby sister. It was sheer luck and God's providence that she was also a match and was willing to give the blood. Now they had blood and they had the money. All would be well.

***

It was still another two weeks before Fumike would return home, but nothing was the same. She had seen, felt and learnt things she would forget in no hurry, of the twisted natured of professed love. The next time she set eyes him, she greeted him with her knees kissing the ground, calling him Baba like the other children, not Baami as she fondly did in the past. When he too heard this, he knew that he had wounded her in ways that could not be remedied and it was now the turn of his own heart to break.

*buka is the slang used in Nigeria for a roadside shed where locally made food is sold and could be eaten.

**Omo mi translates to my child in English language



# Tribute To My Chemicals

Shout-out to  thyroxine, that strives
Within the cells to increase my basal
Metabolic rate.

Everyone wonders about my sudden growth,
Not realizing this astute growth hormone rarely sleep.

Like early morning sun, my cortisol rises with big grin
To shield me from the stress peeping.

So thankful for this ebony skin,
That dances to the music of melanin.

Everyone uses microphone except me,
Definitely, testosterone deserves the credit.

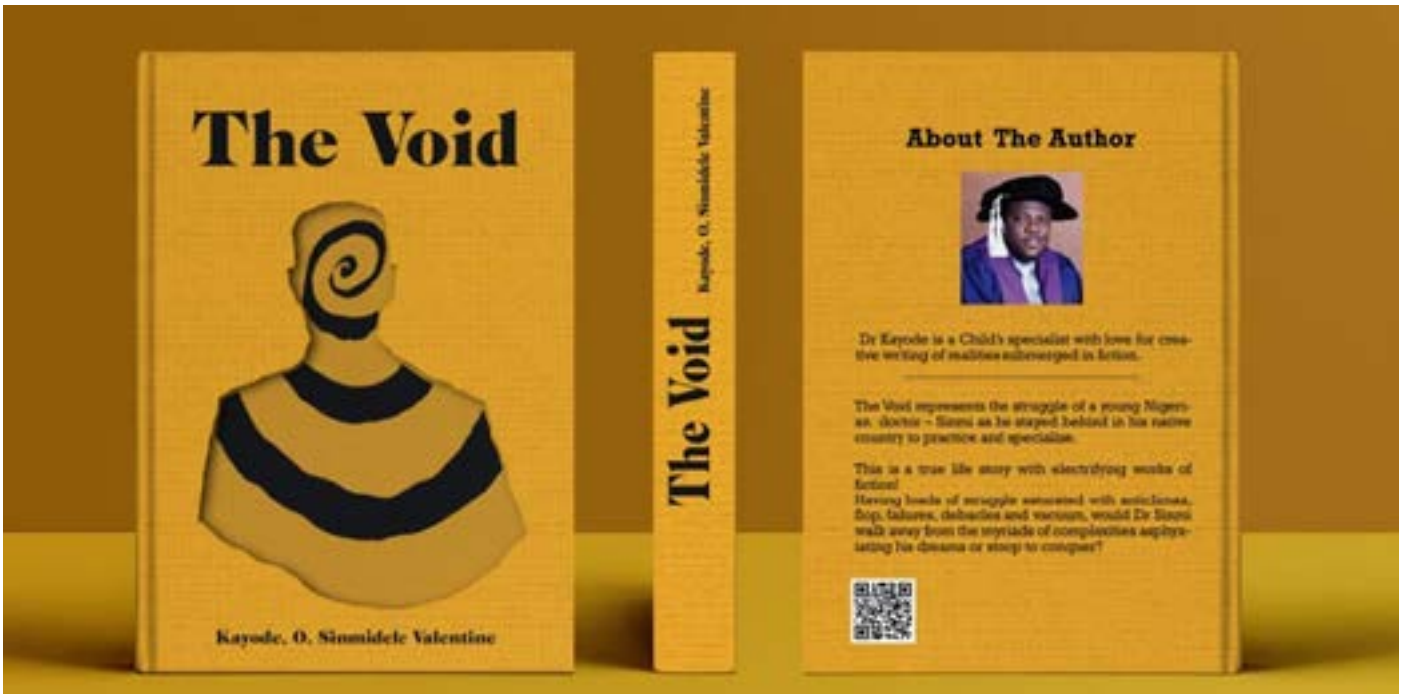
These precious pearls are stealing my gaze,
Oestrogen won't stop beautifying these ladies.

Oh!  my Adrenaline (fright and flight hormone),
that can make me fly during an uprising.

Progesterone, the gestational soldier,
Coupled with oestrogen to ease the nine months affair.

Sending love and light to many other hormones , their
Actions can never be overemphasized.

Yusuff Abdul Quadri.
Ladoke Akintola University of Technology.
yusuffabdulquadri@gmail.com.




## BOOK REVIEW: THE VOID

The Void is an outstanding literary work which is based on the author's true-life experience. It is an inspiring narrative about Sinmi, a young medical graduate, as he navigates through the difficult terrain of post-graduate medical training in Nigeria. While pursuing his aspiration to become a paediatrician, Sinmi encounters multiple obstacles including difficulty in securing a training position, challenging clinical encounters, failure in postgraduate examinations and financial embarrassment.

The story exposed the sad realities of a developing healthcare system with its catastrophic out-of-pocket expenditures, poor infrastructures, perennial staff shortages and poor renumeration orchestrated by visionless political leadership that prioritises white elephant projects above human capital development. Nevertheless, the clinical encounters demonstrate the resilient spirit of many healthcare professionals and their patients, as they strive to make the best out of the near-comatose system.

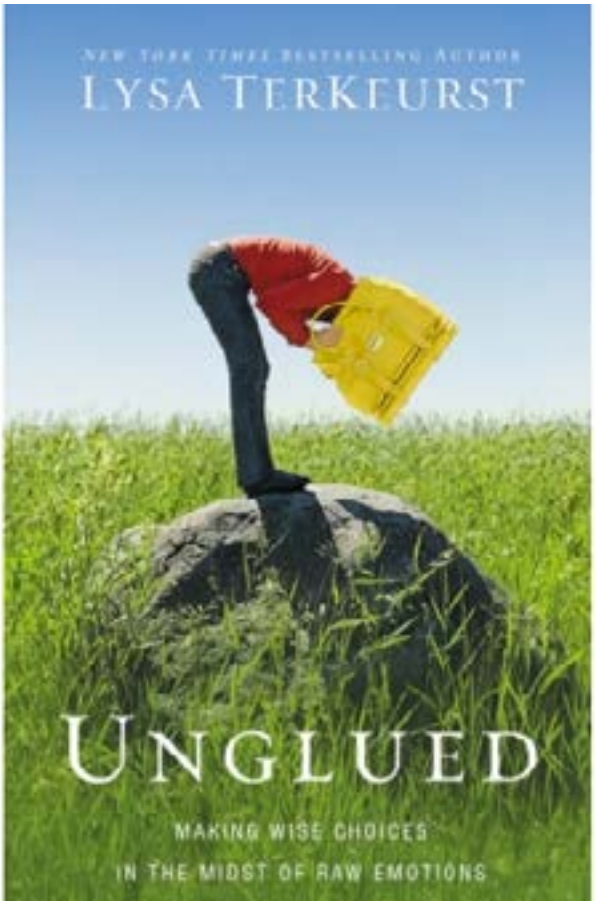
This story summarises "Tropical Paediatrics" in an impressive way, ranging from "pneumonia, diarrhoea, malaria, measles and malnutrition – our pentalogy of Africa's nightmares" as well as birth asphyxia to the rare pseudotumor cerebri. The author was quite sensitive to explain all medical terms, so that non-medically trained readers could also enjoy the story. Similarly, all local slangs and slogans, included to give more life to the story, were explained for the benefits of non-native readers. It is also fascinating how the book integrates health promotion into each clinical encounter. For example, the author admonishes caregivers to "stop gagging the mouth when the child convulses"

Patient confidentiality is generally a concern with many books written by frontline medics, including the Void. Many healthcare professionals shy away from telling their stories as it is often difficult to share personal experience of medics without including patient encounter. It is however pleasing that the author recognised this grey area and included a disclaimer that "all the patients' names, physical environment and scenarios mentioned in this book are pseudonyms and fictional"

Overall, the Void is a literary masterpiece which combines entertainment and education. It is impossible to read the Void without becoming emotional. Therefore, I would like to recommend this 199-page bright yellow hardcover book - containing an inspiring account of a medic's journey towards self-actualization, which ended in self-realisation – coming to terms with the eternal truth that only God can fill the Void.

**Dr Fadahunsi, K. Philip**
**Imperial College, London,**
**United Kingdom.**






# BOOK REVIEW: UNGLUED BY LYSA TERKEURST

Ehinbare Miracle

**Bibliography; Lysa TerKeurst: UNGLUED. Grand Rapids, Michigan: Zondervan, 2012.**

Emotions! Raw emotions (anger, tears, retaining hurts, blaming etc.) puts us in a hard place and injures relationships. But, we should know that "our emotions were given so we can experience life, not destroy it." As hard as it may be for a day to go by and not have a sour experience of an unruly friend or a disappointment, a simple disobedience from siblings or criticisms from outsiders, a foul statement online, an angry lecturer etc. Our response in whatever situation matters the most. Lysa (author) in Unglued recognises with everyone by making personal examples and how we can make our emotions work for us and not against us.

Right from the beginning, readers are made to understand that controlling our emotional response is an imperfect progress – like learning to walk. In making this journey of imperfect progress we need to change out thought patterns, the basic of which is, you can be better. We must remove labels and submit our lives as God's workmanship that He might chisel us into a masterpiece.

In dealing with our emotional response, we need to identify the category our unglued reactions fall into. Categories are;

- The exploder who shames themselves: The exploders aren't always loud in conveying their point but they make sure the point is well received by the offender. They feel justified afterwards.
- The exploder who blames others: They put the responsibility of what happens on the one who crosses their path or have been disobedient to them. They explode from built up tensions.
- The stuffer who builds barriers: These set of individuals don't spell out the offense but create a distance and might later on cut off the relationship completely. They say everything is fine when nothing is.
- The stuffer who collects retaliation rocks: They plan for revenge by keeping quiet. Sometimes unintentionally. They collect rocks of hurt from

previous situations and hurl it out in just one moment that the offender is caught off guard. A good example is one who reminds you of an innocent statement you made the previous year.

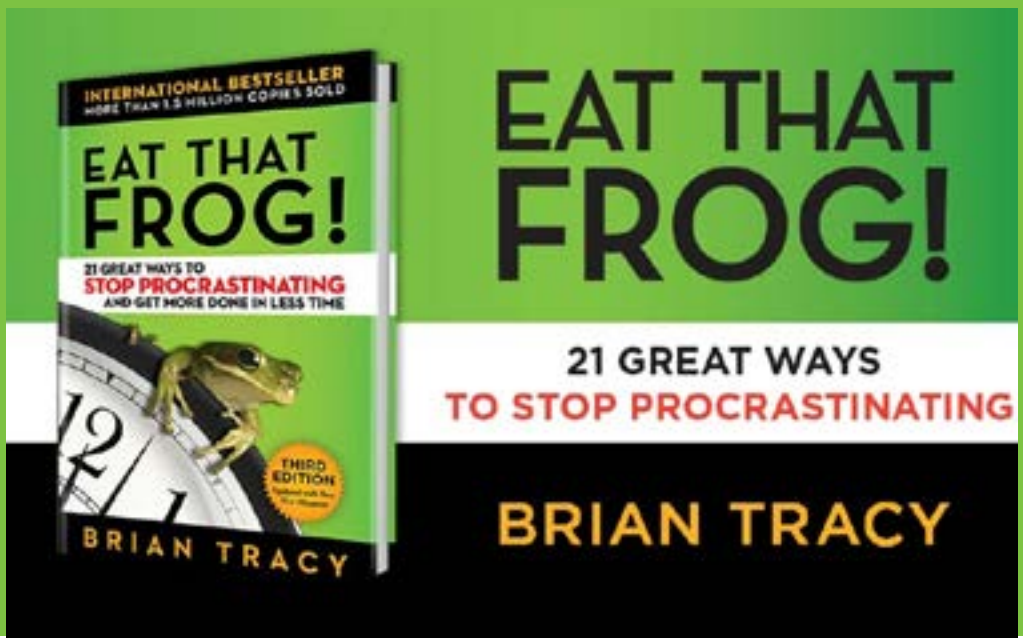
The end result for all categories is regret and fractured relationships. Do note that an individual may belong to more than one category and the offender may also be a determinant. For instance, a mother may shout and blame her kids for making the home a mess but stuff and build barriers with her loose mouthed friend.

How then do we make wise choices in the midst of raw emotions? Lysa explained each category and how to deal with them in the book. In general, here are tips;

- Feelings are indicators not dictators, have self-control.
- Soul integrity; the honesty that is godly. Accept your part and tell the offender their offense in a godly way.
- Be prepared for stimulants i.e. individual who will stir your emotions.
- Have a template for your response instead of exploding.
- Create boundaries not barriers.
- Remember who you are.
- Rely on God's power.
- "The best kind of plan for you is the one you'll follow."

From Unglued I discovered I stuff a lot. I get hurt easily by words and seemingly innocent acts. My reaction most of the time is to keep quiet until the offender is frustrated or stays away. But when I do explode, trust me, I am not nice. From what I have learned in the book, I try my best to keep peace while I also employ wisdom depending on the situation and not allow raw emotions to break me or my relationships. And when I come off Unglued, I do my best to apologize and come out clean. Remember, it's an imperfect progress, but a progress nonetheless.

Genre: Inspirational
Author: Brian Tracy
Publisher: Berrett-Koehler
Chapters: 21
Pages: 130




The need to shun procrastination cannot be overemphasized in the quest for effectiveness amongst the youth and workforce in diaspora. The book 'Eat that frog' explains in a watered down, step by step approach to stop procrastinating.

'Eat that frog' is personified to executing the biggest, most important task which you're likely to procrastinate on if you don't do something about it. The first two chapters center on planning; setting goals and organizing a to-do list on the tasks available for execution. The third chapter explains the 80/20 rule as the most important rule in time and life management.

The art of practising creative procrastination and the preparation for the commencement of a task, using the rule of threes and the organization of the tasks in a scale of importance were explored in other chapters. The concluding part reiterated the essence of taking the tasks one at a time, identifying and upgrading the skills we possess and leveraging on our special talents. It also investigates the need to identify the major constraints to working/studying effectively, and to maximize our resources to work on the tasks at hand till they are done.

The most thrilling part of the book was the seventeenth chapter that talked about getting out of technological time sinks. Technology in this age has invariably taken most time out of our daily schedule which we need to overcome.

The book however seems to be biased with professionals than students in the various life hacks provided to stop procrastination.

Nevertheless, Brian Tracy's 'Eat that frog' is the perfect book for every student and young professional. It's a must have!

**Kayode Ayomide**

# AUTOPSY REPORT

No life is a single story, I once heard it said. Remembered or imagined, the stories of others impact on our own stories and then bear consequences on our lives. A dead man, his death, touches a multitude of proximate strangers. We just don't always know it at the time. Yet, to know is to be informed, to be aware and empowered with what knowledge is.

Of course, the coffee-sipping pathologist knows more than most of his peers in some regards. Relying largely on what his sense of sight can afford him, he lies awake trying to find that bigger picture to an unknown cause of death. Daily, he is forced into the practice of soul-searching as he cuts through sections of organs from human remains. He broods on the many questions of why, how, when and what, as he attempts to diagnose the incurable by looking through thin visceral slides with microscopic eyes. This is how he connects the dots. From what insight affords through the eyepieces of a microscope while looking on a sectioned liver tissue, he links with the past life of a chronic alcoholic whose body was wheeled into the morgue a week before. At 58, Demuren, a successful economist and banker, was said to have died from shock imposed by a massive bleeding through his mouth. This is what is known. But not entirely all, as the pathologist would later reveal that Demuren, a man sold out to philanthropy during his life, was also large in heart as much as he is flabby. A double jeopardy.

But to what end is an autopsy report if it can not be recounted to a dead man so that he realizes how he could have made better choices, befriended moderation or embraced the practice of self-preservation before his river of time flowed gently to its destination? Or to what end is an autopsy report if it is recounted to the living, so that lifestyle can impact health positively?

All advocacy is, at its core, an exercise in empathy.

Perhaps, we must consider the lives of others (more).

*Ayotunde OJO*

# WEIRD FEELINGS

Thoughts and voices bubble in my mind
I wish I could share
I let out a sigh in hopes of how it would be
If my thoughts could be words

For the dialogue I seek might bring relief
And my inner soul will be at peace

**THE WORLD SHE DREAMS**
For the imaginary world I see might be unclear
But I'm willing to risk it
If it adds a smile to my lips

The fantasy I wish
Might not exist
But I will make it into what I dream

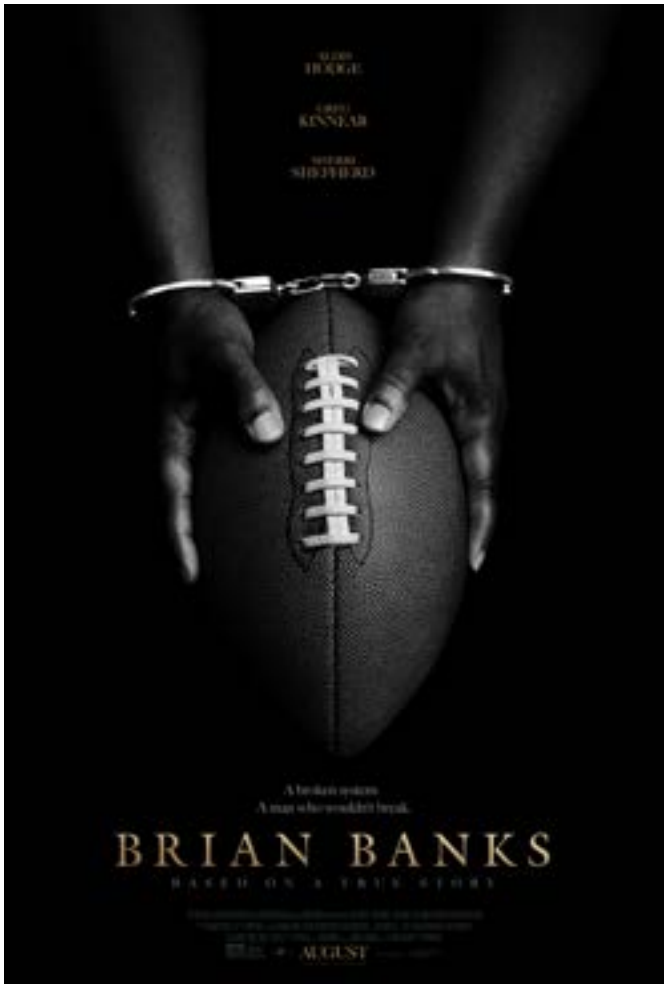
For it's an illusion
A phantom I believe
That will exist
With my dreams at the lead

**FINDING MYSELF**
I get lost a few times
But I ain't loosing myself
In what I've once been lost in
Because when you get stuck in the loop
You loose your you

Never be scared to loose yourself
Because from that loss
You tend to gain
A gain with pain
Might leave a stain
But what you will gain
Shall forever remain

*Koathar Oyeniran*

# BRIAN BANKS




This movie depicts the true life story of former football player, BRIAN BANKS, who was falsely accused of rape by his then high school classmate, WANETTA GIBSON, who claimed he dragged her down a stairway across a corridor with classrooms full of people.

He was arraigned before a court in San Diego and accepted a plea bargain at the insistence of his lawyer with the hope of getting a parole. It was denied.

Brian was sentenced to 6 years in prison with 5 years of probation, and while he was observing his sentence, Gibson and her mother filed a lawsuit against the school, winning 1.5 million dollars in the process.

After being released from prison, while observing his parole, Brian approached the California Innocence Project (CIP) a nonprofit law school clinic, which was started with the aim of exonerating wrongly accused people, to take his case.

At first, he was declined, due to insufficient evidence and the fact that he had already been released from prison. But due to his persistence and wanting to clear his name, he was able to get Justin Brooks, the head of the organization to take his case.

As the story unfolded, Gibson contacted Brian on Facebook and admitted before a friend that the whole story was fabricated while being secretly recorded. The video was inadmissible in court since it was recorded without her consent, but it helped convince the District Attorney take his case back to court and put together new evidence which helped exonerate him.

Gibson never admitted to fabricating the story that got Brian in Prison, due to the fear of having to return the money she and her mother had won from the case years back.

He was released from his sex offender's status and had his parole violations including his records cleared and on August 8, 2013. Brian finally fulfilled his dream of playing in the NFL at the age of 28.

It was a tear jerking movie, as an innocent man who was wrongly incarcerated, worked so hard to regain his freedom yet kept experiencing several setback. Still he never gave up.

Brian continues to work with the CIP and speaking on behalf of the innocent. Since it's establishment in 1999, the CIP has been able to successfully exonerate 27 innocent people, imprisoned for crimes they did not commit.

Adekanle Oluwatosin



# An Attention to Prevention

Bridget Boluwatife Ogungbemi

When I was starting out as a medical student there was an eagerness to step into a hospital. It felt like that was the anointing I need to be a medical doctor. Coincidentally, I visited my aunt who is a matron at a hospital in Ondo state. I followed her to the hospital during my stay with her. After spending some time at the Obstetrics and Gynecology nurses' station, while my aunt attended to some patients, she took me to the dialysis center where I was going to see a dialysis machine. She introduced me to the matron, nurses and the two doctors we met there and left me in their care. I felt like the world was finally seeing me.

There were three patients being dialyzed when I entered. I waited for one of the doctors who I perceived to be friendly to talk me through what was happening. There was another lady standing at the doorway joking with the doctors. I could properly see her when she entered into the room and her cheerful personality made me decide she would be a cool lady to be friends with. Lilian was taller than me, not much older than I was, slim and wore an Ankara top with black jean trousers. The way she talked and laughed was fascinating. She was so alive and I was almost envious of her confidence.

There was something about the way the doctors related with her that suggested that she was not a colleague. They were too friendly with her and it was almost as if she was labeled 'handle with care', but as she lied down on the empty hospital bed at the corner of the room. I realized that she was a dialysis patient and my heart dropped inside my stomach. I spent minutes trying to figure out how she became a dialysis patient. I wanted to move close as the doctors attended to her but I wanted to give her some privacy though I stole glances at her from where I was sitting. Throughout her dialysis session I couldn't muster up the courage to ask the doctors about her so I waited until I saw my aunt at the close of work to ask her about LIlian.

My aunt told me that Lilian developed chronic kidney disease (CKD) one year earlier due to uncontrolled type 1 diabetes and how dialysis and kidney transplantation were her only options; both of which are quite expensive. It was exactly seven months after I left my aunt's place that she informed me of Lillian's death. I didn't really know her but the news was devastating.

I couldn't help but think of the kind of life Lilian would have had if her diabetes had been properly controlled. What if she had visited her doctor to complain about it and her insulin dose with had been increased? What if her blood pressure had been properly controlled? What if she had followed a diabetes meal plan? What if she had limited salt and sodium intake? What if she drank enough water and exercised regularly? WHAT IF?

# The Dangerous Helper

Odedele Temitayo

There was once a time in a dense forest where there was lake. All the animals used to drink water from the lake. The water of this lake was so sweet that many fish lived there for long time. In this lake there lived a crab. The crab's best friend was a swan. That swan was in the same lake. They were happy in the company of each other. Their happiness lasted until one day a snake made its home near the lake.

Every day the swan laid an egg. The snake would come and eat it up. "I have to find a way to save my eggs," thought the swan.

One day, he went to the crab and said, "Please help me, dear friend. My eggs are under threat. That cruel snake eats all the eggs in the nest. What can I do?"

The crab decided to help its best friend. The crab thought for a while. Then he said, "I have an idea. Let us catch some fish from this lake and scatter them from the snake's house till the mongoose house." That mongoose lived in the nearby tree.

Then, the crab and the swan caught some fish and dropped them from the mongoose's house all the way to the snake's house. Then both of them hid behind a tree and watched. They waited for some time. After a long time, the mongoose came out. He saw the fish and was overjoyed. "Mymm! Fish right outside my own house!" he said, smacking his lips. He happily ate all of them one by one. As he ate he kept following the fish trail to the snake's house. Finally, the mongoose reached the house of the snake. Both the crab and the swan were watching all these events, waiting behind the tree.

When the snake saw the mongoose, he thought, "That mongoose is here to attack me. I had better fight with it." After some time, the snake started to fight with the mongoose. They fought for some time. After a fierce battle, the mongoose killed the snake.

Watching this behind the tree, the swan and the crab heaved a sigh of relief. But her joy was short-lived. The next day, the mongoose, looking for more fish, came upon the swan's nest. There the mongoose found more eggs of the swan. He immediately ate all of them. The swan and the crab now felt helpless. They had brought this new threat upon themselves. They did not know that the mongoose was the dangerous helper. "Our thoughtlessness has got us a new enemy. Even more dangerous than the previous enemy," cried the two friends.

After few days, they decided to form one more plan to get rid of the dangerous helper - mongoose. It is must to be careful while fighting with an enemy.

# INTERVIEW with DR. OLAMIDE S. VALENTINE KAYODE

**CAN WE MEET YOU SIR?**

I am Olamide S. Valentine Kayode. An MBBS graduate of the greatest Citadel of Learning - Ladoke Akintola University of Technology, Ogbomosho. I am a Consultant Paediatrician by God's Grace.

I am a fellow of the West African College of Physician – Faculty of Paediatrics and a fellow of the National Postgraduate Medical College of Nigeria. I have interest in Neonatology and Neurology. I was the 2019 Best Graduating Fellow in Clinical Paediatrics at the Faculty of Paediatrics in National Postgraduate Medical College of Nigeria (The Dr Adedayo and Adeoye Adeniyi 2019 Prize).

I am also a creative writer and Author of 2 books: Pearls and Pebbles; and The Void. My 3rd book is under writing. I was the pioneer Editorial Board Chairman of this excellent journal in 2019. It was a great outing and I love the team I worked with which was led by "the girl child Advocate" – Dr Ajufo Somtolise.

**HOW'S BEEN YOUR JOURNEY AS A MEDICAL DOCTOR AND A WRITER SIR?**

Well, I didn't know I would be a serious non - academic writer. (If I say I didn't know I would be a writer, that would be wrong because doctors are writers any day, they write every day in the hospital case notes, in journals and clinical presentations. All these are academic or clinical writings). As a 200L medical student, I wrote an article for my fellowship editorial unit and it was rejected on the ground that the writing was too aggressive and verbose so I stopped writing any piece at that point. I reflect a lot because I am a super critic of myself. Thus, I looked at the elements of tenacity in the said piece and let it sink it: you cannot be an aggressive writer, I told myself.

By 400l, a national essay competition was held in Nigeria by the Federation of Catholic Medical and Dental Students (FECAMDS) and I submitted my article. A friend of mine From LAUTECH Medical School Submitted too (Dr Fadahunsi, Kayode Philip) and LAUTECH submissions came as First and Third among over 40 entries! My friend came first and I came third. That was the point of no return for many reasons. Firstly, I submitted reluctantly because I was initially afraid of other medical students in Nigeria. Secondly, Dr Fadahunsi was our Editor in Chief in our fellowship for a long time. He is a brilliant and fervent reader. I had a "local threat" initially with fears of being submerged by too many competitors. And lastly, the competition panel of judges did not mind that 2 candidates from the same school were carting away their gifts because they recognised good writings. Although, the clinical arm of medical school was more challenging I was writing my poems and keeping them thereafter. Many of those poems are still unpublished till date.

**DOES WRITING INFLUENCES MEDICINE AND VICE VERSA?**

Without making a stereotype really or without being overly generalising and to a great extent, medicine brings about the reflective aspects of her students. Thus, I would rather say, medicine influences writing first and then vice – versa. The fact that when medical students read, they must understand and find ways to explain deeper things to themselves in order to have clarities, they become reflective over time and can say many things they have read in different ways. When they put it into writing, it would flow. But I must not forget to add that there is an intrinsic gift or factor in many doctor - writers. Thus, besides their medical school's reflective mode being activated over time, creative writers in medicine have that innate desire to write for education of others and for leisure. In other words, medicine may be an acquired factor for a creative writer, there was a congenital factor that loves writing in such persons ab initio; an inner fire ignited by the deep reading and meditation that come with medicine. .

**DID YOUR TIME IN THE UNIVERSITY PREPARE YOU FOR A FUTURE AS A RESEARCHER?**

Yes and to a remarkable extent.



**IF YES, HOW? IF NO, HOW CAN A STUDENT IMPROVE ON THIS.**

The final year Community medicine project

gave us a very big clue as the teachers in that wonderful department gave us their best. In fact, all our teachers in medical school were selfless. There was also an Obstetrics and gynaecology research topic in our penultimate year it was thorough and was an exposé. Nonetheless, the effects of all these preparations in medical school also depend on the students' readiness. For example, in our research and project group that time, many of us were actively involved while some were not just interested. We would try to drive them to participate but a few weren't just ready to be in the front group. They would contribute whatever we ask for financially and move on. This means that each student needs to apply himself or herself to every assignment or project in order to make a useful impact.

**AS A MEDICAL RESEARCHER AND WRITER, WHAT DO YOU DO, WHAT ARE THE KIND OF PROBLEMS YOU FACE OR DECISIONS YOU MAKE?**

Truthfully, it's very difficult to be a researcher, a clinician – You forgot to add that and then a (creative) writer. The one you love the least among these might suffer for it. The problem is time. You have to find time to do your research and write it out academically. At the same time, you want to educate and entertain people. You must deliver or you lose your pedigree. And you must continue to be a safe clinician while treating your patients. Other problems include but not limited to - not finding a good academic journal to publish your work after the research and not finding a good platform to sell your creative writings. Those are the basic issues and one of the reasons why LAUMED Journal must not die. It serves both purposes for young researchers to learn from their senior colleagues and also publish their works.

It is a colossal advantage and an enchanting feeling to have LAUMED indeed as a worthy platform to showcase the works of many researchers and creative writers.

**WHY DID THIS TYPE OF WORK INTEREST YOU AND HOW DID YOU GET STARTED?**

Well, like I said earlier on, the 400l competition was the last spark because we got star prizes. I got a very beautiful 3 loader DVD/CD video player that year – it was a new electronic gadget that year. My friend got a home theatre like electronic device with large speakers. That event made me to keep at it.

Moving forward, as a resident doctor in Osun State we had financial challenges in term of non – payments of salaries as the governor then was owing us months and months of salaries even till date. You can see how sad that is. I was almost becoming depressed as we worked hard as doctors and yet receiving less than a third of our salaries compared to our friends in other places. At that point, I started writing about our challenges and other medically related issues. Those writings became my anti-depressant. Each time I write and my friends liked it and talked about it makes me feel good and my endorphin levels were coming up. In fact, at a time, my friends would ask me for my write up if they didn't see it. That meant so much to me! And made me to keep at it till it birthed two self-authored books.

**CAN YOU SUGGEST SOME WAYS A STUDENT WHO'S WILLING TO FOLLOW THIS PATH CAN GET STARTED?**

Very well. The first is the passion for writing. We can do nothing without the passion. Every other gift will fail without the innate passion or drive to do it.

Secondly, being focused is an integral part of any success story. Then readiness to learn and be mentored. Many young people don't have mentors and that is a great setback for succeeding quickly. Mentors do not allow you to learn from your mistakes. They guide you and make your mistakes very less. For example, Prof O. Akinbami – Professor of Paediatrics in FMC Balyesa - is one of my stongest mentors. Professor Emeritus Oyedeji, Professor Adebami, Associate Professors Okeniyi, Oyedeji, Oninla and other teachers are mentors in one way or the other. I learnt greatly from these people great clinicians and academic writers.

This is closely followed by proper time management. Doing the right thing at a time. For example, when I have exams those times all non academic writings must stop. This is because no one would like to read from a failure that remains a failure. In order words, even if you fail once or twice or thrice you are not a failure until you stop trying. In order to succeed in trying, you must drop all distractions when you have a specific project or exams. You cannot be writing fictional work when your non-fictional exam Knocks on your door. Lastly, but not the least is your spirituality or mental health. For the theists, the latter and former are related. I don't joke with prayers because that has been my large strength for a long time. Writers as Muslims, Christians, free thinkers or traditionalist must find their spiritual strength and use it wisely. The GOD factor is a very big plus for me.

**WHAT PARTS DO YOU FIND MOST SATISFYING OR CHALLENGING AS A RESEARCHER?**

I like conceptualisation of the research work, the field works too are fine by me. It's the aspect of writing out my research works into Journal that I find really challenging. The processes seem too bogus for me. I am trying to love it so that I can persevere to get going.

**WHAT ARE THE VARIOUS OPPORTUNITIES AVAILABLE FOR RESEARCHERS?**

There are potentials of being known in the science circle or your field of specialisation. There are many times we cite our works and see a specific author coming up so many times from different angles . That is a "wow – wow" experience for me when a person is said to be an authority in his or her field. Also, there research grants that assist researchers to do many exploits in their fields but research grants are limited these days with many conundrum of inaccessibility. Other opportunities as researchers are the fulfilment of being happy succeeding in your choice and the sense of responsibility it births.

**DO YOU THINK LAUTECH PROVIDES ENOUGH INCENTIVES FOR RESEARCHERS TO EASILY CARRY OUT THEIR WORK?**

That is a difficult one with the financial challenges and the owner state tussles. It has been difficult for our teachers even as a resident before I left after Residency. We are hearing and we are hoping things would take a good turn with the relatively new visitors on board.

**HOW DO YOU PROGRESS AS A RESEARCHER OR BECOME BETTER?**

By reading more. One needs to read more works of your teachers and non teachers researchers; your colleagues and younger colleagues.

**WHAT PERSONAL QUALITIES DOES ONE NEED TO SUCCEED IN THIS FIELD?**

Inquisitiveness births research. Asking questions, being thematic and focused. Be disciplined with your time and values. Relax your mind and engage your friends in your free moments . Find a good hobby, be passionate and responsible. You must also strive to be truthful. Plagiarism is a serious offence. People must take credit for their works. Do not copy people's works without proper referencing. It hurts creativity and dedication of researchers and creative writers.

**WHO ELSE CAN WE LOOK UP? PEOPLE THAT ARE MAKING WAVES IN THE FIELD OF MEDICAL RESEARCH**

Many of our teachers are doing great. In fact, I became the Co – Authors for many of my teachers as a resident because they would always make us participate in their research works and that was so encouraging. Your teachers should be your first motivation first therefore. For creative writers, I found John Grisham as my enigma and some of Chimamanda N. Adichie's books were top notched.

I also like reading from columnists a lot. They inspire me and I learn new words from them.

Ms Abimbola Adelakun and Mr Tunde Odesola are my favourite columnists. I mailed " Broda" (I call him that even though we have never met physically) Tunde Odesola once or twice after reading one of his excellent treatises and he replied. I was ecstatic.

Are there any words of encouragement for anyone pursuing this line of work?

Sure! There are may words of encouragement indeed. It was Sinara that said this and I quote:" Success is the best revenge."

Do you feel cheated? Go and succeed. Do you feel pushed down by others psychologically? Just strive and succeed. It doesn't have to be big things. Succeed in any where you are told to function or any positive task you are giving , succeed!

Don't ever take a no for an answer unless it is a potentially dangerous feat. If not, try it again. Let no one tell you what you can never be. Tell them what you are capable of. If any one says, your writing is unacceptable, say to them, thank you,I am just starting, you will still read the acceptable one from my winning archives.

Impossible is the miraculous word that makes you special if you would work add and separate it using the letter A: I am possible!

A big thank you to LAUMED Editor in Chief, Mr Nicholas, for his doggedness and brilliance. You are all amazing and you are a team to be proud of any day. Keep up the zeal and ring out your talents as you present LAUTECH as still the Best among equals.

I. Love. You. All.

# LAUMED JOURNAL CLUB REPORT

## Here is How to Publish a Journal during a Pandemic

2019 was an amazing year for the LAUMED Journal Club because, we made history. Despite all odds, the club under the leadership of its third generation of leaders was finally able to actualize the dreams of its founders which still resonates sonorously in the hearts of its current members. There were so much praises and accolades and songs of triumph on the occasion of the publishing of the club's first journal issue. But then, this remarkable landmark, it's sustenance was going to be the Achilles' heels of the next generation of leaders. Or maybe not. A seed had been planted and took years and sweat to finally sprout. If it dies, the gardeners are in trouble but if it grows like the cedars of Lebanon, all efforts would have been worth it. So much emphasis was placed on the continuity of the club at the first set of meetings held in 2020. Everyone was quick to understand the stakes at hand and were prepared for what's to be achieved.

It was apparent as the year began that 2020 was going to offer change. At the onset, the change was driven towards efficiency and participation of all members. With different members having different skill sets, engaging each member individually based on their peculiarities will not only improve them but also add colour to the club's activities. So gradually, we started walking uncharted waters, expressing ourselves in different ways. With a particular central theme for each month as a guide, contents were created in various media. Tosin Salam's short clip, Omooba Adetunji's fiction and Racheal's photo story formed content for February 2020's cancer awareness. These were supplemented by pasting of infographics at the hospital and colleges, efforts coordinated by Timi Amao in the clinical arm and Tosin Adekanle in the pre-clinical arm. Many of these were published on the club's medium account.

While content development was underway, training of members with appropriate skill set for academic research and writing didn't stop. A consultant Public Health Physician, Dr. Olagunjoye was on ground to give a memorable talk on Research in Medicine. One of the statements he made still resonates in the minds of those present, 'In Academic Medicine, you either publish or Perish.'

The very first meeting was held on the 19th of January 2020, a beautiful Sunday evening at the Ophthalmology shed of the LAUTECH Teaching Hospital, Osogbo with nine members of the clinical arm. The latest album (LUCID) of Franco-Nigerian singer, Asa was reviewed brilliantly by all present. The club was exploring its full description and it felt really nice. All the while the most senior members were at the thick of preparations for their final exams. They were all eager to become alumni of the club and only success in the exams would guarantee that. They all passed and have long since joined the club's esteemed alumni doing wonderfully well in their endeavors. With a smooth transition, members in the two remaining classes of the clinical school and the entire preclinical school continued to flourish and with commitment and hard work, the club would surpass its own imaginations. Really, what could go wrong? Nothing, until tragedy stuck! COVID reared its head and a new change emerged.

What do you feel when you see your plans float away slowly in a gentle breeze and all the while you could not lift a finger? Now, the most senior members were supposed to begin their part 2 MB; BS professional exam in the last week of March. The 300level students were also preparing for their fast approaching part 1 MB; BS professional exam. Plans had been made on continuation of club's activities with weekly schedules for content submission. A tentative deadline was fixed for April for the submission of Articles for the year's journal. So much consultation was going. We could see ourselves getting things done. It really was a good time but like everyone else in the world we were shaken. No one knew for sure what to expect. At first, we were dealing with a disease that could be unpredictable in action later we had to deal with the fallouts of the interruption in our lives. For us as medical students, it was a major upset. We already lost much time to industrial actions and it was quite depressing to deal with the feeling of stagnation. Most of the plans were put on hold. No one was sure when the pandemic would be over, if it will be over. At the time, a journal wasn't on the priority lists at all especially when we have all been scattered across different towns and cities and constricted by physical

interactions.

However, from the ashes, the phoenix rose and hope began to seep through the crack. We were regaining our strength and started to learn to deal with the pandemic and its fallouts. The club got to its feet and by May, we were publishing brilliant weekly articles and having virtual meetings. In June, Toni Olutayo Oluwatuminu of the University of Ibadan gave an expose on Rudiments of Editing. Some days later, Dr. Chiagozie Jideana a former Editor-in-Chief of IFEMED Journal Club took a session on Review Articles. In July, A Rwandan Physician Dr. Pacifique Hagenimana led a virtual training session. As in, we went global!

By now, the club was in its full swing. There was no stopping. In August, new deadlines were set for the journal release. And by the 10th of August, calls for articles had been sent; garnering submissions from all over the world.

Truly, no manual on 'How to Publish a Journal in a Pandemic' would be complete if it doesn't touch on commitment, resilience, hard work, discipline and optimism. The year might have been rough and unprepared for but, the LAUMED JC survived and came out stronger and better. Verily, if we can find safe harbor baring the odds, you can pull through whatever storm life throws at you too.

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